MEDICAL TECHNOLOGY LICENSURE EXAM REVIEW

CLINICAL CHEMISTRY

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EQUIPMENTS AND SUPPLIES

TYPES OF GLASS MATERIAL - Primarily characterized in terms of thermal and chemical resistance	
1. **Borosilicate** *(silica-containing)*	- Tolerates heating and sterilization for lengthy periods - May cloud with strong alkali and is subject to scratching - Most common brand: *Pyrex*; Other: Kimax
2. **Aluminosilicate** *(silica-containing)*	- Has higher silica content than Borosilicate; *6x stronger* than borosilicate - Better able to resist scratching and alkali attack - Most common brand: *Corex*
3. **Soft glass** *(Boron-free glass)*	- Low thermal resistance (cannot be subjected to heating) - Can be used with strong acids and alkali
4. **Flint glass** *(Soda lime glass)*	- Most inexpensive (cheapest); commonly used to manufacture disposables - Made of the oxides of silicon, calcium, and sodium - Disadvantage: releases alkali causing errors in certain determinations
5. **Low actinic glass** *(Amber-colored/ coated glass)*	- Reduces light transmission - Used to contain photosensitive substances

*Silica: provides high thermal resistance or heat tolerance

TYPES OF PLASTIC MATERIALS Primarily characterized in terms of thermal and chemical resistance		
1. **Teflon** *Polytetrafluoroethylene*	- Most excellent temperature tolerance and unparalleled chemical resistance	Can be autoclaved
2. **Polypropylene**	- Resistant to most chemicals - Used for pipet tips, test tubes	Can be autoclaved
3. **Polycarbonate**	- Clear; stronger than polypropylene (resistant to shattering) and has better temperature tolerance - Chemical resistance is not as good - Used for centrifuge tubes and graduated cylinders	Can be autoclaved
4. **Polyethylene**	- Resistant to most chemicals (except concentrated acids) - Used for disposable transfer pipets, test tubes, bottles	Cannot be autoclaved
5. **Polystyrene**	- Rigid, clear; used for test tubes, graduated tubes	Cannot be autoclaved
6. **Polyvinyl chloride**	- Soft and flexible but porous - Frequently used as tubing	Cannot be autoclaved

GLASS PIPETTES

PIPET	TYPE	DESIGN	CALIBRATION MARKS	USE
1. **Volumetric** *(most accurate)*	Transfer pipets	TD/SD	Single calibration mark	Nonviscous samples; standards
2. **Ostwald-Folin**		TD/BO		Viscous fluids
3. **Serologic**	Graduated or Measuring pipets	TD/BO	Graduations down to tip	Serial dilution; measuring reagents
4. **Mohr**		TD/SD	Graduations between two marks	

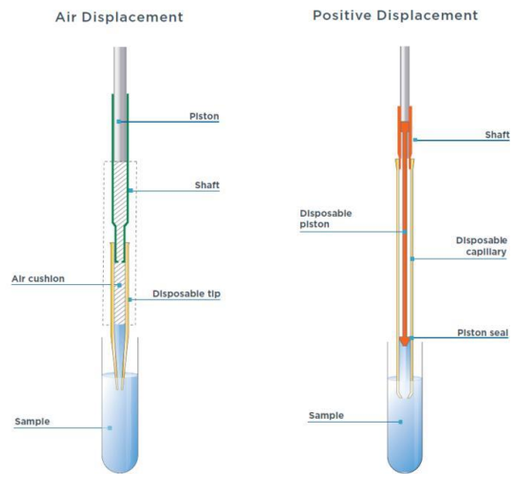
Additional:
- To contain = hold but do not deliver the volume they contain because of the tendency of liquids to cling to the inside wall of the glass
- Tolerance value is proportional to capacity and inversely proportional to accuracy
- Transfer pipet = only 1 calibration mark; no subdivisions
- Graduated/measuring pipet = there are subdivisions; you can measure and deliver fractions of the capacity
- Self-draining pipet (SD): contents drain by gravity; without the need of aspirator bulb
- Blow-out (BO): any amount left in the tip after drainage is blown out to the receiving vessel; has etched rings near the mouth

SEMI-AUTOMATIC MICROPIPETTORS

Types	1. ***Air displacement*** - Uses suction to draw sample into a disposable polypropylene tip - Piston does not come in contact with liquid (air gap) 2. ***Positive displacement*** - Operates like hypodermic syringe - Piston is in direct contact with the liquid being aspirated - Tips are reusable but must be rinsed out - Components: disposable piston, disposable capillary (glass or Teflon), piston seal
Calibration *(similar to volumetric glasswares)*	1. ***Gravimetric*** - Weight of *distilled water* delivered using an analytical balance; most accurate - Calibration temperature: 20°C 2. ***Spectrophotometric*** - Absorbance of *colored solution* delivered - Examples of colored solution: Potassium dichromate or para-nitrophenol

SEMI-AUTOMATIC MICROPIPETTORS

Air-Displacement Pipettes	Positive-Displacement Pipettes
- Always have *air cushion (dead volume)* between the pipette piston and the liquid sample	- Piston is in direct contact with the sample, therefore, there is *no air cushion* and the *aspiration force remains constant*
- The piston is integrated into the lower part of the pipette. When the push button is pressed, air is displaced by the piston. The volume of air displaced is equivalent to the volume of liquid aspirated.	- The disposable piston is part of the tip (along with piston seal and disposable capillary). As the push button is released, the piston moves up and the ambient pressure forces the desired volume of liquid through the orifice into the capillary.
- Recommended for aqueous samples and for general laboratory work.	- Recommended for *very viscous or high density samples* such as glycerol and blood and other problem samples e.g. volatile, radioactive, corrosive, hot and cold samples



CENTRIFUGES

Types	1. ***Horizontal or Swinging bucket*** - Tubes attain a horizontal position during spinning and a vertical position when at rest - Capable of speeds up to about 3000 rpm (limited by heat buildup due to air friction) - Flat, tightly packed sediment 2. ***Fixed-angle or angle-head*** - Tubes are at fixed angle when rotating - Capable of higher speeds with much less heat build-up - Capable of speeds up to 7,000 rpm or 9,000 g (RCF) - Sediment is slanted - Disadvantage: not recommended for procedures which require decantation 3. ***Ultracentrifuge*** - Used to separate layers of different specific gravities - Usually refrigerated to counter the heat produced due to friction - Capable of speeds up to 100,000 rpm or 165,000 g (RCF) 4. ***Cytocentrifuge*** - Used for body fluid cell counts - Speed = 200-2,000 rpm (low speed centrifuge) - Meant to increase the cell yield, especially for CSF analysis
Units used	- rpm: revolutions per minute - RCF (g): relative centrifugal force (gravity units) - S: Svedberg unit; unit used in ultracentrifugation - rpm to RCF = *1.118×10^{-5} x radius of centrifuge head in centimeter x rpm^2*
Quality control	- **Parameters** ○ Speed: checked using tachometer or strobe light ○ Timer: checked using a stopwatch ○ Temperature: if refrigerated; checked using thermometer - **Frequency** ○ Every three months or quarterly

REAGENTS

CHEMICALS	
1. Analytic Grade Reagent (ACS)	- High degree of purity suitable for use in most analytical procedures
2. Ultrapure Grades *(Spectrograde, Nanograde)*	- Purer than ACS; for procedures which require maximum purity and minimal interference - For specific procedures such as chromatography (HPLC), AAS, immunoassays, molecular diagnostics, and standardization techniques
3. Chemically Pure Grade	- Impurity limitations not stated - Not acceptable for research and laboratory techniques unless further purification is carried out or a reagent blank is included during analysis
4. United States Pharmacopeia/National Formulary Grade	- Primarily used to manufacture drugs - Purity standards may not meet assay requirements - Not recommended for routine use
5. Technical or Commercial Grade	- Should not be used in the clinical laboratory - For industrial use only

REAGENT WATER	
1. Type I	- Purest type (has been subjected to 2 or more purification techniques such as distillation, deionization, reverse osmosis, or filtration) - Recommended for procedures which require maximum purity and minimal interference; *for standard preparation* and for other procedures which require high degree of accuracy and precision
2. Type II	- Acceptable for most laboratory procedures including reagent preparation
3. Type III	- Can be used for some qualitative tests but not for routine analyses and reagent preparation - Water source for the preparation of Type I or II water and for washing glassware (should be given a final rinse with either type I or type II)

TYPES OF REAGENT WATER

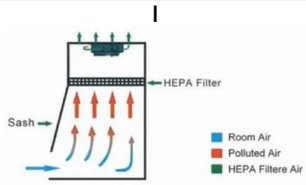
Characteristic	Type I	Type II	Type III
Maximum colony count *(CFU/mL)*	<10	1,000	Not specified
pH	Not specified	Not specified	5.0-8.0
Silicate *(mg/L SiO_2)*	0.05	0.1	1.0
Uses	Procedures that require maximum water purity e.g. preparation of standards, buffers, controls, and in quantitative analytical procedures, electrophoresis, toxicology screening tests, and HPLC	General laboratory tests that do not require type I water e.g. qualitative chemistry procedures and most procedures done in hematology, immunology, microbiology, and other clinical test areas	Some qualitative laboratory tests e.g. general urinalysis; water source for preparation of type I or type II water; and washing and rinsing of glassware

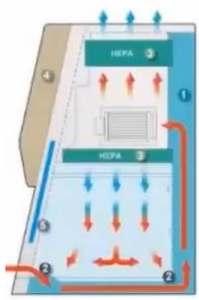
LABORATORY SAFETY

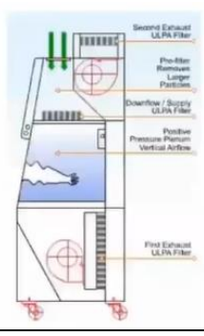
CONTROL OF BIOHAZARDS

1. **Handwashing**
   - Most important means of preventing the spread of infection
   - If hands are visibly soiled (directly came in contact with potentially biohazardous substance), wash hands for *1 to 2 minutes*, using *soap and water*
   - If not visibly soiled (after removing gloves), wash hands for *at least 30 seconds* or use *alcohol-based hand cleansers*
2. **Personal protective equipment**
   - Perform handwashing before donning
   - Sequence of donning: gown → mask → goggles → gloves
   - Sequence of doffing: gloves → goggles → gown → mask
3. **Decontamination of Body Fluid Spills**
   - Use *5.25% sodium hypochlorite* or *10% bleach* (1:10 dilution; 1-part household bleach + 9 parts water)
   - Contact time of at least 20 minutes (also applies to the decontamination of glasswares used for procedures involving biohazardous substances or body fluids)
     - HBV: inactivated after 10 minutes
     - HIV: inactivated after 2 minutes

4. Biological Safety Cabinet

Class	Description	Use
I 	- Air passes through *1 HEPA (High efficiency particulate air) filter* before being exhausted - Open front	- Minimal personnel protection and environmental protection. - Doesn't protect work surface (air that enters the work surface is unfiltered)

II 	- Air passes through *2 HEPA filters*
	- **A1**
	  ○ 70% of air is recirculated, 30% exhausted into the room
	  ○ 75 lfm
	- **A2**
	  ○ Recommended BSC class for COVID-19 RT-PCR testing
	  ○ Same as A1 but with higher intake air velocity (100 linear feet per minute or lfm)
	- **B1**
	  ○ 30% recirculated, 70% exhausted
	  ○ Hard ducted (air expelled outside the building)
	- **B2**
	  ○ 100% exhausted to the outside of the building (hard ducted)
		- Most commonly used in micro labs; provides protection for both worker and work

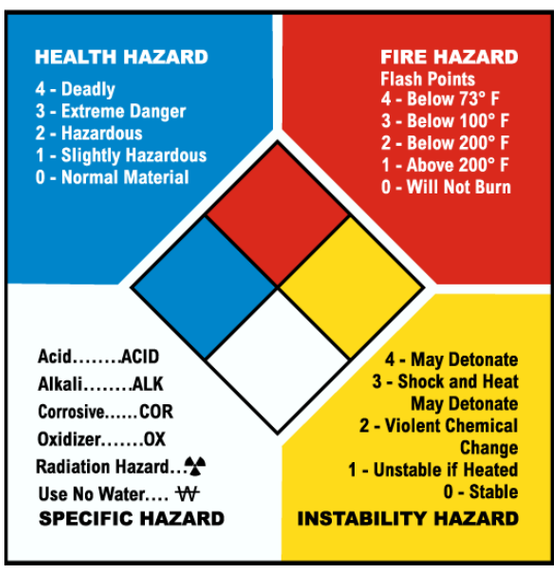
III 	- Completely enclosed with glove ports; no opening; gas tight chamber - Equipped with negative pressure - 3 HEPA or ULPA filters - HEPA filter: has 99.97% efficiency (at 0.3 microns) - ULPA: Ultra Low Penetration Air; has 99.99% efficiency (at 0.12 microns)	- For extremely hazardous organisms - Recommended for BSL 4 organisms (for extremely hazardous organisms); for culture and isolation of SARS-CoV-2

CHEMICAL HAZARD

- ***MSDS (Material Safety Data Sheets)***
  - Must be available for reference by the staff
  - Major source of safety information for employees who may use hazardous materials in their occupations
  - Contain everything you need to know about a chemical reagent
    - Product name and identification
    - Hazardous ingredients
    - Permissible exposure limit
    - Physical and chemical data
    - Health hazard data and carcinogenic potential
    - Primary routes of entry
    - Fire and explosion hazards
    - Reactivity data
    - Spill and disposal procedures
    - PPE recommendations
    - Handling
    - Emergency and first aid procedures
    - Storage and transportation precautions
    - Chemical manufacturer's name, address, and telephone number
    - Special information section

CHEMICAL STORAGE REQUIREMENTS	
SUBSTANCE	**STORED SEPARATELY**
Flammable liquids	Flammable solids
Mineral acids	Organic acids
Caustics	Oxidizers
Perchloric acid	Water-reactive substances
Air-reactive substances	All other chemicals
Heat-reactive substances requiring refrigeration	

- ***TLV (Threshold Limit Value)***
  - Allowable exposure value during an 8-hour shift
  - Inversely proportional to toxicity
  - ↑ Toxicity = ↓ TLV
- ***NFPA (National Fire Protection Agency)***
  - Standard hazards identification scheme

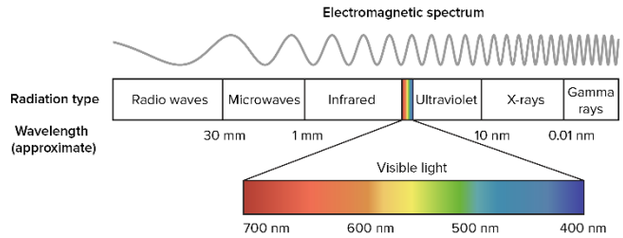


DEGREES OF HAZARD (GENERAL)	
0	Non-hazardous
1	Slightly hazardous
2	Moderately hazardous
3	Seriously hazardous
4	Extremely hazardous

RADIATION HAZARD

- Non-ionizing forms of radiation emitted by equipment
- Non-ionizing radiation: less hazardous than ionizing radiation (ex. X-rays and gamma rays)

Type	Wavelength	Example
Ultraviolet	<400nm	Germicidal lamps used in biologic safety cabinets
Visible Spectrum	400-700nm	General illumination
Infrared	>700nm	Heat lamps, lasers
Microwaves	3 um – 3mm	Microwave energy beam used to accelerate tissue staining
Low frequency	>1 cm	Radiofrequency coil in ICP-MS (Inductively-coupled plasma – Mass spectrometry)



COLOR CODING SCHEME FOR HEALTH CARE WASTES	
Type of Waste	**Color of Container/Bag**
1. Non-infectious dry waste	**Black**
2. Noninfectious wet waste	**Green**
3. Infectious and pathological wastes	**Yellow**
4. Chemical and Pharmaceutical wastes	**Yellow with black band**
5. Radioactive wastes	**Orange**
6. Sharps and pressurized containers	**Red**

FIRE HAZARD

Class	Combustible Materials	Examples	Extinguishers
A	Ordinary combustibles	Wood, paper, cloth, rubber, etc.	Pressurized water (A) Foam Dry chemical *(multipurpose; ABC powder)*
B	Flammable liquids and gases	Grease, oil, paint, solvents, gasoline	Foam Dry chemical (ABC) Carbon dioxide (BC) Halon
C	Electrical equipment	Electrical panel, motor, wiring, etc.	Dry chemical (ABC) Carbon dioxide (BC) *(also good for extinguishing computers because it does not leave residue)* Halon *(recommended for computers)*
D	Combustible metals	Magnesium, aluminum, etc.	Special dry chemical Sand
K	Oil, grease	Cooking oils, animal fats, vegetable oils	Wet chemical *(saponifying agents)*

- Actions to take in the event of fire:
  - **R**escue, **A**larm, **C**onfine/**C**ontain, **E**xtinguish/**E**vacuate
- Correct use of fire extinguishers:
  - **P**ull pin, **A**im Hose/Nozzle, **S**queeze Handle, **S**weep from Side to Side

LABORATORY MATHEMATICS

Applications:
1. Calculation used in reagent preparation
2. Converting values from conventional to SI unit and vice versa

UNITS OF MEASURE

SELECTED SI AND NON-SI UNITS		
BASE	**QUANTITY NAME**	**SYMBOL**
Length	Meter	m
Mass	Kilogram	kg
Time	Second	s
Electric current	Ampere	A
Thermodynamic temperature	Kelvin	K
Amount of substance	Mole	mol
Luminous intensity	Candela	cd
SELECTED DERIVED UNITS		
Frequency	Hertz	Hz
Force	Newton	N
Celsius temperature	Degree Celsius	°C
Catalytic activity	Katal	kat
SELECTED ACCEPTED NON-SI		
Time, minute	60s	min
Liter (volume)	1 $dm^3 = 10^{-3}\ m^3$	L
Angstrom	$0.1nm = 10^{-10}m$	Å

PREFIXES USED WITH SI UNITS		
FACTOR	**PREFIX**	**SYMBOL**
10^{-18}	atto	a
10^{-15}	femto	f
10^{-12}	pico	p
10^{-9}	nano	n
10^{-6}	micro	u
10^{-3}	milli	m
10^{-2}	centi	c
10^{-1}	deci	d
10^{1}	deka	da
10^{2}	hector	h
10^{3}	kilo	k
10^{6}	mega	M
10^{9}	giga	G
10^{12}	tera	T
10^{15}	peta	P
10^{18}	exa	E

EXPRESSIONS OF CONCENTRATION

Molarity	$\frac{grams\ (g)\ of\ solute}{gram\ molecular\ weight\ x\ liter\ of\ solution}$	number of moles of solute per liter (L) of solution
Molality	$\frac{grams\ (g) of\ solute}{molecular\ weight\ x\ kilogram\ of\ solvent}$	number of moles of solute per kilogram (kg) of solvent
Normality	$\frac{grams\ (g) of\ solute}{equivalent\ weight\ x\ volume\ (L)}$	number of equivalent weights of solute per liter (L) of solution
% w/v	$\frac{\%\ solution\ desired\ x\ volume\ desired}{100}$	amount of solute per 100 total units of solution; often expressed as g/100 mL or g/dL
Dilution	$\frac{Volume\ of\ solute/sample}{Total\ Volume\ of\ solution}$	volume of sample divided by total volume of solution; expression of relative concentration
Ratio	$\frac{Volume\ of\ solute}{Volume\ of\ solvent}$	amount of something relative to another; usually expressed as part per part

CALCULATIONS USED IN REAGENT PREPARATION

To determine the molarity of solution given the w/v (g/L):	M or $\frac{mol}{L} = \frac{g/L}{MW}$
To determine the normality of solution given the w/v (g/L):	N or $\frac{Eq}{L} = \frac{g/L}{EW}$ EW = $\frac{MW}{valence}$
To determine the molarity of solution given the normality:	M = $\frac{N}{V}$; N ≥ M N > M: Valence is greater than or equal to 2 N=M: Valence is equal to one
To determine the normality of solution given the molarity:	M x V
To determine the w/v (g/L) given the molarity:	$\frac{g}{L}$ = M x MW
To determine the w/v (g/L) given the normality:	$\frac{g}{L}$ = N x EW
To compute for volume or concentration of diluted solutions:	$C_1V_1 = C_2V_2$

SAMPLE PROBLEMS:

What is the molarity of a 1-liter solution containing 73 g hydrochloric acid?

(H=1, Cl=35.5)
MW 36.5 g/mol

$M = \frac{73\ g/L}{36.5\ g/mol} = 2$

What is the molarity of a 1-liter solution containing 10 g sodium hydroxide?

(Na=23, O=16, H=1)

$M = \frac{10\ g/L}{40\ g/mol} = 0.25$

What is the normality of a 1-liter solution containing 24.5 g H_2SO_4 (sulfuric acid)? What is its molarity?

(H=1, S=32, O=16)
MW = 98 g/mol
EW = 98/2 = 49

$N = \frac{24.5\ g/L}{49\ g/Eq} = \boxed{0.5}$

$M = 0.5/2 = 0.25$

What is the molarity of 0.3N H_3PO_4?

$M = N/V = 0.3/3 = 0.1$

What is the normality of 3 M sulfuric acid?

$N = 3 \times 2 = 6$

How many milliliters of 70% ethanol are needed to prepare 35 mL of 40% ethanol?

$V_1 = \frac{C_2V_2}{C_1} = \frac{40(35)}{70} = \frac{1,400}{70} = 20\ mL$

How much serum is needed to prepare 2 mL of a 1:10 dilution?

$\frac{x}{2} = \frac{1}{10}$

$x = \frac{2}{10} = 0.2\ mL$

A mixture consists of 4 parts of Solution A and 6 parts of Solution B. How many mL of Solution A and B are needed to prepare a 3000 mL of the mixture?

Sol A $= \frac{4}{10} \times 3,000 = 1,200\ mL$

Sol B $= \frac{6}{10} \times 3,000 = 1,800\ mL$

UNITS AND CONVERSION FACTORS FOR SELECT CHEMISTRY ANALYTES			
ANALYTE	**CONVENTIONAL UNIT**	**RECOMMENDED SI UNIT**	**CONVERSION FACTOR**
Glucose	mg/dL	mmol/L	0.0555
BUN			0.357
Uric acid			0.0595
Cholesterol			0.026
Triglyceride			0.0113
Calcium			0.25
Phosphorus			0.323
Magnesium *(Mg^{2+})*	mEq/L		0.5
$Na^+/K^+/Cl^-/HCO_3^-$			1
Creatinine	mg/dL	µmol/L	88.5
Bilirubin			17.1
Ammonia	µg/dL		0.588
Iron			0.179
Cortisol			0.0276
T_4		nmol/L	12.9
T_3	ng/dL		0.0154
Folic acid	ng/mL		2.27
Vitamin B_{12}	pg/mL	pmol/L	0.738
Proteins (TPAG)	g/dL mg/dL *(for fibrinogen and transferrin)*	g/L	10 0.01

- **Conventional unit**
  - Also known as traditional unit
  - Often expressed in weight over/per volume
  - Commonly used expression: mg/dL
- **SI unit**
  - Molecular weight (molar units) of the compound is taken into consideration
  - Commonly used expression: mmol/L
- **CU to SI**: multiply the value by the conversion factor
- **SI to CU**: divide the value by the conversion factor
- Additional notes:
  - Cortisol is synthesized from cholesterol, that is why they have almost similar molecular weights and conversion factor

SAMPLE PROBLEMS:

1. 90 mg/dL glucose = 5 mmol/L
2. 2.26 mmol/L triglyceride = 200 mg/dL
3. 10 mg/dL calcium = 2.5 mmol/L = 5 mEq/L
4. 135 mEq/L sodium = 135 mmol/L
5. 0.2 g/dL IgD = 2 g/L IgD
6. 500 mg/dL fibrinogen = 0.5 g/dL = 5 g/L
7. 600 mg/dL protein = 0.6 g/dL = 6 g/L

LABORATORY STATISTICS

MEASURES OF CENTER

- ***Mean ($\bar{x}$)***
  - average or arithmetic mean
  - commonly used in QC as a measure of center
  - calculated by dividing the summation of observations by the number of observations
  - corresponds to the center of the Levey-Jennings chart or to the peak of a bell curve or Gaussian curve

$$\bar{x} = \frac{\sum x}{n}$$

- ***Median***
  - Central value or midpoint of a data set after the values have been ranked or ordered
- ***Mode***
  - most frequently occurring value in a data set
  - can be multimodal, bimodal, or no mode

- TAKE NOTE! If the data set is normally distributed, these three are essentially or statistically the same or identical

MEASURES OF SPREAD/VARIATION

- ***Standard Deviation (s)***
  - Distribution of data points around the mean
  - Most commonly used measure of spread especially when constructing QC charts
  - Square root of variance; **variance** is the square of SD

$$s = \sqrt{\frac{\sum(x-x)^2}{n-1}}$$

- ***Coefficient of Variation***
  - Best indicator of precision
  - Used in lipoprotein analysis
  - CV and precision has an inverse relationship; a low CV indicates high precision
  - Standard deviation divided by the mean and multiplied by 100
  - **Sample problem!** Determine the coefficient of variation for a set of data where the mean is 89 mg/dL and the 2 standard deviation (2SD) value is 14 mg/dL

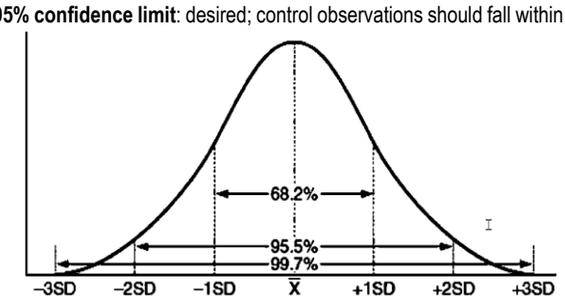
$$CV = \frac{s}{\bar{x}} x100$$

$$CV = \frac{SD}{\bar{x}} \times 100 = \frac{7}{89} \times 100 = 7.9\%$$

- ***Range***
  - Difference between the highest and lowest values
  - Not usually used since it is greatly affected by the presence of outliers
  - Reference range: discouraged from use
  - Reference interval: represents the 95% confidence limit

GAUSSIAN/NORMAL DISTRIBUTION

- Data points are distributed symmetrically around the mean (bell curve) with most values close to the center; mean, median, and mode are identical
- **95% confidence limit**: desired; control observations should fall within this limit



- 68-95-99 rule (if the data set is normally distributed)
  - 68% of values fall within 1SD
  - 95% of values fall within 2SD
  - 99% of values fall within 3SD

REFERENCE INTERVAL STUDIES

- ***Verifying a reference interval***
  - Done to confirm the validity of an existing or published RI for an analyte
  - Requires at least 20 study individuals
  - RI is adopted if <10% of the subjects fall outside the range
- ***Establishing a reference interval***
  - Done when there is no existing RI for an analyte or when transference studies fail
  - Requires at least 120 study individuals
  - RI is set as $\bar{x} \pm 2s$

METHOD EVALUATION

- ***Precision study***
  - First step in method evaluation
- ***t test*** (Comparative statistics)
  - Compare the *means* of two groups of data or the *accuracy* of two methods
- ***F test*** (Comparative statistics)
  - Compares the *standard deviation* of two groups of data or the *precision* of two procedures
- ***Linear Regression*** (Correlated statistics)
  - Independent variable/Reference method → plotted along the X-axis
  - Dependent variable/New method → plotted along the Y-axis

MEASURES OF DIAGNOSTIC EFFICIENCY

Diagnostic sensitivity	- ability of a test to detect a given disease or condition - proportion of individuals who have the disease which also tested positive	$\frac{TP}{TP+FN} x\ 100$
Diagnostic specificity	- ability of a test to detect the absence of a given disease or condition - proportion of individuals who do not have the disease which also tested negative	$\frac{TN}{TN+FP} x\ 100$
Positive predictive value	- probability that a positive test indicates disease - proportion of individuals who tested positive and who truly have the disease	$\frac{TP}{TP+FP} x\ 100$
Negative predictive value	- probability that a negative test indicates absence of disease - proportion of individuals who tested negative and don't have the disease	$\frac{TN}{TN+FN} x\ 100$

Result	Disease	No disease	Total
Positive	True Positives	False Positives	TP + FP
Negative	False Negatives	True Negatives	FN + TN
Total	TP + FN	FP + TN	

- ***Analytical sensitivity***
  - Ability of a method to detect the smallest concentration of an analyte
  - Desirable in screening tests
  - Not prone to false negatives
- ***Analytical specificity***
  - Ability of a method to detect only the analyte of interest
  - The ability of a method to specifically determine the concentration of a given analyte in the presence of interfering substances in a sample
  - Desirable in confirmatory tests
  - Not prone to false positives (cross reactions)

QUALITY MANAGEMENT

- Overall process that encompasses quality assurance, benchmarking, and other aspects that provide for quality improvement as a means to meet set standard
- QM > QA > QC
- **Quality system**: refers to all of the laboratory's policies, processes, procedures, and resources needed to achieve quality testing
- **Quality assurance/assessment**: process by which the laboratory ensures quality results by closely monitoring pre-analytical, analytical, and post – analytical stages of testing; under QA is QC

QUALITY ASSURANCE

- 3 Phases of QA
  - ***Pre-analytical***: include test requisition, patient preparation, patient identification, specimen collection, labeling, specimen transport, handling, and processing
  - ***Analytical***: include reagents, preventive maintenance of equipment, calibration, analysis of samples, and quality control
  - ***Post-analytical***: include verification of calculations and reference ranges, flagging and notification of panic/critical values, delta checks, reporting of results
    - ***Delta Check***: an algorithm in which a current laboratory result is compared with results obtained on a previous specimen from the same patient; important for detecting pre-analytical and analytical errors

QUALITY CONTROL

- Goal of QC: ***ensure reliability***; to maintain both accuracy and precision
- **Accuracy**
  - closeness of the result to the true or actual value
- **Precision**
  - ability to produce a series of results that agree closely with each other
  - commonly expressed in terms of coefficient of variation; inversely proportional with CV
  - also called reproducibility
- ***Internal/Intralaboratory QC***
  - Involves the analysis of at least 2 levels of control every 24 hours for at least 20 days
  - Important for the daily monitoring of accuracy and precision of analytical methods
- ***External/Inter-laboratory QC***
  - Proficiency testing (EQAS or NEQAS in the Philippines)
  - Involves testing samples of unknown concentration of analytes sent periodically by regulatory agencies (national reference laboratories) to participating laboratories
    - Lung Center of the Philippines – NRL for Clinical Chemistry
  - **SDI >2.0** indicates poor performance; outliers; not in agreement with other participating laboratories

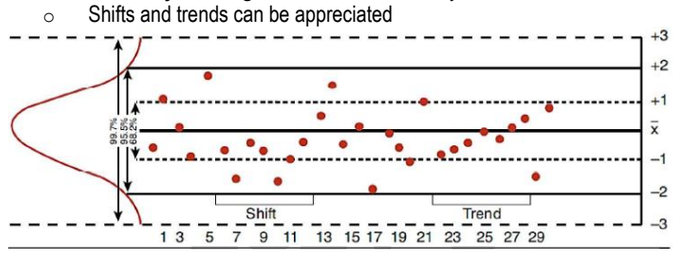
$$SDI = \frac{Lab\ mean - Peer\ group\ mean}{Peer\ group\ SD}$$

ANALYTICAL VARIATION *TYPES OF ERROR IN LABORATORY TESTING*	
RANDOM ERROR	**SYSTEMATIC ERROR**
- due to chance or an unpredictable cause - **affects precision** - affects few control observations or samples	- influences observations consistently in one direction - **affects accuracy** - persists until troubleshooting is performed - affects all observations in a run - shift and trend
- Examples: o instrument instability o dirty optics o improper mixing of sample and reagent o temperature and voltage fluctuations o operator variability e.g. pipetting error	- Examples: o deterioration of reagents or change in reagent lot, o unstable reagent blanks o calibration error o changes in the standard concentration o contaminated control solutions o diminishing lamp power or deteriorating light source o pipette or maintenance error

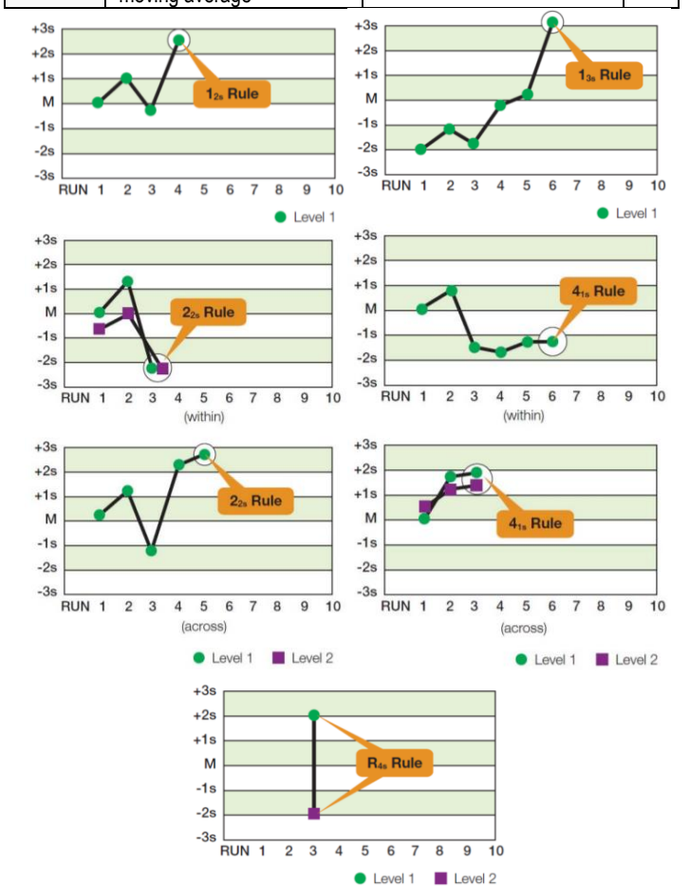
- Patterns in QC charts that indicate errors
  - **Shift**
    - formed by control values that distribute themselves on one side or either side of the mean for 6 consecutive days; abrupt or sudden change
    - most common cause: calibration error
  - **Trend**
    - formed by control values that continue to increase or decrease (gradually) over a period of 6 consecutive days passing through the mean
    - most common cause: deterioration of reagents
  - **Outliers**
    - highly deviating control values caused by random or systematic errors; loss of precision

INTERPRETATION OF QUALITY CONTROL RESULTS

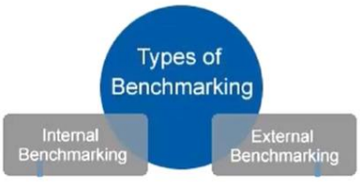
- **Shewart Levey-Jennings Chart**: most commonly used QC chart
  - Shifts and trends can be appreciated



WESTGARD CONTROL RULES			
Rule	**Description**	**Use**	
1_{2s}	One control value exceeds the $\bar{x} \pm 2SD$	Warning rule; trigger the inspection of the control data Accept the run if there is no other violation Not recommended for use as a rejection rule because of high false alert rate	Detect random error
1_{3s}	One control value exceeds the $\pm$ 3SD	Detects imprecision or systematic bias	
R_{4S}	Range between two levels of control within a run exceeds 4SD	Detects imprecision or random error	
2_{2s}	Two consecutive values exceed the $\pm$ 2SD in the same direction	Detects bias	Detect systematic error
4_{1s}	Four consecutive control values exceed $\bar{x} \pm 1SD$	Detects bias trend	
8_{1s}	Eight consecutive observations for the same level of control exceed 1SD in the same direction		
10_x	Ten consecutive control values are on the same side of the mean	Detects bias; but not recommended	
CUSUM	Cumulative sum of SDI	Detects bias trend	
EWMA	Exponentially weighted moving average		



QUALITY MANAGEMENT APPROACHES



- **Internal benchmarking**
  - setting standards and try to meet or exceed such standards for your organization/lab
- **External Benchmarking**
  - compares a laboratory 's productivity with that of other laboratories to find out what processes or resources have been used to achieve high productivity
  - when you adopt best practices used by other laboratories

PDCA/DEMING CYCLE

- Plan, Do, Check, Act
- Most commonly used and simplest approach used in lab quality management

CQI

- TQM tool that strives to continually improve practices and not just meet established standards

SIX SIGMA

- applied in the clinical laboratory to reduce the frequency of test errors or defects per million opportunities (DPMO)
- 3.4 defects per million laboratory tests

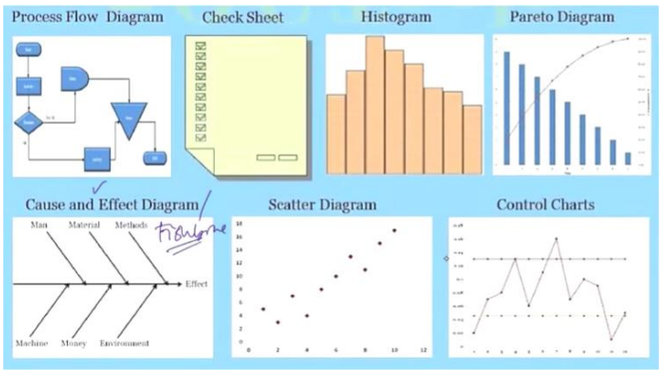
- Lean - Six Sigma Colored Belt Designations
  - **Master**
    - Lean Six Sigma advisers/coaches
    - Black belt with at least 2 years of experience
  - **Black belt**
    - Lead and manage projects
    - Coaches/leaders
    - Dedicate as much as 100% of their time to QI projects
    - With at least two years of experience
  - **Green belts**
    - Lead or support projects
    - Contribute approximately 20% of their time to quality improvement projects while delivering their normal job functions
  - **Yellow belt**
    - Understands the basics of lean six sigma

LEAN

- designed to reduce waste (unnecessary steps) and improve a process such as TAT (turnaround time)

7 BASIC QUALITY TOOLS

- Use of Pareto charts (in strategic planning), fishbone diagrams (in analyzing TATs; cause and effect), and flow charts (for improving TATs)



BLOOD COLLECTION AND SPECIMEN CONSIDERATIONS

VENIPUNCTURE

- ***GENERAL PRECAUTIONS AND TECHNIQUES***
  - Sites of puncture:
    - Veins in the antecubital fossa
      - Median cubital
      - Cephalic
      - Basilic (close to the median nerve and brachial artery)
    - Alternatives: dorsal hand, wrist, ankle, and foot veins
  - Tourniquet application: *<1 minute*
    - If BP cuff is used, inflate to *40-60 mm Hg*
  - Antiseptics:
    - 70% isopropyl alcohol
    - Ethanol assay (alcohol must not be used; may falsely increase results in osmometry and glass chromatography): Benzalkonium chloride or Zephiran
    - Culture (double antisepsis): 70% alcohol → Iodophor/Chlorhexidine
- ***VENIPUNCTURE METHODS***
  - *Needle and Syringe*: not recommended by the CLSI due to safety and specimen quality issues
    - Standard needle: 21 g for adults, length = 1-1.5in
    - Recommended angle between needle and vein = 15-30°
  - *Butterfly/Winged Infusion Set*: recommended for drawing blood of infants and children, hand veins, and difficult draw situations
  - *Evacuated Tube System*: commonly used; addressed the issues in needle and syringe method; consists of double-headed needle, adapter, color-coded evacuated tube

CATEGORIES OF ADDITIVES

	ADDITIVE/S	COLOR CODE/S	COMMENTS
Anticoagulants	EDTA	Lavender *(Hematology)* Pink *(Crossmatching)*	Removal of calcium *(EDTA and citrate both chelate calcium while Oxalate precipitates calcium)* Contraindicated for: F↑: Sodium and potassium F↓: Calcium, Magnesium, Iron, Phosphate, CK, ALP, AMS
	Citrate	Light blue (Coagulation) Black (ESR)	
	Oxalate	Gray	
	Heparin	Green	Inhibits thrombin (most commonly used anticoagulant in CC)
Clot activators	Silica, celite, diatomite	Red plastic tubes	STAT serum tests
	Thrombin	Orange	
Separator tubes	Gel + clot activator (SST)	Red/Black Gold	Contraindication: TDM (gel absorbs certain drugs; causes falsely decreased values)
	Gel + anticoagulant (PST)	Light Green Green/Gray	
Antiglycolytic agent	Sodium fluoride	Gray	Prevents glycolysis Primarily for glucose but may also be used for lactate and ethanol Inhibits urease and uricase (may falsely decrease BUN and uric acid levels respectively)

ORDER OF DRAW IN MULTIPLE COLLECTION USING ETS

Mnemonic	Tube	Inversions
Stop	SPS/Sterile tube for culture	8-10x
Light	Light blue (Coagulation)	3-4x
Red	Red (Plastic)	5x (clot activators)
Stay	SST (Serum Separator Tube)	
Put	PST (Plasma Separator Tube)	8-10x
Green	Green (Heparin)	
Light	Lavender (EDTA-containing tubes)	
Go	Gray (Fluoride/Oxalate)	

VENIPUNCTURE NEEDLES		
GAUGE	**NEEDLE TYPE**	**TYPICAL USE**
20 (yellow)	Multisample (ETS), hypodermic (needle and syringe set)	Sometimes used when large-volume tubes are collected or large-volume syringes are used on patients with normal-size veins
21 (green)		Considered the standard venipuncture needle for routine venipuncture on patients with normal veins or for syringe blood culture collection
22 (black)		Used on older children and adult patients with small veins or for syringe draws on difficult veins
23 (blue)	Butterfly	Used on the veins of infants and children and on difficult or hands veins of adults

- Needles for drawing blood range from 19 to 23 gauge with lengths of 1-1.5in
- The most common needle size for adult venipuncture is 21 gauge with a length of 1 inch which has the advantage of providing better control during venipuncture
- Butterfly needles are typically ½ to ¾ in. long

CAPILLARY PUNCTURE

- Indications:
  - Newborn screening (blood spots), daily monitoring of DM patients, POCT
- General precautions and techniques
  - Sites of puncture
    - Plantar heel surface (heelstick; medial or lateral part): Newborns and infants
    - 3rd or 4th finger (adults; palmar surface of the hand that is less used), big toe (children), ear lobe (site of choice for arterialized capillary puncture)
  - Recommended length of the lancet blade for heelsticks: *1.75 mm*
- Consequences of squeezing (milking) the site: hemolysis and dilution with tissue juice
- Capillary Order of Draw:
  - Blood gases (infants)
  - Preparation of slides/smears
  - EDTA tube
  - Other additive tubes (green, mint green, gray)
  - Serum tubes (gold, red tube)

ARTERIAL PUNCTURE

- Indications:
  - Arterial blood gas analysis
  - Absence of palpable veins
- General precautions and techniques
  - Sites of puncture: radial (45-60° angle), brachial (45-60° angle), femoral arteries (90° angle)
  - *Modified Allen Test* should be done before using the radial artery; this ensures that the ulnar artery can very well supply the palm/hand even when the radial artery is damaged
    - Involves compressing the radial artery near the thumb and the ulnar artery near the smallest finger
    - Depriving the palm/hand with oxygenated blood, after which, the ulnar artery is released and complete blood return is evaluated/observed
  - Amount of anticoagulant (heparin): 0.05mL/mL of blood (excessive amount leads to falsely decreased pH)

SPECIAL SITUATIONS IN PHLEBOTOMY

- IV
  - Alternative: use the opposite arm, perform fingerstick
  - Turn off the IV line for 2 minutes to avoid contamination and dilution
- CVA (Central Venous Access Collection)
  - Discard the first 3-5mL of blood which are likely to contain the infused medications
- Cannula/Fistula, Mastectomy
  - Draw blood from the opposite arm
- Burns, edema, scars, sclerosed veins, tattoos
  - Select another site
- Hematoma
  - Draw below

PHLEBOTOMY COMPLICATIONS

- Vascular: bleeding from venipuncture site and hematoma formation
- Infections: second most common complication
- Anemia (iatrogenic): results from blood loss for testing; particular problem with pediatric patients
- Neurologic: post-phlebotomy seizure or pain
- Cardiovascular: orthostatic hypotension, syncope, shock, and cardiac arrest
- Dermatologic: allergic reaction to iodine, adhesives, or latex

PREANALYTICAL VARIABLES	
FACTOR	**EXAMPLES OF ANALYTES AFFECTED**
Diurnal variation/ Circadian rhythm	↑ AM: ACTH, Cortisol, Aldosterone, Iron ↑ PM: GH, TSH, PTH, ACP
Posture	Standing: ↑ Albumin, cholesterol, calcium, renin, aldosterone Supine to upright (hydrostatic pressure increases): ↑ proteins (albumin), enzymes, protein-bound substances such as calcium, cholesterol, triglycerides, and bilirubin
Activity	Ambulatory: ↑ TPN, CK Exercise: ↑ Lactic acid, CK, LD, AST, Aldolase, Creatine
Stress	↑ ACTH, cortisol, prolactin, catecholamines such as epinephrine and norepinephrine
Diet	Requires fasting: fasting blood sugar, triglyceride, lipid profile, OGTT, Gastrin, Insulin Collected in the *basal state*: Cholesterol, electrolytes
Smoking	↑ Ammonia, carbon monoxide, BUN (enzymatic methods)
IV contamination	↑ Glucose, electrolytes, drugs ↓ Dilutional effect on certain analytes such as BUN
Hemoconcentration	↑ Potassium, lactate, proteins, protein-bound substances
Hemolysis	↑ Intracellular ions such as potassium, magnesium, phosphate, iron, LD, AST, ALT, ALP, CK ↓ Lipase and glucose (hexokinase)
Handling	Require ice/chilling: lactate, ACP, Ammonia, ABG, RA (renin activity), ACTH Anaerobic collection/transport (kept capped): ACP, Ionized calcium (not necessarily on ice), ammonia Protected from light: Bilirubin, beta-carotene, porphyrin, Vitamin K, Vitamin B_6 and B_{12}

- Physiologic factors and related activities that must be controlled/minimized when preparing a patient for phlebotomy: diurnal variation, exercise, fasting diet, ethanol consumption, smoking, drug ingestion, and posture
- Other factors classified as physiologic but cannot be controlled: age, sex/gender

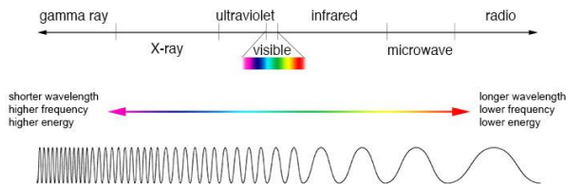
FACTORS THAT AFFECT EVACUATED TUBE QUALITY

- **Ambient temperature**
  - ↑ temperature = ↓ draw volume
  - ↓ temperature = ↑ draw volume
- **Altitude**
  - ↑ (>5,000 ft) = ↓ draw volume
- **Humidity**
  - Affects plastic tubes only (permeable to water vapors)
  - ↑ Humidity = migration of water vapor inside a tube
  - ↓ Humidity = escape of water vapor from a tube containing a wet additive
- **Light**
  - Affects glass CTAD (Citric Acid/Citrate, Theophylline, Adenosine, Dipyridamole) tube, a photosensitive additive for coagulation testing which minimizes platelet activation after blood collection
  - Dipyridamole prevents platelet activation but is photo sensitive

ANALYTICAL TECHNIQUES AND INSTRUMENTATION

SPECTROPHOTOMETRY

- Measurement of the amount of light from a specific wavelength of the electromagnetic wavelength
- Regions of Interest in Electromagnetic Spectrum:
  - Ultraviolet light (*<400nm*)
  - Visible light *(400-700nm)*; violet, blue, green, yellow, orange, red
  - Infrared (*>700nm*)



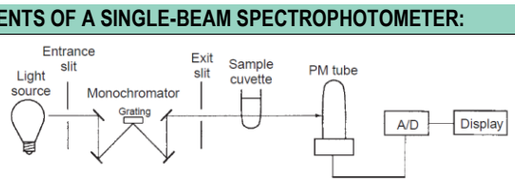
- Relationship between wavelength and energy (inverse)
  - Wavelength: distance between 2 successive peaks
  - In the image above:
    - Left to Right: Increasing wavelength but decreasing energy and frequency
      - Gamma rays and X-rays are called ionizing radiation which damage DNA because they are highly energetic
    - Right to left: Shorter wavelength, but higher energy and frequency
      - Microwave and radio waves
- **Beer's Law (Beer-Lambert's Law)**: The concentration of a substance is directly proportional to the amount of light absorbed and is inversely proportional to the logarithm of transmitted light
  - **Absorbance (A) = abc**
    - a = molar absorptivity
    - b = light path (diameter of cuvette) in centimeters; 1cm
    - c = concentration
  - **A = 2 — log %T** (percent transmittance)
    - $\%T = 10^{(2-A)}$
  - Given that A = abc, the concentration of an unknown analyte can be obtained by running a standard whose concentration is known, $\mathbf{c_u = A_u c_s / A_s}$

$$C_u = \frac{A_u\ (absorbance\ of\ the\ unknown\ sample)\ x\ C_s(known\ conc.\ of\ the\ standard)}{A_s\ (absorbance\ of\ the\ standard)}$$

SOLUTIONS USED IN SPECTROPHOTOMETRY

- ***Blank***
  - Distilled water, reagent, or sample used to subtract absorbances not due to the analyte of interest
  - Sets the spectrophotometer to 0 absorbance
- ***Standard/Calibrator***
  - Substance of known purity and concentration used to calibrate an assay method or determine the concentration of the unknown analyte (for a single analyte only)
- ***Control***
  - Solution containing various analytes with known target values
    - Assayed = published reference range of target values
    - Unassayed = laboratory should set or determine the target values
  - Monitor the performance of an assay to ensure accuracy and precision
  - Bovine origin; preferred because of biohazard issues associated with human sera
  - Must not be hemolyzed, Non-lipemic/non-lactescent, Non-icteric

COMPONENTS OF A SINGLE-BEAM SPECTROPHOTOMETER:



- **Light source**
  - provides the light which the analyte in the sample will absorb (polychromatic; consists of various wavelengths)
  - Ex. Deuterium, xenon, tungsten (for visible and near infrared regions)
- **Entrance slit**
  - ensures that the light that enters the monochromator only comes from the light source; prevents stray light
- **Monochromator**
  - selects a specific wavelength of the light emitted by the source (makes polychromatic light monochromatic)
  - Prisms and diffraction gratings (holographic and diffraction gratings; diffraction gratings are commonly used)
- **Exit slit**
  - further prevents stray lights and controls the band pass
  - ensures that only a narrow fraction of the spectrum will strike the sample
- **Cuvet**
  - Should be transparent to the wavelength of interest
  - Light should pass through and not absorb any part of it
  - Can be quartz, plastic, or glass.
    - **Quartz** is for UV and IR
    - **Plastic** is for UV and visible light
    - **Glass** is for visible light
      - If the wavelength is UV, don't use glass, because glass absorbs UV
- **Photodetector/Photomultiplier**
  - Light transmitted is the light which was not absorbed
  - Transforms light into an equivalent amount of electrical energy which will be quantified by the meter (part of the read-out device)
  - Most commonly used and most sensitive: photomultiplier

QUALITY ASSURANCE

- **Wavelength accuracy**: the wavelength set is the actual one selected by the monochromator (using didymium or holmium oxide)
- **Absorbance check**: done using glass filters and solutions that have known absorbance values
- **Linearity**: a change in concentration results in a straight-line calibration curve (Beer's Law); conformity with Beer's law = as the concentration increases, absorbance should also increase (using neutral density filters and dichromate solution)
- **Stray light**: any wavelength outside the band of interest; causes absorbance error and loss of linearity; detected using sharp cutoff filters (ex. $NiSO_4$, $NaNO_2$, and acetone)

DOUBLE-BEAM SPECTROPHOTOMETRY

- Designed to compensate for variations in intensity of the light source by splitting the light beam from the lamp and directing one portion to a reference cuvet and the other to the sample cuvet
- Only 1 light source but split into two (strikes the reference cuvet and sample cuvet); meant to compensate for variations in the intensity of the light source
- 2 types:
  - **Double-beam-in-space**: 2 cuvets are struck by light at the same time; 2 photodetectors are used, one for each cuvet

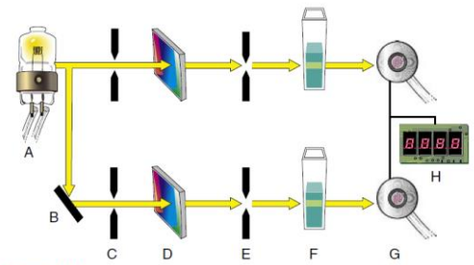
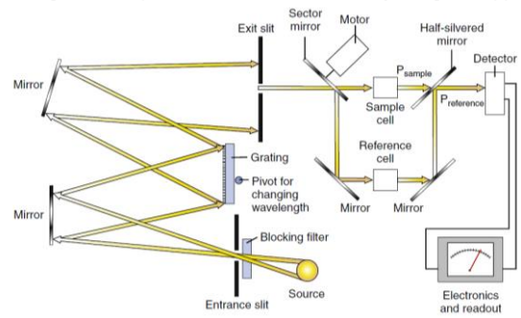


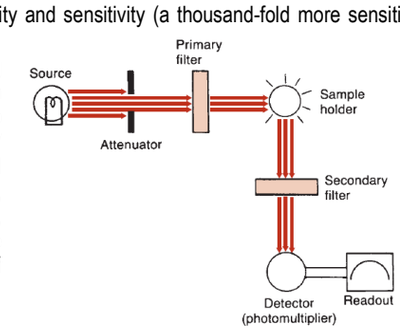
Figure 4.11 Double beam in space design of spectrophotometer. **A,** Exciter lamp. **B,** Mirror. **C,** Entrance slits. **D,** Monochromators. **E,** Exit slits. **F,** Cuvettes. **G,** Photodetectors. **H,** Light-emitting diode (LED).

  - **Double-beam-in-time**: 1 photodetector; light is alternately passed through the sample and the reference cuvet by using a chopper



FLUOROMETRY

- Also known as **Molecular Emission Spectroscopy**
- Principle: Measurement of the amount of light emitted by excited molecules
- Important components: gas discharge lamps (source); 2 monochromators or filters (primary and secondary monochromator); Photomultiplier tube (detector)
  - Primary monochromator: selects wavelength of light from the source (excitation energy)
- Advantages: High specificity and sensitivity (a thousand-fold more sensitive than spectrophotometry)
- Disadvantage: **Quenching** (any decrease in fluorescence due to prolonged exposure to UV light, increased temperature, pH changes, presence of contaminants, very high analyte concentration [loss of linearity])



CHEMILUMINESCENCE

- Principle: Measurement of light emitted by a chemical reaction; involves oxidation of an organic compound catalyzed by an enzyme, a metal, or hemin
- Substrates used: luciferin, luminol, acridinium ester
- Advantages: subpicomolar detection limits (sensitive), speed, ease of use, simple instrumentation

SCINTILLATION COUNTING

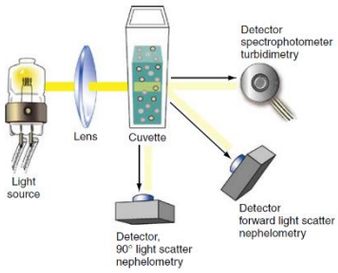
- Detection of scintillations (flashes of light) using a PM tube and counting of the electrical impulses
- Label is a radioisotope
  - **Crystal scintillation**: gamma counter; ligands labelled with iodine 125 or iodine 131
  - **Liquid scintillation**: beta counter; ligands labelled with hydrogen 3 of carbon 14

OTHER IMMUNOASSAY TECHNIQUES

- **Microparticle capture enzyme immunoassay (MEIA)**
  - uses microparticles as solid phase, an enzyme label, and a fluorogenic substrate
  - heterogenous immunoassay technique
  - separation technique is used
  - application: for high molecular weight substances
  - Microparticle enzyme immunoassay: technique that is similar to ELISA in that there is a double antibody system that forms a sandwich with the analyte of interest e.g. hormones
- **Fluorescence polarization immunoassay (FPIA)**
  - uses polarized light to excite a fluorescent label
  - rate of rotation is inversely proportional to the degree of polarization and analyte concentration
  - homogenous immunoassay is used
  - separation is not needed
  - application: low molecular weight substances
- Both used in therapeutic drug monitoring (specially for immunosuppressants)

TURBIDIMETRY AND NEPHELOMETRY

- Common application: measurement of immune complexes (antigen-antibody complexes)
- Measurement of the amount of light blocked (**turbidimetry**) or of the amount of light scattered (**nephelometry**) by a suspension of particles



MASS SPECTROMETRY

- Involves sample ionization and separation of ions based on *mass-to-charge ratio*
- Definitive identification of samples eluting from GC or HPLC columns
- Applications:
  - **GC-MS or HPLC-MS**: Involves sample ionization and separation of ions based on mass-to-charge ratio; GC/MS: Gold standard for drug analysis
  - **Tandem MS**: used for IEM detection (inborn errors of metabolism; reference method)
  - **IDMS**: reference method for non-protein nitrogenous compounds
  - **MALDI TOF MS**: used for the analysis of biomolecules e.g. peptides and protein signatures, also used for microbial or bacterial identification

NUCLEAR MAGNETIC RESONANCE SPECTROSCOPY

- Non-destructive method for determining the structure of organic compounds
- Used in lipoprotein particle measurements

AUTOMATION

- **Continuous flow**
  - Pumps liquids through a system of continuous tubing; significant carry-over problems and wasteful use of continuously flowing reagents
  - e.g. SMA (Simultaneous Multiple Analyzer), Technicon
  - **Air bubbles**: serve as separating and cleaning media
  - **Coiled tubing/glass coils**: meant for mixing
  - Issues: carryover and wasteful use
- **Centrifugal analysis**
  - Uses centrifugal force to transfer liquids in separate cuvettes for measurement; most capable of batch analysis
  - Batch analysis: running multiple samples, one test at a time
  - Has low throughput rate
  - e.g. Cobas Bio (Roche), IL Monarch
- **Discrete analysis**
  - Places each sample and accompanying reagents in separate containers; batch analysis, random access, and stat capabilities
  - e.g. Vitros, Beckman, Dupont ACA

POINT OF CARE TESTING (POCT) DEVICES

- Advantages: reduced turnaround time and connectivity (electronic documentation of testing - LIS)
- Blood glucose monitors (glucometers) — most commonly used POCT
- Lacks quality control (Recommendation: QC must be done once for each day of use)

ELECTROCHEMICAL TECHNIQUES

- Used for determining electrolytes and blood gases
- Includes potentiometry, amperometry, polarography, and coulometry

CARBOHYDRATES

HORMONAL REGULATION	
Insulin	- *The only **hypoglycemic agent***; lower plasma glucose level - Pancreatic hormone (produced by the beta cells of the islets of Langerhans) - Promotes cellular uptake of glucose, glycolysis, glycogenesis, and lipogenesis - Inhibits glycogenolysis
Glucagon	- ***Major hyperglycemic hormone***; increases glucose level - Pancreatic hormone (produced by the alpha cells of the islets of Langerhans) - Promotes **gluconeogenesis** (synthesis of glucose from non-carbohydrate sources) and **glycogenolysis** (reverse of glycogenesis) - Inhibits glycolysis
ACTH	- Promotes gluconeogenesis and glycogenolysis - Anterior pituitary hormone
Growth hormone	- Inhibits glycolysis - Anterior pituitary hormone
Thyroxine	- Promotes intestinal absorption of glucose - Produced by the thyroid
Cortisol	- Promotes gluconeogenesis and lipolysis - Produced by the adrenal cortex
Epinephrine	- Promotes glycogenolysis and lipolysis - Produced by the adrenal medulla

HYPOGLYCEMIC DISORDERS

- Hypoglycemia: plasma glucose level **≤ 50 mg/dL**
- Panic value for hypoglycemia: **<40mg/dL**
- Symptoms (appear at plasma glucose levels between 50-55mg/dL)
  - **Neurogenic:** tremulousness, palpitations, anxiety, diaphoresis, hunger, and paresthesias
  - **Neuroglycopenic:** dizziness, tingling, blurred vision, behavioral changes, seizure, and coma
- Types
  - **Post-absorptive/Fasting**: occurs after 10 hours without food; secondary to hyperinsulinism (insulinoma; endogenous hyperinsulinism), hormonal deficiencies, genetic disorders, autoimmunity or drug-induced
  - **Post-prandial/Alimentary/Reactive:** occurs usually within 4 hours after eating a meal
- Diagnosis
  - **Whipple's triad of hypoglycemia**:
    - Symptoms of hypoglycemia
    - Low plasma glucose level
    - Relief of symptoms following correction
  - **Diagnostic Criteria for Insulinoma**:
    - Signs and symptoms of hypoglycemia
    - Plasma glucose level ≤ 50 mg/dL
    - Increased insulin, pro-insulin, and C-peptide levels
      - Insulinoma = inappropriate production of insulin by tumors (no feedback mechanism)
      - C-peptide is part of the proinsulin molecule that is cut off for proinsulin to become biologically active insulin
    - ↓ β-hydroxybutyrate level: low ketone level

DIABETES MELLITUS

- **Type I DM**
  - autoimmune or idiopathic beta cell destruction leading to absolute insulin deficiency
  - Insulin-dependent
- **Type 2 DM**
  - insulin resistance with progressive insulin deficiency
- **Gestational DM**
  - glucose intolerance during pregnancy that disappears post-partum but may convert to type 2 DM in 30-40% of cases within 10 years
  - due to metabolic and hormonal changes
- **Other types**
  - Endocrinopathies that present with glucose intolerance or hyperglycemia
  - May be genetic or drug-induced
- Additional Notes!
  - Type I, II, IDDM, NIDDM nomenclature is no longer used

	TYPE 1	TYPE 2
Frequency	- <10%	- >90% - More common in women, blacks, and hispano-americans
Age of Onset	- Childhood/Juvenile	- Adulthood

Risk factors	- Autoimmunity - Genetic predisposition (inheritance of **HLA-DR3/DR4)**	- Family history of DM - Advanced age - Overweight tendencies (Overweight = BMI of 25-29.9 kg/m^2; Obese = >30 kg/m^2) - Habitually physically inactive - High-risk minority population - Hypertension (≥140/90 mmHg) - Dyslipidemia (HDL: <35mg/dL, TAG >200mg/dL) - History of CVD - HbA1C >5.7% - IGT or IFG - History of GDM or delivering a baby weighing more than 9 lb - PCOS - Other clinical conditions associated with insulin resistance (e.g. severe obesity and acanthosis nigricans)
Clinical findings	- Opposite of insulinoma - C-peptide levels low then undetectable - Autoantibodies - Diabetic Ketoacidosis (increased beta-hydroxybutyric acid and other ketone bodies)	- Nonketotic hyperosmolar state (coma) possible (plasma levels become uncontrolled) - Panic value for hyperglycemia: >500 mg/dL
Therapy	- Insulin injection	- Lifestyle changes - Oral hypoglycemic agents - May require insulin

TESTS AND DIAGNOSTIC CRITERIA FOR DM

	RPG *(Random Blood Glucose/ Casual Plasma Glucose)*	**FPG** *(Fasting Plasma Glucose/Fasting Blood Sugar)*	**2-hr PG** *(OGTT)*	**HbA_{1C}**
Important considerations	Significant results must always be in conjunction with symptoms of DM: - Polyuria - Polyphagia - Polydipsia ***Indicative but not diagnostic***; must be confirmed by a 2hr post load glucose	Fasting of at least 8 hours is required Levels that exceed the cutoff at 2 occasions = DM is confirmed Preferred as confirmatory test	CHO intake of >150g/day for 3 days Fasting for 8-14 hours Standard load of 75 grams or 1.75g/kg body weight 2 samples are collected	Part of the diagnostic criteria Specimen: EDTA whole blood Sample is a lysate from the whole blood
Normal	<200mg/dL	<100mg/dL	<140 mg/dL	---
Pre-diabetes	---	100-125 mg/dL *(Impaired fasting glucose/Pre-DM)*	140-149 mg/dL *(Impaired glucose tolerance)*	5.7-6.4%
DM	**>200mg/dL + symptoms**	**≥126 mg/dL**	**≥200 mg/dL**	**≥6.5%**

- **Gestational DM** (100 g; 3-hour OGTT)
  - Screening test: 50 g; 1-hour OGTT
  - Diagnosis is made if any 2 or more of the following glucose levels are met or exceeded:
    - FPG ≥ 95 mg/dL
    - 1-h PG ≥ 180 mg/dL
    - 2-h PG ≥ 155 mg/dL
    - 3-h PG ≥ 140 mg/dL

MEASURES OF GLYCEMIC CONTROL

1. *GLYCOSYLATED HEMOGLOBIN/ HbA_{1c}*

- Hb A with glucose irreversibly attached to one or both N-terminal valines of the β-chains
- Provides an index of average blood glucose levels over the past 2-3 months
  - Long term monitoring: 2-3 months (lifespan of RBCs)
- Unreliable in patients with hemolytic disorders and hemoglobinopathies (shortened RBC lifespan)
- Sample: hemolysate (from EDTA whole blood)
- Methods: Electrophoresis, chromatography, immunoassays
- Glycemic goal:
  - **ADA: ≤ 7%** (Monitoring; maintained by DM patient)
  - **AACE: < 6.5%**

2. *FRUCTOSAMINE*

- Used to assess glycemic control over the past 2-3 weeks in cases when HbA_{1c} is unreliable
- Represents glycosylated albumin
- Has the advantage of using serum samples
- Unreliable when serum albumin level is very low or < 3g/dL
- Reference values: ***205-285 umol/L***

3. *WHOLE BLOOD GLUCOSE*

- Difference is significant between plasma glucose and whole blood glucose
  - Whole blood glucose is **10-15% lower** than plasma glucose
  - When interpreting blood glucose levels, you should not use reference interval for plasma glucose
  - Plasma glucose is **less than 5%** than serum glucose
  - Capillary blood glucose is higher than venous blood glucose by 2-5mg/dL
- Most commonly analyzed using POCT devices at home or at the patient's bedside
- Recommendation: 3-4 times per day including pre-prandial and post-prandial;
- ADA glycemic goals:
  - **Pre-prandial: 70-130 mg/dL**
  - **Post-prandial: < 180 mg/dL**

METHODS OF GLUCOSE MEASUREMENT

SPECIMEN CONSIDERATIONS

- Use of **NaF (Sodium Fluoride) or SST** to prevent glycolysis (only used when prompt analysis is not possible; if specimen will stand for a considerable amount of time)
  - Gray top: Sodium fluoride
  - Otherwise, glucose will be lost to glycolysis at a rate of:
    - ***7 mg/dL/hour*** at room temperature
    - ***2 mg/dL/hour*** at 4°C (refrigerator temperature)
  - Additional! High WBC count will also lead to a substantial decrease in plasma glucose
- Effect of dextrose contamination
  - **10% contamination with 5% dextrose** will increase plasma glucose by **500mg/dL**

A. CHEMICAL METHODS
(Non-specific; Based on Redox and Condensation reactions)

1. *COPPER REDUCTION*

- **Redox**; based on Benedict's principle, which involves the reduction of cupric ions to cuprous ions by glucose in the presence of heat and alkaline environment; cuprous ions are oxidized by PMA, AMA
- Any reducing agent may replace glucose; falsely increased when there are other reducing substances
  - **Folin-Wu** (**Colorimetric; blue**)
    $Cu^+ + \text{phosphomolybdic acid} \longrightarrow \text{phosphomolybdenum}$
  - **Nelson Somogyi** (**Colorimetric; blue**)
    $Cu^+ + \text{arsenomolybdic acid} \longrightarrow \text{arsenomolybdenum}$
  - **Neocuproine** (**Colorimetric; Yellow-orange**)
    $Cu^+ + \text{neocuproine} \longrightarrow \text{Cu-neocuproine complex}$

2. *FERRIC-REDUCTION*

- Hagedorn-Jensen Method
- **Redox**; there is loss of color, reverse colorimetry; The degree of decrease in color intensity is proportional to the glucose concentration
  $$\text{ferricyanide} \xrightarrow{glucose} \text{ferrocyanide}$$
  - Ferricyanide is yellow orange, but becomes colorless when reduced into ferricyanide

3. *CONDENSATION*

- **Dubowski reaction**, intensity of color reaction is proportional to glucose concentration; **colorimetric - green**)
  $$\text{glucose} + \text{o-toluidine} \longrightarrow \text{Schiff's base}$$
  - Reaction of glucose with aromatic amines (ortho-toluidine)

B. ENZYMATIC METHODS (more popular; has high sensitivity and specificity)

1. GLUCOSE OXIDASE

$$\beta\text{-D-glucose} + O_2 + H_2O \xrightarrow{glucose\ oxidase} \text{gluconic acid} + H_2O_2$$

- 2 major isomers in plasma: 65% is β-D-glucose while 35% is α-D-glucose
- Glucose oxidase is only specific to β-D-glucose, so α-D-glucose is converted to beta by mutarotase
- Detection Methods:
  - **Peroxidase-coupled/Trinder reaction** — subject to many interferences

    $$H_2O_2 + \text{chromogen} \xrightarrow{peroxidase} \text{oxidized chromogen} + H_2O$$

    - Aka. Trinder assay
    - Colorimetric, spectrophotometric
    - Nonspecific may be affected by interfering agents
    - Oxidizing agents: false increase
    - Reducing agents: false decrease
      - Chromogen is added which reacts with H_2O_2, producing a colored compound (oxidized chromogen)
  - **Polarographic** (Amperometric) - measurement of the degree of O_2 consumption using a pO_2 (Clark) electrode; requires addition of molybdate and iodide or catalase and ethanol to prevent reformation of hydrogen from hydrogen peroxide; the amount of oxygen detected is inversely proportional to glucose

2. HEXOKINASE

$$\text{glucose} + \text{ATP} \xrightarrow{hexokinase} \text{glucose-6-PO}_4 + \text{ADP}$$

$$\text{glucose-6-PO4} + \text{NADP} \xrightarrow{G-6-PD} \text{6-phosphogluconolactone} + \text{NADPH}$$

- **Reference method**; not subject to interferences and the coupling is highly specific
- G6PD is highly specific
- Increase in absorbance at 340nm is proportional to the level of NAPDH and level of glucose
- Sources of errors:
  - **Hemolyzed sample**: falsely decreased result; due to the interference of hemoglobin with the activity of the enzymes in the assay
  - **Icteric samples**: false decrease

INBORN ERRORS OF CARBOHYDRATE METABOLISM

- **GALACTOSEMIA**
  - due to deficiency of the enzyme ***GALT (Galactose-1-phosphate uridyltransferase)***
  - one of the most common enzyme deficiencies
  - affected children present with mental retardation, failure to thrive, and galactosuria (screened using Clinitest; positive since galactose is a reducing sugar)
  - part of newborn screening
- **GLYCOGEN STORAGE DISEASES/GLYCOGENOSES**
  - **Hepatic glycogenosis**
    - usually manifest with hepatomegaly, hypoglycemia, and growth retardation
    - **von Gierke disease (GSD Type Ia)**: most common; due to glucose-6-phosphatase deficiency
      - presents with hypoglycemia, ketonuria, lactic acidosis, dyslipidemia, and hyperuricemia
  - **Muscle glycogenosis**
    - Manifest with exercise intolerance, muscle cramps, fatigue and weakness; ↑CK, AST, LD

GSD Type	Enzyme deficient
0	• Glycogen synthase • No hepatomegaly
Ia (Von Gierke)	• Glucose-6-phosphatase
Ib	• Glucose-6-phosphatase translocase • With neutropenia resulting to recurrent bacterial infections
II (Pompe disease)	• Lysosomal acid alpha glucosidase • Muscle glycogenosis
IIIa (Cori-Forbes disease)	• Glycogen debranching enzyme • Involves both liver and muscle
IIIb	• Glycogen debranching enzyme • Hepatic glycogenosis only
IV (Anderson's disease/ Amylopectinosis)	• Glycogen branching enzyme
V (McArdle disease)	• Muscle phosphorylase • Muscle glycogenosis
VI (Hers disease)	• Glycogen phosphorylase
VII (Tarui disease)	• Phosphofructokinase • Muscle glycogenosis
IXa	• Phosphorylase kinase • Hepatic glycogenosis
IXb	• Phosphorylase • Both Hepatic and Muscle glycogenosis
XI (Fanconi-Bickel syndrome)	• Glucose transporter 2 • Glucosuria, aminoaciduria, and electrolyte wasting

LIPIDS AND LIPOPROTEINS

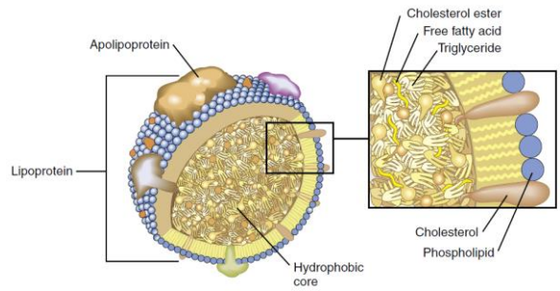
COMPONENTS OF LIPOPROTEINS

- SURFACE

- **Apolipoproteins**
  - Protein moieties of lipoproteins (allow their transport in plasma)
  - Lipids are water-insoluble so the particles must contain protein moieties to facilitate their transport in plasma
  - Basic role:
    - Structural support; primary/basic role of apolipoproteins, to maintain the integrity of the particle
    - Lipoprotein metabolism (enzyme cofactor or inhibitor)
    - Ligands for cell receptors
- **Phospholipids**
  - Primarily makes up the surface layer along with cholesterol; apolipoproteins are just embedded
  - Amphipathic (has polar/hydrophilic and non-polar groups)
- **Non-esterified cholesterol**
  - 30% of cholesterol; also amphipathic

- CORE

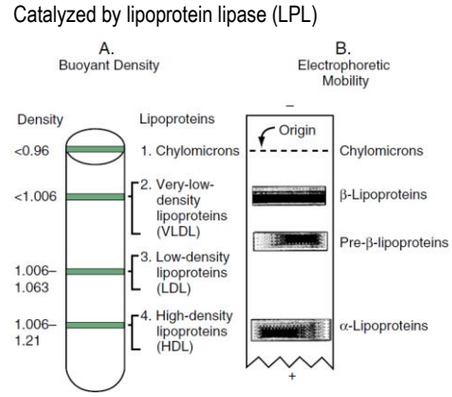
- **Triglycerides**
  - Hydrophobic; insoluble in water
- **Cholesterol esters/Cholesteryl esters**
  - A result of the attachment of fatty acid to non-esterified cholesterol
  - The formation is catalyzed by LCAT
  - Esterified or Non-esterified
    - 70% of cholesterol: esterified
    - When a fatty acid is attached to the cholesterol, it attaches to the polar group, therefore the polarity is lost (which explains why esterified cholesterol is found in the core)
- **Free fatty acids**
  - Non-esterified fatty acids
  - Fatty acids that are not forming parts of phospholipids and triglycerides; they form part of the core



SIGNIFICANT APOLIPOPROTEINS

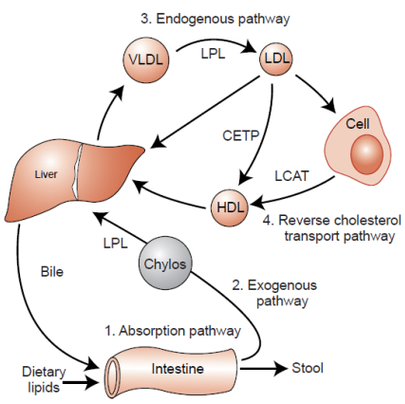
Apo	Plasma Conc. (mg/dL)	Main distribution	Function	
A-I	100-200	HDL	Structural	LCAT activator
A-II	20-50			
A-IV	10-20	CM, VLDL, HDL		
B-100	70-125	VLDL, IDL, LDL		LDL-R ligand
B-48	<5	CM		Remnant receptor ligand
C-I	5-8	CM, VLDL, HDL		
C-II	3-7			LPL cofactor
C-III	10-12			LPL inhibitor
E	3-15	CM, VLDL, IDL, remnants and HDL	Recognition factor that targets CM and VLDL remnants to hepatic receptor	

- **Apo B-100**
  - Major apolipoprotein in LDL, VLDL, IDL
  - Represents the sequential conversion of:
    - VLDL → IDL → LDL
    - LDL is the product of the hydrolysis of VLDL
    - IDL is intermediate between VLDL and LDL
    - Catalyzed by lipoprotein lipase (LPL)



MAJOR CLASSES OF LIPOPROTEINS					
	Density	**Electrophoretic mobility**	**Major lipid component**	**Apolipoproteins**	**Other comments**
CHYLOMICRON	<0.95 (float) Density lower than water	Origin	TG-rich	B48, C, E	Synthesized in the small intestine "Floating creamy layer" after standing; milky, lactescent, or lipemic plasma Transports dietary or ***exogenous TGs***
VLDL	0.95-1.006	Pre-beta		B100, C, E	Synthesized in the liver Transports dietary or ***endogenous TGs*** Remain suspended in plasma even after centrifugation
LDL	1.019-1.063	Beta	Cholesterol-rich	B100	May be atherogenic Transports cholesterol to peripheral tissues
HDL	1.063-1.21	Alpha	Protein-rich	AI, AII, AIV, C	Reverse cholesterol transport (tissues to liver)

- TG: Triglyceride
- IDL is excluded because it is just a minor component
  - Density of IDL: 1.006-1.019
- Ultracentrifugation: reference method for protein analysis
- Note:
  - Higher lipid content, lower density
  - Higher protein content, higher density
- Floating creamy layer formed by chylomicrons is especially evident in non-fasting samples
- LDL is atherogenic because a significant part of LDL is transported to tissues and may be deposited there in excess amounts; bad cholesterol
- HDL is supposed to sweep away the deposited cholesterol by LDL; High LDL and Low HDL is bad



CHYLOMICRONS

- Are the largest lipoproteins and have the lowest density.
- They are formed in the intestines and transport triglycerides after a meal (nonfasting), giving serum a turbid appearance. Because of their low density, chylomicrons will float to the top and form a creamy layer when plasma is stored overnight.
- Chylomicrons are composed of 86% triglyceride, 5% cholesterol, 7% phospholipid, and 2% apolipoprotein.
- Chylomicrons have apoproteins B-48, mainly, and lesser amounts of A-I, C-I, C-II, and C-III on their surface.
- In normal lipid metabolism, chylomicrons enter the circulation and are metabolized to remnant particles for uptake and further modification by the liver.

VERY-LOW-DENSITY LIPOPROTEIN (VLDL)

- Carries endogenous triglycerides synthesized in the liver. VLDL molecules are composed of 55% triglycerides, 19% cholesterol, 18% phospholipid, 8% apolipoprotein and have apolipoproteins B-100, mainly, and C-I, C-II, C-III, and E on their surface.
- In normal lipid metabolism, VLDLs are secreted into the blood by the liver for metabolism in peripheral tissues.

INTERMEDIATE-DENSITY LIPOPROTEIN (IDL)

- A transitional form, as it is formed from VLDL and then further modified in the liver to LDL.
- IDLs carry endogenous triglycerides and cholesterol esters.
- IDL molecules are composed of 23% triglycerides, 38% cholesterol, 19% phospholipid, 19% apolipoproteins and have apolipoproteins B-100, mainly, and some E on their surface.

LOW-DENSITY LIPOPROTEIN (LDL)

- The body's major cholesterol carrier and transports a large amount of endogenous cholesterol. Known as "bad cholesterol," LDL is easily taken up by cells, so elevated levels are associated with increased risk for atherosclerosis (atherogenic).
- LDLs are composed of 50% cholesterol, 22% phospholipids, 6% triglycerides, and 22% protein and have apoprotein B-100 on their surface.
- In normal lipid metabolism, this lipoprotein brings cholesterol to peripheral cells for membrane synthesis and formation of adrenal and reproductive hormones.

HIGH-DENSITY LIPOPROTEIN (HDL)

- Is also known as "good cholesterol."
- HDL is synthesized in the intestine and liver cells.
- HDL molecules are recycled chylomicron and VLDL molecules.
- HDL is composed of 50% protein, 28% phospholipids, 19% cholesterol, and 3% triglycerides. HDL has apoproteins A-I, mainly, and A-II on its surface.
- In normal lipid metabolism, HDL removes excess cholesterol from peripheral tissues and transports it to other catabolic sites. This function has an anti-atherogenic effect.

MINOR AND ABNORMAL LIPOPROTEINS

- **IDL** *(Minor lipoprotein)*
  - VLDL remnants that normally exist transiently during the conversion of VLDL to LDL through the action of lipoprotein lipase
  - Density, electrophoretic mobility, and protein and lipid contents are intermediate between those of VLDL and LDL
- **β-VLDL** *(Abnormal lipoprotein)*
  - ***Floating beta-lipoprotein***
  - Migrates along with beta-lipoprotein but floats in ultracentrifugation
  - It floats because it has higher cholesterol content than normal VLDL
  - Seen in type 3 hyperlipoproteinemia
- **Lp(a)** *(Abnormal lipoprotein)*
  - ***Sinking pre-beta lipoprotein***
  - Contains the prothrombotic Apo (a)
  - Sinks because of Apo-a in addition to B100
  - Heavier than normal LDL
  - Associated with cardiovascular disorders such as stroke, MI, CHD
  - Competes with plasminogen or plasmin core binding sites; has homology with plasmin or plasminogen
- **LpX** *(Abnormal lipoprotein)*
  - Abnormal lipoprotein associated with obstructive biliary disease and LCAT deficiency
  - **LCAT (Lecithin-cholesterol acyltransferase)**: catalyzes the esterification of cholesterol
  - Consists mostly of phospholipids and non-esterified cholesterol

FREDRICKSON CLASSIFICATION OF HYPERLIPOPROTEINEMIAS				
Type	**Particle**	**Representative disorder/s**	**Standing plasma test**	**Lipid profile** *(TC, TG, LDL, HDL)*
1	CM	***Familial hyperchylomicronemia*** (due to LPL Deficiency or Apo C-II deficiency)	+ Clear	↑↑ TG
2a	LDL	***Familial hypercholesterolemia*** Apo-B defect; Usually due to mutation in the LDL receptor, impeding the uptake of LDL particles by hepatocytes Remember! LDL is cholesterol-rich	- Clear	↑↑ LDL, TC
2b	LDL, VLDL	***Familial combined hyperlipoproteinemia***	- Turbid	↑↑ LDL, TC, TG
3	IDL, β-VLDL	***Familial dysbetalipoproteinemia***		↑ TC, TG
4	VLDL	***Familial (endogenous) hypertriglyceridemia***		↑↑ TG
5	CM, VLDL	***Mixed hypertriglyceridemia*** Acquired dyslipidemia Chylomicrons, exogenous VLDL, endogenous May be acquired secondary to other underlying disease states (i.e. DM, nephrotic syndrome, alcoholism, etc.)	+ Turbid	

- Legend:
  - TC: Total cholesterol
  - TG: Triglycerides
  - ↑↑: Very high
  - ↑: High

- **Standing plasma test**
  - Primarily detects chylomicrons and VLDL
    - Chylomicrons: floating creamy layer (positive)
    - VLDL: turbid plasma after standing
    - If LDL is present in elevated amounts = plasma is clear; because LDL does not alter the clarity of plasma

HYPOLIPOPROTEINEMIAS

- Involves either apo-B or apo-B containing lipoprotein or HDL
- ***ABETALIPOPROTEINEMIA***
  - also known as **Bassen-Kornzweig Syndrome**
  - Absolute absence of Apo B-containing lipoproteins
  - Is a rare, autosomal recessive disorder
  - Involves mutations in the MTTP gene located on chromosome 4 with absolute nonexistent levels of apo B-48 and apo B-100
  - Inability to synthesize apo-B48 and apo-B100
  - Low levels of total cholesterol and normal to low triglycerides (nearly undetectable)
- ***HYPOBETALIPOPROTEINEMIA***
  - Autosomal dominant disorder; nonsense or missense mutation in the Apo-B gene
  - inability to synthesize Apo B-100 and Apo B-48
  - low total cholesterol level and normal to low triglyceride level

HDL ASSOCIATED DISEASES

- ***TANGIER DISEASE***
  - Inherited disorder (Autosomal recessive)
  - Resulting from mutations in the ABCA1 gene
  - Absence of HDL, apo A-I and apo A-II
- ***HYPOALPHALIPOPROTEINEMIA***
  - Autosomal dominant disorder
  - Low HDL in the presence of normal VLDL and LDL cholesterol levels

PLASMA LIPID ANALYSIS

ANALYTICAL METHODS

- Lipid profile: 12-14 hours fasting
- Cholesterol: no fasting

CHOLESTEROL

A. CHEMICAL METHODS (COLORIMETRIC)

- The 4-step method involve the following steps:
  - **Saponification**: ***KOH (Potassium hydroxide)***
    - Hydrolyzes esters; cholesterol is measured as free type only
    - Majority of cholesterol molecules are in esterified form
  - **Extraction**: ***Bloor's reagent*** (3:1 mixture of ethanol and ether)
    - Extracts lipids and precipitates proteins so that they will not interfere in the colorimetric reaction
  - **Purification**: ***Digitonin***
    - Precipitates free cholesterol
    - Allow the reaction of fat only
  - **Color development**: uses ***H_2SO_4 (sulfuric acid), acetic acid***, and either ***acetic anhydride or $FeCl_3$ (ferric chloride)***
    - Acetic anhydride: is **Liebermann-Burchard** reaction producing an end product that is characteristically **green**
    - $FeCl_3$ or **Salkowski's method**: with end product (cholester-dienyl disulfuric acid) of **red coloration**
- **Abell-Kendall method**: consists of saponification, extraction, and color development using Liebermann-Burchard
- **Modified Abell-Kendall**; used to be the CDC reference method which involves:
  - saponification with alcoholic KOH
  - extraction with petroleum ether (N-hexane)
  - colorimetry using the Liebermann-Burchard reaction which comprises sulfuric and acetic acids and acetic anhydride
- **GCMS (Gas Chromatography-Mass Spectrometry)**
  - Current ***reference method***; specifically measures cholesterol and does not detect related sterols
  - Shows good agreement with the gold standard Definitive method using IDMS

B. ENZYMATIC METHOD

- peroxidase-catalyzed dye oxidation

$$\text{Cholesterol ester} + H_2O \xrightarrow{\text{cholesterol esterase}} \text{cholesterol} + \text{fatty acid}$$

$$\text{Cholesterol} + O_2 \xrightarrow{\text{cholesterol oxidase}} \text{cholestenone} + H_2O_2$$

$$H_2O_2 + \text{chromogen} \xrightarrow{\text{peroxidase}} \text{oxidized chromogen} + H_2O$$

- The enzyme cholesterol esterase (cholesterase) catalyze the hydrolysis of cholesterol esters liberating free cholesterol
- Once, the cholesterol is liberated, it is oxidized in the presence of cholesterol oxidase producing cholestenone and H_2O_2 (hydrogen peroxide).
- Once we have H_2O_2, the reaction can be coupled with a peroxidase catalyzed reaction.
- Oxidation of a chromogen (4-aminophenazone) by H_2O_2 (one of the products of preceding reaction) producing an oxidized chromogen (quinoneimine; red-purple; increased absorbance at 500nm) that is colored and whose color intensity is proportional to the level of cholesterol.
- C/S (Colorimetric/Spectrophotometric) method [TRINDER REACTION]
- Reference Values
  - Desirable: <200 mg/dL
  - Borderline high: 200-239 mg/dL
  - High: ≥240 mg/dL
- Recommended cutoff points according to age:

Age (years)	Moderate risk for CHD	High risk for CHD
2-19	≥ 170 mg/dL	≥ 185 mg/dL
20-29	≥ 200 mg/dL	≥ 220 mg/dL
30-39	≥ 220 mg/dL	≥ 240 mg/dL
40 and above	≥ 240mg/dL	≥ 260 mg/dL

- CHD: Coronary Heart Disease

CLINICAL SIGNIFICANCE OF ABNORMAL RESULTS

- ***HYPERCHOLESTEROLEMIA***
  - Hyperlipoproteinemia types 2a, 2b, 3, nephrotic syndrome, poorly controlled diabetes mellitus, hypothyroidism
- ***HYPOCHOLESTEROLEMIA***
  - Severe liver disease, malnutrition, malabsorption, hyperthyroidism

TRIGLYCERIDES

A. CHEMICAL METHODS

- alcoholic KOH is used to catalyze the reaction

$$\text{triglyceride} + H_2O \xrightarrow{\text{alcoholic KOH}} \text{glycerol} + 3\ \text{fatty acids}$$

- **COLORIMETRIC**
  - Traditional methods include ***Van Handel and Zilversmith method***, is a color reaction where the end-product has characteristic color which is then measured spectrophotometrically
  - Original method: **(+) Blue**
  - CDC method (Modified Van Handel and Zilversmith method): **(+) Pink** – commonly used
- **FLUOROMETRIC**
  - **Hantzsch's method** is characterized by production of **yellow fluorescent** LUTIDINE compound

B. ENZYMATIC METHODS

- lipase is used to catalyze the reaction

$$\text{triglyceride} + H_2O \xrightarrow{\text{lipase}} \text{glycerol} + 3\ \text{fatty acids}$$

- We start with the hydrolysis of triglycerides, once you have glycerol.
- Either proceed with a **peroxidase catalyzed reaction** or **phosphorylation reaction** then measure the reaction at 340nm
- Peroxidase catalyzed dye oxidation (colorimetric) or measurement of the decrease in absorbance at 340 nm
  - C/S: peroxidase-coupled enzymatic reaction
  - Measurement of the decrease in absorbance 340nm (NADH oxidized to NAD)
- Only differs in terms catalyst of the hydrolytic reaction

- Reference Values:
  - Normal: < 150 mg/dL
  - Borderline high: 150—199 mg/dL
  - High: 200—499 mg/dL
  - Very high: ≥ 500 mg/dL

- Appearance of plasma (reflects the level of TGs in the sample):
  - Clear: <200 mg/dL
  - Hazy or turbid: >300 mg/dL
  - Opaque or milky: >600 mg/dL

CLINICAL SIGNIFICANCE OF ABNORMAL RESULTS

- ***HYPERTRIGLYCERIDEMIA***
  - Hyperlipoproteinemia types 1, 2b, 3, 4, 5 (all except 2a), nephrotic syndrome, alcoholism, pancreatitis, hypothyroidism
- ***HYPOTRIGLYCERIDEMIA***
  - Malnutrition, malabsorption syndrome, hyperthyroidism

LIPOPROTEIN CHOLESTEROL

HDL-C

- Cholesterol content of HDL which is routinely measured
  - ***Ultracentrifugation – reference method***
  - Direct Method – mostly based on immunoassay; homogenous assays
  - Two-step method (routinely done):
    - **Precipitation of apoB-containing lipoproteins by polyanion-divalent cations** (e.g. heparin sulfate-Mn^{2+}, dextran sulfate and sodium phosphotungstate-Mg^{2+}, heparin-Ca^{2+})
    - apoB is precipitated and removed from the sample
    - Lipid profile requires fasting sample → we're getting VLDL, IDL, LDL, HDL
    - Purpose of fasting is to clear chylomicrons from the plasma
    - After centrifugation, HDL is suspended in the supernatant while other lipoproteins are in the sediment → supernatant is taken for cholesterol determination.
    - **POD (Peroxidase)-coupled reaction**

- Reference values:
  - High risk for CHD: <40 mg/dL
  - Desirable: 60 mg/dL (the higher, the better)

LDL—C

- Ultracentrifugation – reference method
- Direct method – when the calculation is unreliable because of high levels of triglycerides > 400 mg/dL
  - Errors are noticeable at TG of > 200 mg/dL
  - Unreliable and unacceptable when the TG level exceeds > 400 mg/dL
- Total Cholesterol = HDL + LDL + VLDL
- **Friedewald equation**: LDL = TC – (HDL + TG/5)
  - Use 5 for mg/dL
  - Use 2.175 for mmol/L
  - This equation is more commonly used
  - VLDL = TG/5 or 2.175
- **De Long equation**: LDL = TC – (HDL + TG/6.5)
  - Use 6.5 for mg/dL
  - Use 2.825 for mmol/L
- Reference values:
  - Optimal: < 100 mg/dL
  - Near to above optimal: 100 - 129 mg/dL
  - Borderline high: 130 - 159 mg/dL
  - High: 160 - 189 mg/dL
  - Very high: ≥190 mg/dL

NCEP Guidelines for Acceptable Measurement Error

Analyte	Total Error	Bias	CV*
Cholesterol	≤9%	≤3%	≤3%
Triglyceride	≤15%	≤5%	≤5%
HDL-cholesterol	≤13%	≤5%	≤4%†
LDL-cholesterol	≤12%	≤4%	≤4%

NONPROTEIN NITROGEN COMPOUNDS

UREA (45-50%)

- Most abundant NPN

PHYSIOLOGY

- NPN present in highest concentration in the blood
- Formed in the liver from CO_2 and ammonia (deamination of amino acids) – product of the detoxification of ammonia
- Major excretory product of the metabolism of proteins and other nitrogen-containing chemicals
- Major factors that affects plasma urea and BUN concentration
  - Protein content in the diet
  - Renal function; freely filtered but 40-60% is reabsorbed
  - State of hydration
  - Rate of protein catabolism (proteins are broken down into amino acids, amino acids are deaminated and ammonia is formed, ammonia is converted to urea through the urea cycle in the liver)
- Concentration is expressed in terms of nitrogen content:
  BUN (Blood Urea Nitrogen) x 2.14 = Urea
  - If BUN to Urea: Multiply by 2.14
  - If Urea to BUN: divide by 2.14

  UREA: $CO(NH_2)_2$ with a molecular weight of 60 g/mol
  - Taking only Nitrogen:
    - N_2 = 28 g/mol
    - 60 g/mol divided by 28 = 2.14

CLINICAL APPLICATION: ANALYTICAL METHODS

- ***Reference method: Isotope Dilution Mass Spectrometry (IDMS)***

A. CHEMICAL

- Direct, Diacetyl Monoxime method, or Fearon Method; Non-specific

$$\text{diacetyl monoxime} + H_2O \rightarrow \text{diacetyl} + \text{hydroxylamine}$$
$$\text{diacetyl} + \text{urea} \longrightarrow \text{diazine}$$

- Diacetyl directly reacts with urea → diazine making a yellow color end-product which can be measured spectrophotometrically.
- No hydrolysis
- Unaffected by smoking

B. ENZYMATIC (INDIRECT)

$$\text{urea} + 2\ H_2O \longrightarrow 2\ NH_4^+ + CO_3^{2-}$$

- Indirect because enzymatic methods detect ammonia/ammonium and not really UREA; more specific
- Urea is hydrolyzed, in the presence of urease, to ammonium and carbonate ions → the product ammonium will be detected and made to react with some reagents
- False increase: smoking
- False decrease: fluoride tubes

1. COLORIMETRIC

Nesslerization $NH_4^+ + K_2(HgI_4) \longrightarrow$ dimercuric ammonium iodide

Berthelot $NH_4^+ + NaOCl + \text{phenol} \xrightarrow{Na\ nitroprusside}$ indophenol

- Nesslerization: Yellow
- Berthelot: blue
- GLDH-Coupled Enzymatic

$$NH_4^+ + \text{2-oxoglutarate} + NADH + H^+ \xrightarrow{GLDH} \text{glutamate} + NAD$$

- NAD is measured as decrease in absorbance at 340nm
- Not colorimetric since it uses UV
- NAD does not absorb light at 340nm
- The reduced form NADH is the one that absorbs light maximally at 340nm

2. CONDUCTIMETRIC

- measurement of the increase in conductivity due to NH_4^+ and CO_3^{2-} ions
- Non-ionized is hydrolyzed and ions is liberated that will increase the conductivity that is proportional to the urea in the sample

CLINICAL SIGNIFICANCE OF ABNORMAL VALUES

- **Reference intervals: 6 - 20 mg/dL (BUN)**
- **Panic value: ≥80 mg/dL**
- ***AZOTEMIA***
  - **Prerenal**: ↓ renal blood flow/volume (dehydration, hemorrhage, shock), ↑ protein diet, ↑ protein catabolism, dehydration (relative ↑ in hematocrit)
  - **Renal**: ↓ excretion (glomerulonephritis, renal failure)
  - **Postrenal**: urinary tract obstruction (nephrolithiasis, tumors, severe infection)
- ***UREMIA/UREMIC SYNDROME***
  - Very high plasma urea concentration accompanied by renal failure
  - Associated with hematologic abnormalities
  - Production of urine smelling sweat

CREATININE (5%)

PHYSIOLOGY

- Constitutes 5% of total plasma NPNs
- Formed from creatine in the muscles and excreted into plasma at a constant rate (body surface area is considered in testing for creatinine)
- Factors that affect concentration:
  - Muscle mass (males have larger values than females)
  - Renal function (glomerular filtration function)
- Removed from the circulation by glomerular filtration; not reabsorbed and minimally secreted by the tubules
- A better marker of GFR than Urea
- Concentration is **inversely related** to GFR (renal function)
  - Plasma creatinine levels is always inversely proportional to creatinine clearance/GFR
  - Creatinine clearance (CC) is a routine procedure for estimating GFR
  - Thus, ↑Plasma Creatinine (PC), ↓ CC
  - If the rate of removal is low, creatinine remains in the circulation and it accumulates (inverse relationship).
- Blood volume is directly proportional to cardiac output which is proportional to renal blood flow which is also proportional to GFR
- Increased creatinine clearance: high cardiac output
- Decreased creatinine clearance: impaired kidney function such as in cases of renal failure, hemorrhage, shock, dehydration

CLINICAL APPLICATION: ANALYTICAL METHODS

- CHEMICAL (JAFFE)

- non-specific, colorimetric, endpoint:

$$\text{creatinine} + \text{alkaline picrate} \longrightarrow \text{red-orange tautomer}$$

- Alkaline picrate: saturated picric acid and sodium hydroxide
- False increase: plasma chromogens
- Inexpensive/cheap
- Techniques to improve specificity:
  - **Kinetic**: timed-rate; taking absorbance measurement intervals or by using a continuously recording spectrophotometer
  - **Use of adsorbents**: by removing interfering substances using adsorbents like:
    - Lloyd's (sodium aluminum silicate)
    - Fuller's Earth (magnesium aluminum silicate)

- ENZYMATIC/CREATININASE METHOD

$$\text{creatinine} + H_2O \xrightarrow{creatininase} \text{creatine}$$

- Superior in terms of sensitivity and specificity

Creatininase–POD coupled

$$\text{creatine} + H_2O \xrightarrow{creatinase} \text{sarcosine} + \text{urea}$$

$$\text{sarcosine} + O_2 \xrightarrow{sarcosine\ oxidase} CH_2O + \text{glycine} + H_2O_2$$

$$H_2O_2 + \text{chromogen} \xrightarrow{POD} \text{oxidized chromogen} + H_2O$$

Creatininase–CK coupled

$$creatine + ATP \xrightarrow{CK} creatine\ phosphate + ADP$$

$$ADP + phosphoenolpyruvate \xleftrightarrow{PK} pyruvate + ATP$$

$$pyruvate + NADH \xleftrightarrow{LDH} lactate + NAD$$

- NAD → decreased absorbance of NAD at 340nm (UV)
- Reference intervals
  - **Jaffe:**
    - M 0.9—1.3 mg/dL
    - F 0.6—1.1 mg/dL
  - **Enzymatic:**
    - M 0.6—1.1 mg/dL
    - F 0.5—0.8 mg/dL
  - Panic value: ***≥5mg/dL***
  - Males have higher values due to bigger muscle mass and body surface area

METHODS FOR ESTIMATING GFR

- **BUN/creatinine ratio**: 10-20: 1
- **Creatinine clearance**: $\frac{UV}{P}\ x\ \frac{1.73}{A}$
  - Most widely used clearance test
  - Creatinine is endogenous; not significantly reabsorbed nor secreted; that is why CC is the recommended method for estimating GFR
  - In case where the patients' body surface area deviates greatly from the average (obese, big, child) then use a correction factor by multiplying UV/P to 1.73/A
    - U = urine creatinine in mg/dL
    - V = volume of 24-hour urine in mL/min
    - P = plasma/serum creatinine in mg/dL
  - Drawbacks: it has substantial individual variation; quantity of meat ingestion can influence the total daily production of creatinine; interfering chromogens; there is partial secretion of creatinine in the renal tubules but can be prevented by using certain drugs which prevents tubular secretion such as Cimetidine
  - Reference intervals:
    - Male: 85-125 mL/min
    - Female: 75-115 mL/min

OTHER CLEARANCE TESTS

Inulin	**Exogenous**	○ Plant polysaccharide ○ Gold standard for measuring GFR ○ Procedure requires IV infusion and timed urine collections which makes it clinically impractical
Radioisotopes	**Exogenous**	○ Involves administration of ^{125}iodine-iothalamate and ^{99m}Technetium – DTPA (diethylene-triamine-pentaacetate) ○ Most important consideration: hazard posed
Cystatin C	**Endogenous**	○ Only measured in plasma/serum, no need for urine collection ○ Completely reabsorbed by the proximal tubule and destroyed preventing reentry into the circulation ○ Plasma concentration is inversely related to GFR ○ Levels rise more quickly than creatinine in cases of Acute renal failure ○ Unaffected by diet, gender, race, age, and muscle mass ○ Glucocorticoids may reduce its production and may overestimate renal function in kidney transplant patients who are regularly given glucocorticoids
β trace protein	**Endogenous**	○ Not affected by glucocorticoids ○ Originates from the brain and freely filtered by the glomeruli ○ Reabsorbed completely and catabolized by tubular cells
B2-microglobulin	**Endogenous**	○ A component of MHC-Class I and present in all nucleated cells ○ Dissociates from the MHC at a constant rate ○ Falsely elevated: multiple myeloma, lymphoma, and malignancies ○ A low molecular weight protein (11,800 daltons) ○ Freely filtered, reabsorbed, and metabolized completely by the PCT ○ Has a tendency to fold into a beta sheet configuration which results to amyloid formation (dialysis associated amyloidosis)

GFR ESTIMATES

- Cockcroft-Gault formula

$$= \frac{(140 - age)(weight\ in\ kg)}{72\ x\ Serum\ creatinine\ (mg/dL)} x0.85\ (if\ female)$$

  - Women have 15% lower CC than men
  - Urine collection is not needed
- Modification of Diet in Renal Disease formula
  - Original formula in 1999 and modified in 2000
  - Original MDRD formula: Creatinine, BUN, Albumin, Age, Race, and Gender
  - Newer formula: only Creatinine, Race, Age, and Gender

▫ MDRD formula:

$$GFR_{(mL/min/1.73\ m2)} = 170 \times Cr^{-0.999} \times Age^{-0.176} \times 1.180\ (\text{if Black}) \times 0.762\ (\text{if female}) \times BUN^{-0.170} \times Alb^{0.318}$$

$$GFR_{(mL/min/1.73\ m2)} = 186 \times Cr^{-1.154} \times Age^{-0.203} \times 1.212\ (\text{if Black}) \times 0.742\ (\text{if female})$$

▫ CKD-EPI formula:

$$eGFR_{(mL/min/1.73\ m2)} = 141 \times \min(SCr/k,1)^{a} \times \max(SCr/k,1)^{-1.209} \times 0.993^{Age} \times 1.018\ (\text{if female}) \times 1.159\ (\text{if Black})$$

- CKD-EPI (2009) formula is better; has less bias;
  - k = 0.7 in females and 0.9 in males
  - alpha = -0.329 in females and -0.411 for males

CLINICAL SIGNIFICANCE OF ABNORMAL VALUES

Increased (↑) creatinine	muscular diseases, impaired renal function (low GFR), chronic nephritis
BUN/Creatinine ratio > 20:1 with normal creatinine	pre-renal (dehydration)
BUN/Creatinine ratio > 20:1 with elevated (↑) creatinine	renal or post-renal azotemia
BUN/Creatinine ratio <10:1	repeated dialysis, liver disease, low protein diet

URIC ACID (10-20%)

PHYSIOLOGY

- constitutes 10-20% of plasma NPNs
- product of the catabolism of purines (adenine and guanine) in nucleic acids or nucleoproteins; primary source of uric acid is purine catabolism from cell turnover/apoptosis
- relatively insoluble in plasma
- can be deposited in the joints and tissues at concentrations ***greater than 6.8 mg/dL*** causing inflammation
- Freely filtered by the glomeruli and almost completely (98-100%) reabsorbed
- Uric acid in urine represent tubular secretion of uric acids
- Factors that affect concentration:
  - The rate of cell turnover or apoptosis
  - Diet
  - Renal function (glomerular filtration, reabsorption, secretion)

CLINICAL APPLICATION: ANALYTICAL METHODS

A. CHEMICAL (PHOSPHOTUNGSTIC ACID METHOD)

- Catalyst: Sodium carbonate (used in Caraway, Henry, Archibald method) or sodium cyanide (used in Folin, Benedict, and Newton method)
- Colorimetric: Tungsten (blue); the intensity is monitored spectrophotometrically
- False increase: turbid sample

$$\text{uric acid} + H_3PW_{12}O_{40} + O_2 \xrightarrow{Na2CO3\ or\ NaCN} \text{allantoin} + \text{tungsten} + CO_2$$

B. ENZYMATIC (URICASE METHOD)

$$\text{uric acid} + O_2 + H_2O \xrightarrow{uricase} \text{allantoin} + CO_2 + H_2O_2$$

- Detection methods: measurement of the ↓A_{293nm} (decrease in absorbance; UV method) or peroxidase-coupled colorimetric reaction (C/S)
- Uric acid maximally absorbs light at 293nm as it decomposes to allantoin there will be a proportional decrease in the absorbance.
- Reference intervals
  - M: 3.5 - 7.2 mg/dL
  - F: 2.6 - 6.0 mg/dL
  - Panic values: ***≥13 mg/dL***
  - Conversion factor: 0.0595

CLINICAL SIGNIFICANCE OF ABNORMAL VALUES

- ***HYPERURICEMIA***
  - gout, leukemia, chemotherapy, chronic nephritis, Lesch-Nyhan syndrome (inborn error of metabolism resulting from the deficiency of HGPT – hypoxanthine guanine phosphoribosyltransferase presenting as orange sand in diaper), alcoholism, lactic acidosis, and von Gierke's disease
- ***HYPOURICEMIA***
  - Fanconi syndrome (failure/decrease of tubular reabsorption), severe liver disease, purine inhibitors, allopurinol (drug use to treat hyperuricemia)

AMMONIA (0.2%)

PHYSIOLOGY

- Smallest in concentration; only comprises 0.2% of total plasma NPNs
- Ammonia is neurotoxic and the body does not allow its accumulation in the blood
- Produced from the catabolism of amino acids
- Toxic compound metabolized exclusively in the liver (via the Krebs Henseleit or urea cycle)
- Liver function test (hepatic detoxification function), not a renal function test

CLINICAL APPLICATION: ANALYTICAL METHODS

- The preferred specimen: arterial blood
- Although venous blood is not recommended, if used, tourniquets should be used minimally, and fist clenching and relaxing avoided during collection
- Specimens should be kept in ice water until separation of cells from plasma occurs

A. CHEMICAL

- **Potentiometric**: measurement of pH change as NH_3 diffuses through a selective membrane
- **Colorimetric**: using an indicator dye or nesslerization and berthelot

$$NH_3 + \text{bromphenol blue} \longrightarrow \text{blue dye}$$

B. ENZYMATIC (GLDH METHOD)

- NAD is measured as the decrease in absorbance at 340nm
- GLDH: Glutamate dehydrogenase

$$NH_3 + \text{2-oxoglutarate} + NADH + H^+ \xrightarrow{GLDH} \text{glutamate} + NAD$$

- Reference intervals: 19—60 ug/dL
- ***Panic value: >109 ug/dL***

CLINICAL SIGNIFICANCE OF HYPERAMMONEMIA

- Liver disorders like hepatic failure
- Reye's syndrome
- Urea cycle enzyme deficiencies
- Cirrhosis
- Hepatoencephalopathy leading to coma and death

Additional info!

CHRONIC GLOMERULONEPHRITIS

- Lengthy glomerular inflammation (slow developing disease) leading to glomerular scarring and the eventual loss of functioning nephrons
- Gradual development of uremia (or azotemia, excess nitrogen compounds in the blood) may be the first sign of this process

NEUTROPHIL-ASSOCIATED LIPOCALIN (NGAL)

- Early predictor of acute kidney injury
- A 25-kDa protein expressed by neutrophils and epithelial cells including those of the proximal tubule
- It can be measured in plasma and urine and is elevated within 2-6 hours of AKI

PROTEINS

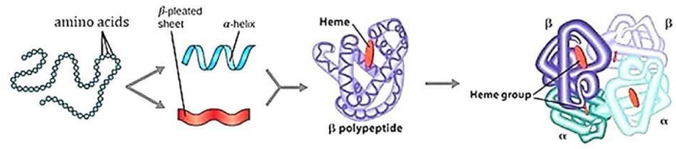
GENERAL PROPERTIES

- Net charge and electrophoretic mobility are determined by the acidic and basic amino acid monomers
- Useful in laboratory analysis specially in fractionation
- **Isoelectric point (pI)** – the pH at which a particular protein has net charge equal to zero
- **Solubility**: lowest at pI or insoluble at pI; hydrophilic and more soluble when charged or when they carry net charge that's also related to the amphoteric nature of proteins
- Nitrogen content of plasma proteins: approx. 16%

AMINO ACIDS	
ESSENTIAL *Cannot be synthesized by the body*	**NON-ESSENTIAL** *Can be synthesized by the body*
- Arginine - Histidine - Isoleucine - Valine - Lysine - Methionine - Phenylalanine - Threonine - Tryptophan	- Alanine - Asparagine - Aspartic acid - Cysteine - Glutamic acid - Glutamine - Glycine - Proline - Serine - Tyrosine

*Glutamine: most abundant amino acid in the body; synthesized from glutamic acid; has a role in the transport of ammonia to the liver for metabolism

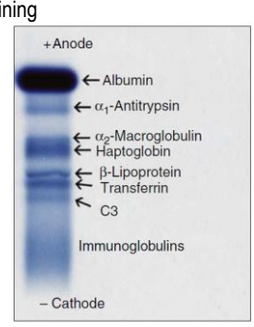
LEVELS OF STRUCTURE



- **Primary** – number and types of amino acids in a specific amino acid sequence
- **Secondary** – regularly repeating structures stabilized by hydrogen bonds between the amino acids (beta pleated sheets and alpha-helix)
- **Tertiary** – overall shape or conformation of the protein molecule; 3-dimensional
- **Quaternary** – results from the interaction of more than one protein molecule or subunit (held together by noncovalent forces)

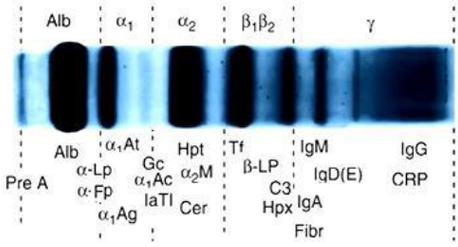
PROTEIN ELECTROPHORESIS

- Serum samples are applied close to the cathode end (sample wells) of a support medium that is saturated with an alkaline buffer (pH 8.6) at which pH proteins carry a net negative charge and migrate toward the anode (+)
- pH > pI = negative charge; pH < pI = positive charge; pH = pI = zero charge
- **Support media**:
  - Agarose
  - Cellulose acetate
  - PAGE (polyacrylamide gel)
- **Stains used**:
  - Coomassie Brilliant Blue
  - Amido black
  - Ponceau S
  - Lissamine green
- **Densitometer** – specialized colorimeter designed to scan and quantitate electrophoretic pattern
  - The concentration of individual fraction is determined based on the intensity of staining



HIGH-RESOLUTION PROTEIN ELECTROPHORESIS

- uses a higher voltage coupled with a cooling system and more concentrated buffer
- separates proteins into 12 or more bands
- detects unusual monoclonal bands
- In stained gel, you can only judge the concentration based on the intensity of staining. However, it is subjective that is why we need to perform densitometric scanning to determine the relative concentration of the major bands.
- The darker the band, the greater the intensity of staining, the taller the peak



- Enable us to see minor and unusual band

SERUM PROTEIN FRACTIONS

PRE-ALBUMIN

- Also known as Transthyretin or Thyroxine-binding pre-albumin (TBPA)
- Indicator of malnutrition (better than albumin and transferrin)
- Binds/transports thyroid hormone (primarily T_4) and Vitamin A associated with Retinol-binding protein
- Forms a distinct band in CSF electrophoresis (Immunofixation electrophoresis; 2nd most abundant protein in CSF)
- Negative acute phase reactant

INCREASED (↑)	DECREASED (↓)
Steroid therapy, chronic renal failure, and alcoholism	Liver disorders, inflammation, malignancy, and poor nutrition

ALBUMIN

- Major contributor to osmotic or oncotic pressure (most abundant plasma protein)
  - Osmotic pressure holds the blood or plasma within the confines of the blood vessel walls
  - ↓ Albumin = ↓ Osmotic pressure → edema and transudative effusion
- General transport protein
  - Transport variety of substances like unconjugated bilirubin, certain drugs (primarily acidic drugs such as penicillin), steroids, free fatty acids, and ions such as calcium and magnesium, and etc.
- Negative acute phase reactant – decreases in inflammatory states

INCREASED (↑)	DECREASED (↓)
- There are no physiological diseases that cause increased production of albumin by the liver. - Elevated serum albumin is only associated with *dehydration*. It is a relative increase that will return to normal when fluids are administered to alleviate the dehydration.	- Liver disorders because of decreased production, - Gastrointestinal disease associated with malabsorption, - Muscle-wasting diseases, severe burns caused by loss, - Protein-losing conditions or Renal disease caused by loss (nephrotic syndrome, glomerulonephritis) - Inflammation, Starvation, and malnutrition

- ***Bisalbuminemia***. Serum protein electrophoresis with normal pattern in right lane and bisalbuminemia consisting of two equal intensity bands of albumin in left lane. The additional band of albumin in this patient migrated more slowly than normal albumin into the α_1-region.

ALPHA 1

1. *α – Fetoprotein*

- Synthesized during gestation in the yolk sac and liver of the fetus, peaking at 13 weeks and declining at 34 weeks.
- Levels in adult serum are negligible (Normally, adult levels are very low).
- Only significant in pregnant women and in adult serum in the context of screening for certain tumors
- **Maternal serum AFP:** is measured between 15 and 20 weeks of gestation and is reported as multiples of the median (MoM).

	INCREASED (↑)	DECREASED (↓)
Maternal serum/Amniotic fluid	Neural tube defects, anencephaly, spina bifida, and fetal distress	Down syndrome, trisomy 18
Adult serum	Hepatocellular carcinoma and gonadal tumors	

2. *α_1 – Antitrypsin*

- Most abundant protein in the alpha 1 band
- Positive acute phase reactant – increases during inflammatory conditions
- Protease inhibitor
- Proteolytic inhibitor: It inhibits the self-destruction of one's own tissue by forming inactive complexes with proteolytic enzymes. In this way the enzymes are inhibited, and tissue destruction through self-digestion is avoided.
  - It is an effective inhibitor of the enzymes *chymotrypsin, plasmin, thrombin, collagenase, and elastase.*
- The primary effect of α_1-antitrypsin may be seen in the respiratory tract and the closed spaces of the body where physiological pH values are maintained
- α_1 –Antitrypsin is least effective in the stomach and intestines.
- In cases of antitrypsin deficiency, the alpha 1 band is almost flat

INCREASED (↑)	DECREASED (↓)
- Inflammation	- Emphysema-associated pulmonary disease or pulmonary emphysema - Severe juvenile hepatic disorders that may result in cirrhosis

3. *α_1 – Acid glycoprotein (orosomucoid)*

- Positive acute phase reactant
- Binds progesterone and some drugs (basic/alkaline drugs such as lidocaine)
- Negatively charged even in acid pH – unique characteristic of α1 – Acid glycoprotein
- Has a very high carbohydrate content (orosomucoid)

INCREASED (↑)	DECREASED (↓)
Inflammatory disorders such as rheumatoid arthritis, pneumonia, and conditions associated with cell proliferation	Nephrotic syndrome

4. *α_1 – Antichymotrypsin*

- Can be considered as a positive acute phase reactant
  - However, the increase in levels is note comparable to the other APR because it is just a mild APR.
- Protease inhibitor
- Binds prostate-specific antigen: might be considered when doing TSA assay

5. *Gc-globulin (DBP)*

- Vitamin D binding protein
- Transports Vitamin D
- It has a role in the maintenance of calcium homeostasis
- ↓ level may lead to abnormal calcium levels

ALPHA 2

1. *α_2 – Macroglobulin*

- Protease inhibitor and Proteolytic enzyme inhibitor (inhibits thrombin, trypsin, and pepsin)
- Largest non-Immunoglobulin protein
- Large size has an important role in protein losing condition.
- It somehow compensates for the loss of proteins (only small proteins are lost).
- ↑ ten-fold in nephrotic syndrome and other protein losing conditions

INCREASED (↑)	DECREASED (↓)
Nephrotic syndrome, contraceptive use, pregnancy, estrogen therapy	*Decreased slightly* in: acute inflammatory disorders and prostatic cancer; *Decreased markedly* in: acute pancreatitis

2. *Haptoglobin*

- Positive acute phase reactant (APR)
- Binds free hemoglobin to preserve iron
- Hemoglobin is lost in urine if not bound to haptoglobin

INCREASED (↑)	DECREASED (↓)
Inflammatory conditions, burns, trauma	*Intravascular hemolysis* because of formation of haptoglobin-hemoglobin complex for removal by the liver

3. *Ceruloplasmin*

- Positive acute phase reactant; has oxidase activity
- Copper binding protein

INCREASED (↑)	DECREASED (↓)
Pregnancy, inflammatory disorders, malignancies, and with intake of oral estrogen and oral contraceptives	*Wilson's disease* (copper accumulation in tissues such as the cornea), malnutrition, malabsorption, severe liver disease

BETAGLOBULINS

1. *Transferrin*

- Negative acute phase reactant
- Transports ferric form of iron
- ↑ level seen in Iron Deficiency Anemia (IDA)
  - Pseudoparaprotein
  - Mistaken for monoclonal gammopathy because the position of the peak is close to the peak produced in monoclonal gammopathy
  - One parameter in the differential diagnosis of microcytic, hypochromic anemias
- Also considered as indicator of malnutrition
- The levels of transferrin are conveniently expressed in terms of ***Total Iron Binding Capacity (TIBC)***
  - Tf (mg/dL) = 0.7 x TIBC (mg/dL)
  - Used to differentiate IDA and ACI

INCREASED (↑)	DECREASED (↓)
Iron-deficiency anemia and pregnancy	Infections, liver disease, and nephrotic syndrome

2. *C3 complement*

- Positive acute phase reactant
- Most abundant complement component
- Immune response; convergence point of classical and alternative pathways

3. *Hemopexin*

- Binds heme; back-up mechanism
  - Conserve iron even though hemoglobin is broken down
- ↓ level in intravascular hemolysis

4. *β_2 – Microglobulin*

- a single polypeptide chain that has a small size because it's just a component of Major Histocompatibility Complex (MHC) or Human Leukocyte Antigen (HLA) molecules
- It is found on the surface of nucleated cells and is notably present on lymphocytes.
- Used to measure Glomerular Filtration Rate (GFR) → β_2-microglobulin clearance test
  - But the recommended procedure is creatinine clearance
  - Endogenous, normally present in the body, dissociates from the MHC at a constant rate, and freely filtered by the glomeruli
  - Not a reliable as a measure of GFR in cases of inflammatory disorders or autoimmune disorders (abnormally high)

INCREASED (↑)
Inflammation, renal failure, lymphocytosis, rheumatoid arthritis, and systemic lupus erythematosus

5. *C-reactive protein*

- Most sensitive positive acute phase reactant
- Enhances phagocytosis in inflammatory disease
- Most useful nonspecific marker of inflammation
- ↑ up to 1000 times in inflammatory states
- Gamma migrating beta globulin; a beta globulin but in electrophoresis it migrates to the gamma region
- **High-sensitivity CRP (hs-CRP)**
  - C-reactive protein (CRP): β-globulin that is an acute-phase reactant
  - High-sensitivity CRP refers to the sensitivity of the assay to determine low levels in serum.
  - Clinical significance: Used as a *predictor for cardiovascular risk*; increased levels seen in inflammation, infection, stress, trauma, and AMI
  - Test methodology
    - Quantified by immunoassay; hs-CRP detection limit 0.05 mg/L
    - Reference ranges: Males, 0.3-8.6 mg/L; females, 0.2-9.1 mg/L
    - Cardiovascular risk classification:
      - Low risk <1.0 mg/L; average risk 1.0-3.0 mg/L; high risk >3.0 mg/L

INCREASED (↑)
Tissue necrosis, rheumatic fever, infections, myocardial infarction, rheumatoid arthritis, and gout

TAKE NOTE!

- **Serum Amyloid A** increases 1000 times in inflammatory conditions but at a later time compared to CRP; an essential APR that helps in the transport of cholesterol
- Most plasma proteins are produced by the HEPATOCYTES in the LIVER
- Immunoglobulins are produced by B cells (plasma cells)

GAMMA

1. *Immunoglobulins*

- Antibodies produces by plasma cells
- Include IgG, IgM, IgA, IgE, and IgD
- Extra bands will be seen if the specimen used is:
  - Plasma: extra band between Beta and Gamma region due to **FIBRINOGEN**
    - present in plasma but not present in serum
  - Hemolyzed serum: additional band will be attributed to **HEMOGLOBIN**
    - Additional band in Beta or between alpha 2 and beta region

	INCREASED (↑)	DECREASED (↓)
IgG	Liver disorders, infections, and collagen disease	Increased susceptibility to infection When a monoclonal gammopathy is associated with an increase in another immunoglobulin
IgA	Liver disorders, infections, and autoimmune diseases	Inhibited protein synthesis and hereditary immune disorders
IgM	Various bacterial, viral, and fungal infections and *Waldenstrom macroglobulinemia*	Renal diseases associated with protein loss and immunodeficiency disorders
IgE	Allergies, asthma, and hay fever, and during parasitic infections.	
IgD	Liver disorders, infections, and connective tissue disorders.	

SUMMARY!

Fraction	Specific Proteins
Albumin	Albumin
Alpha-1 globulin	α_1 – Antitrypsin Lipoproteins
Alpha-2 globulin	Ceruloplasmin Haptoglobin α_2 – macroglobulin Lipoproteins
Beta globulin	Transferrin Hemopexin Complement system Lipoproteins
Gamma globulin	Immunoglobulins

Negative acute phase reactants	Positive acute phase reactants
- Prealbumin (Transthyretin and Retinol-binding protein) - Albumin - Transferrin	- α1 – Antitrypsin - α1 – Acid glycoprotein - α1 – Antichymotrypsin (mild) - Haptoglobin - Ceruloplasmin - Fibrinogen - C3 complement - C-reactive protein - Serum Amyloid A

CLINICAL SIGNIFICANCE OF ABNORMAL PROTEIN LEVELS

HYPOPROTEINEMIA

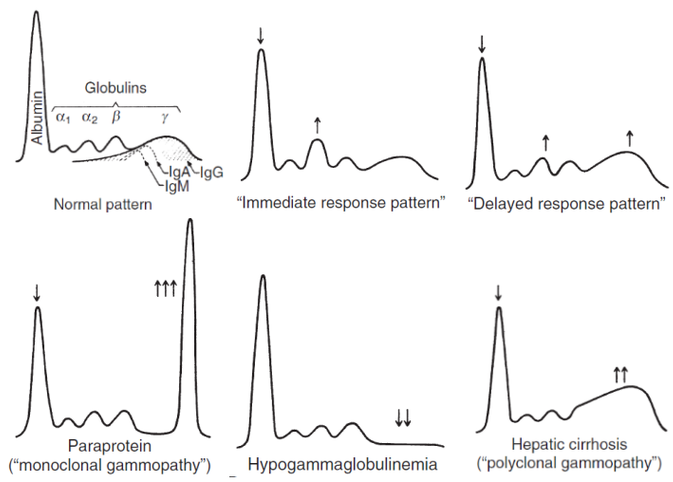
- **Increased protein loss**
  - Nephrotic syndrome, protein-losing enteropathy, blood loss, extensive burns
  - ↓ Alb, ↑ α2, ↓ gamma
  - α2 macroglobulin increase 10 folds in protein losing conditions because of the compensatory mechanism by this protein.
- **Malnutrition/Malabsorption**
  - Indices of malnutrition:
    - Pre-albumin
    - Albumin
    - Transferrin
- **Decreased synthesis**
  - Integrity of the hepatocytes are compromised
  - Hepatic cirrhosis: formation of beta-gamma bridge
  - Polyclonal gammopathy – diffuse increase in beta and gamma region
- **Increased catabolism**
  - infection, inflammation → albumin is slightly low and it is a negative APR which is decreased during inflammatory conditions
  - α1 and α2 - elevated
- **Immunodeficiency**
  - Hypogammaglobulinemia, agammaglobulinemia (Bruton's X-linked agammaglobulinemia; flat gamma curve)

HYPERPROTEINEMIA

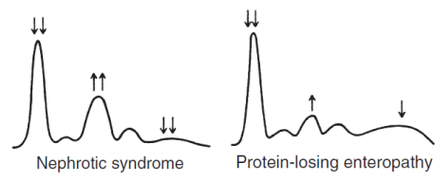
- **Monoclonal gammopathy** *(absolute increase)*
  - Absolute increase in total protein concentration due to inappropriate production of one of the gamma globulins usually IgG or IgM
  - **Multiple Myeloma** = increase IgG
  - **Waldenstrom Macroglobulinemia** = increase IgM
    - Producing a tall gamma peak
    - Low albumin but very high gamma region
- **Dehydration** *(relative increase)* – when the patient is hydrated the protein levels goes back to normal.

SUMMARY!

Acute inflammation	↑ α_1 and α_2
Chronic inflammation	↑ α_1, α_2, and gamma
Cirrhosis	Polyclonal ↑ (all fractions) in gamma with beta-gamma bridging
Monocolonal gammopathy	Sharp ↑ in 1 immunoglobulin ("M" spike), ↓ in other fractions
Polyclonal gammopathy	Diffuse ↑ in gamma
Hypogammaglobulinemia	↓ gamma
Nephrotic syndrome	↓ albumin, ↑ α_2
Alpha-1 antitrypsin deficiency	↓ α_1
Hemolyzed specimen	↑ beta or unusual band between α_2 and beta
Plasma	Extra band (fibrinogen) between beta and gamma



PROTEIN-LOSING CONDITIONS:



PROTEINS IN OTHER BODY FLUIDS

1. URINARY PROTEINS

- Quantification performed on 24-hour urine specimens
- Test methods
  - Sulfosalicylic acid, trichloroacetic acid, benzethonium chloride (turbidimetric)
  - Coomassie brilliant blue (spectrophotometric)
- Reference range urine total protein: 1-14 mg/dL; <100 mg/day
- Clinical significance of proteinuria
  - **Increased protein** in urine may result from tubular or glomerular dysfunction, multiple myeloma, Waldenstrom macroglobulinemia, nephrotic syndrome
  - **Bence Jones protein** may be found in urine of patients with multiple myeloma.
  - Glomerular membrane can be damaged in diabetes, amyloidosis, and collagen diseases.
  - Glomerular dysfunction can be detected in its early stages by measuring microalbumin in urine. **Microalbuminuria** is a condition where the quantity of albumin in the urine is greater than normal, yet it is not able to be detected by the urine dipstick method. The presence of microalbuminuria in a diabetic individual is a concern because it generally precedes nephropathy.
- Methods for quantification: Enzyme immunoassays and immunonephelometric assays
- Reference range for urine albumin: ***<30mg/day***

2. CEREBROSPINAL FLUID (CSF) PROTEINS

- CSF is an ultrafiltrate of plasma formed in the ventricles of the brain.
- Test methods include:
  - sulfosalicylic acid
  - trichloroacetic acid
  - benzethonium chloride
  - Coomassie brilliant blue.
- Reference range: ***15-45 mg/dL***
- Clinical significance:

INCREASED (↑)	DECREASED (↓)
- in viral, bacterial, and fungal meningitis - traumatic tap (bloody) - multiple sclerosis - herniated disk - cerebral infarction	- in hyperthyroidism - central nervous system leakage of CSF

MISCELLANEOUS PROTEINS OF MEDICAL IMPORTANCE

1. MYOGLOBIN

- Tissue location: Found in skeletal and cardiac muscles
- Heme-containing molecule that serves as the primary O_2-carrying protein in skeletal and cardiac muscles
- Used as a marker of myocardial infarction
- ***Detected sooner than troponin following MI but not cardiac (tissue) specific***
- Lacks specificity that is why it is used to rule out MI rather than establish diagnosis.
- Increased in skeletal muscle injuries, muscular dystrophy, and AMI
- Myoglobin is released early in cases of AMI, rising in 1-3 hours and peaking in 5-12 hours, and returns to normal in 18-30 hours.
- It is better used as a negative predictor in the first 2-4 hours following chest pain,
- Test methodology
  - Quantified by immunoassay
  - Reference ranges: Male, 30-90 ng/mL; female, <50 ng/mL

2. CARDIAC TROPONINS

- ***Gold standard for Acute coronary syndrome*** (Myocardial infarction)
  - Cardiac-specific and show sustained elevation
- TnI with greater cardiac specificity than TnT
- **Tissue location:** Troponins T, I, and C form a complex of three proteins that bind to filaments of skeletal muscle and cardiac muscle to regulate muscle contraction.
- **Clinical significance**
  - **cTnT or cTnI** (cardiac troponin T or cardiac troponin I) is used as an **AMI indicator** because of specificity and early rise in serum concentration following AMI.
  - **In cases of AMI, cTnT** increases in 3 - 4 hours following infarction, peaks in 10-24 hours, and returns to normal in 10-14 days.
  - **cTnI** increases in 3-6 hours following infarction, peaks in 14-20 hours, and returns to normal in 5-10 days.
- **Test methodology**
  - Quantified by immunoassay
  - **Reference ranges:** cTnT <0.03 ng/mL; cTnI <0.40 ng/mL (Values vary considerably among laboratories and are dependent on the methodology employed.)

3. NATRIURETIC PEPTIDES

- Structurally related neurohormones that affect body fluid homeostasis and blood pressure
- Three forms: ANP, CNP, BNP
- Although effects are minimal, they function to **promote excretion of sodium and water** by increasing the glomerular filtration rate and decreasing the tubular reabsorption of sodium by the kidneys.
- **B-type (brain) natriuretic peptide (BNP)** is synthesized in and secreted from myocardial ventricles in response to ventricular volume expansion and pressure overload.
  - BNP causes dilation of blood vessels and promotes sodium and water loss, thus reducing fluid load on the heart to improve cardiac function.
  - Specimen: heparinized plasma
- Clinical significance: ***BNP is increased in congestive heart failure*** (CHF)
- **Test methodology**
  - BNP quantified by fluorescence and chemiluminescence immunoassays
  - Reference range: **BNP < 100 pg/mL**
- **ProBNP** assay measures N-terminal proBNP **(NT-proBNP),** which is released when BNP is cleaved from precursor proBNP.
  - NT-proBNP has a longer half-life than BNP.
  - Measurement of NT-proBNP shows no interference from nesiritide (human recombinant BNP) administration to treat CHF.
  - NT-proBNP is measured by electrochemiluminescence.

4. CYSTATIN C

- Endogenous serum marker of GFR
- Useful in cases where creatinine measurement is not appropriate; also associated with an increased risk of CVD and heart failure in the elderly
- Does not require collection of 24-hour urine specimen; uses an immunoassay to determine the cystatin levels in the serum
- Appropriate for patient whose patient is not amenable for 24-hour urine collection

5. β TRACE PROTEIN

- Potential GFR marker
- Correlates with serum cystatin C and urine microproteins (β2 microglobulins)

6. FIBRONECTIN (FN)

- Used as marker for nutritional status
- Demonstrates a wide variety of cellular interactions, including cell adhesion, tissue differentiation, growth, and wound healing
- **Fetal FN** is used to predict the risk of premature delivery
- It is a normal constituent in the placenta and amniotic fluid.
- When stress, infection, or hemorrhage causes leakage of fibronectin into the cervico-vaginal secretions, increased fibronectin is suggestive of risk for premature birth.

7. ADIPONECTIN

- Inversely correlated with BMI (↓ value = ↑ risk of obesity or metabolic syndrome)

8. CROSS-LINKED C TELOPEPTIDES

- Proteolytic fragments of collagen I
- Biochemical marker of bone resorption in serum and urine

9. NEURONAL THREAD PROTEIN

- Biomarker for detecting Alzheimer's disease

10. AMYLOIDS

- Insoluble fibrous protein aggregates formed due to an alteration in their β-pleated sheets; deposition in tissues is termed amyloidosis
- **Amyloid B42 and Tau protein**: supplemental tests used to aid in the differential diagnosis of Alzheimer's disease from other forms of dementia

11. HOMOCYSTEINE

- Clinical significance: Elevated levels cause damage to arterial walls that precedes formation of plaques. It is an indicator of arterial inflammation.
- Test methodology
  - Immunoassay, fluorometric, chromatographic
  - Reference range: 5-15 umol/L

LABORATORY DETERMINATION OF PROTEINS

1. ***KJELDAHL METHOD***
   - Ultimate reference method based on the measurement of nitrogen content
   - It is estimated that the average nitrogen content of protein is 16% of the total weight.
   - Factor: 6.25
   - In the Kjeldahl technique, protein undergoes a digestion process with sulfuric acid through which the nitrogen content of the protein is converted to ammonium ion. The ammonium ion in turn may be reacted with Nessler's reagent, forming a colored product that is:
     - read spectrophotometrically
     - or the ammonium ion may undergo distillation, liberating ammonia that is titrated.

2. ***REFRACTOMETRY***
   - Measurement of refractive index due proteins in solution
   - Is based on the change in velocity of light (light is bent) as light passes through the boundary between air and water, which function as two transparent layers. In protein analysis, the light is bent and such change is proportional to the concentration of the solutes (proteins) present in the water (serum).

3. ***TURBIDIMETRY***
   - For CSF protein analysis
   - Decrease in light transmission
   - Measurement of the change in optical density following precipitation with trichloroacetic acid (TCA) for CSF, Sulfosalicylic acid (SSA) for urine, and Benzalkonium chloride (BC) for automated methods, added to the sample prior to the determination of the change in optical density

4. ***UV ABSORPTION***
   - Measurement of increased absorbance at 280nm owing mostly to Tryptophan, Tyrosine, and Phenylalanine
   - Direct spectrophotometry; no reagents used

5. ***BIURET METHOD***
   - Colorimetric method; Detection of peptide bonds forming a violet color
   - Based on cupric ions complexing with peptide bonds in an alkaline medium to produce a purple-colored complex. The amount of purple complex produced is directly proportional to the number of peptide bonds present and reflects protein concentration.
   - **Biuret reagent**: is a combination of copper sulfate, potassium iodide in sodium hydroxide, and potassium sodium tartrate.
     - The *copper sulfate* is the key to the reaction because it is the cupric ion that complexes with the peptide bonds of protein.
     - To keep the copper in solution until its use, *potassium sodium tartrate* is employed as a complexing agent
     - The autoreduction of copper is prevented by *potassium iodide*.

6. ***FOLIN-CIOCALTEAU (LOWRY) METHOD***
   - Colorimetric method
   - More sensitive than Biuret; Oxidation of phenolic compounds in proteins using phosphotungstomolybdic acid producing a deep blue color

7. ***NINHYDRIN METHOD***
   - Colorimetric method
   - Formation of violet color in the presence of amino groups in amino acids and peptides following chromatography

8. ***DYE-BINDING TECHNIQUE***
   - Measurement of the shift in absorbance when proteins bind to a bye
   - The increase in absorbance is directly proportional to protein concentration.
   - **Dye used for total protein**
     - ***Coomassie brilliant blue (CBB)***
       - stain the dye following electrophoresis
       - it may also be used to determine total protein level based on the principle that the absorbance will change when a protein binds to the dye or when the dye binds to the protein
       - The concentration of total protein in cerebrospinal fluid (CSF) is 15-45 mg/dL. Such a low level of protein requires a method with sufficient sensitivity such as Coomassie brilliant blue.
     - Amido black
     - Ponceau S
   - **Dyes used for albumin** – most routinely performed
     - Methyl orange – nonspecific
     - 2,4-HABA – less sensitive; more specific than methyl orange
     - BCG (Bromocresol Green) – most commonly used dye
     - BCP (Bromocresol Purple) – specific, sensitive, precise

REFERENCE VALUES

TP, Ambulatory	6.5 – 8.3 g/dL
TP, Recumbent position	6.0 – 7.8 g/dL
Albumin	3.5 – 5.5 g/dL
A/G ratio	1.1 - 2

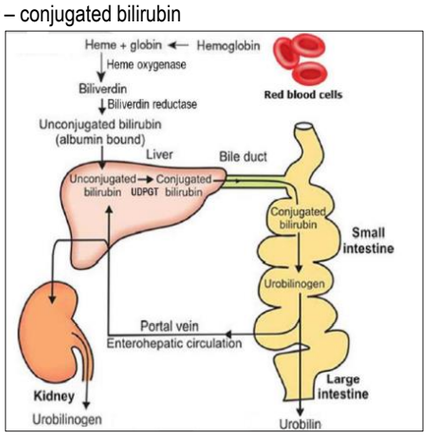
A/G RATIO

- Important parameter especially when distinguishing among the causes of decrease A/G ratio
- Its value is significant in the diagnosis of multiple myeloma and it is also decreased in protein losing conditions and liver disease
- We have to take in consideration the TPAG level [Total Protein, Albumin, Globulin level]
- Globulins are not routinely assayed; they are just calculated from the total protein and albumin. Therefore, we can determine A/G ratio:
  - Albumin ÷ Total Protein – Albumin = A/G ratio
- Significant when the level is low
- Low A/G ratio can be attributed to a decreased albumin or increased globulin
  - ↓ albumin = ↓ A/G ratio
  - ↑ globulin = ↓ A/G ratio
- If AP is high and low A/G ratio, this is consistent with multiple myeloma
  - ↓ A/G ratio, ↑ Globulin, ↑ TP = Monoclonal gammopathy
  - ↓ A/G ratio, ↓ Albumin, ↓ TP = NS, PLE, Liver disease

ASSESSMENT OF LIVER FUNCTION

FUNCTIONS OF THE LIVER

- **Storage** - glycogen, fat-soluble vitamins (ADEK)
- **Synthetic/Metabolism** – lipids, lipoproteins, carbohydrates (through glycogenesis, gluconeogenesis, glycogenolysis), proteins (TPAG and Coagulation factors)
  - Coagulation test such as prothrombin time
- **Detoxification** – drugs, poisons, and metabolic products like unconjugated bilirubin (toxic if it increases because it may cross the blood-brain barrier), and ammonia
- **Secretory and Excretory** – conjugated bilirubin
  - Primarily bile and part of bile is conjugated bilirubin
  - Bile is also related to the synthetic function of the liver because is also synthesized by the liver before it is secreted



BILIRUBIN METABOLISM

- Red blood cells are destroyed releasing hemoglobin which is broken down into ***heme and globin*** by the cells of the reticuloendothelial system
- Globin returns to the amino acid pool, ***iron*** released from heme is carried to storage sites, and heme is acted upon by ***heme oxygenase*** to form ***biliverdin***
- Biliverdin is acted upon by ***biliverdin reductase*** to form ***unconjugated bilirubin (B1/albumin bound)***
- Unconjugated bilirubin is transported in the portal system, bound mainly to albumin
- Once in the hepatocyte, bilirubin is handed off from one protein complex to another and transported to the smooth endoplasmic reticulum where it is conjugated in the presence of the enzyme ***Uridine Diphosphate Glucuronyl Transferase (UDPGT)*** producing ***bilirubin diglucoronide or conjugated bilirubin***.
- The conjugated bilirubin is then secreted to the bile canaliculi and ultimately excreted into the intestinal tract where it is further metabolized by intestinal bacteria to ***Urobilinogen*** *(May grouply refer to Urobilinogen, stercobilinogen, and mesobilirubinogen)*
- The majority of urobilinogen is reabsorbed and re-excreted by the liver.
- ***Stercobilinogen*** is not reabsorbed.
- While urobilinogen is being recirculated, a minor fraction may be excreted in urine. Small amount escapes in urine that explains why a normal urine contains urobilinogen as it is recirculated in the body.
- Ultimately, intestinal urobilinogen and stercobilinogen are oxidized by anaerobic bacteria to stool pigments ***Urobilin*** and stercobilinogen is oxidized to ***Stercobilin*** that give stool its brown color.

FORMS OF BILIRUBIN IN PLASMA

	B1	B2
Origin	- Pre-hepatic/Hemolytic - Produced from the destruction of senescent erythrocytes	- Post-hepatic/ Regurgitative - Levels ↑ when there is backflow when there is obstruction in the passageway
Structure	- Unconjugated	- Conjugated
Solubility	- H_2O insoluble	- H_2O soluble
Polarity	- Non-polar	- Polar
Reaction	- Indirect (needs an accelerator/solubilizer to react with diazo)	- Direct
Excreted in urine	- No	- Yes
Affinity for CNS	- Yes → kernicterus	- No

Delta bilirubin
Conjugated bilirubin (B2) covalently *bound to albumin*
When bound to albumin, its excretion into urine is prevented
Long half-life of 17 days (half-life similar to albumin)
Seen in cases of biliary obstruction (cholelithiasis) B2 is not excreted → Jaundice (accumulation of B2)
Reacts directly with the diazo reagent Performing direct bilirubin assay will measure both free form and delta form

DERANGEMENTS OF BILIRUBIN METABOLISM

Type	Pathologic conditions	B1	B2	Comments		
Pre-hepatic *(always **unconjugated hyperbilirubinemia**)*	**Hemolytic disorders**	↑	N	- ↑ **RBC destruction** → ↑ production of unconjugated bilirubin - Due to intravascular hemolysis - Urine bilirubin is negative, urine urobilinogen is ↑ or 3+ because of failure of Urobilinogen transport		
Hepatic *(**conjugated, unconjugated, or both**)*	**Gilbert syndrome**			- ***Hepatic unconjugated*** *hyperbilirubinemia*	- Transport defect coupled w/ a mild deficiency in UDPGT - Only 30% of enzyme activity is present (but is not serious enough)	
	Lucey Driscoll syndrome				- Presence of circulating inhibitor of bilirubin conjugation - Transient familial neonatal hyperbilirubinemia	Treated with phototherapy to prevent kernicterus (>20 mg/dL)
	Physiologic jaundice of the newborn				- Transient UDPGT deficiency	
	Crigler-Najjar syndrome				- Deficiency (partial; in heterozygotes) or total absence of UDPGT (in homozygotes)	
	Viral hepatitis **Cirrhosis** **Hepatic Carcinoma**	↑		- ***Combined hepatocellular hyperbilirubinemia*** because of the interference on the bile flow (↑ B2) and failure of enterohepatic circulation in relation to decreased conditions - ↑ B_1 = ↓ conjugation, failure of enterohepatic circulation → ↑ urine Urobilinogen - ↑ B_2 = caused by interference in bile flow		
	Dubin-Johnson syndrome **Rotor syndrome**	N	↑	- ***Hepatic conjugated*** *hyperbilirubinemia*	- Secretory defect → failure to convey the conjugated bilirubin to the bile canaliculi - Dubin-Johnson: deficiency of canalicular transport protein; hepatocytes contain dark stained granules or pigmented lysosomes - Rotor syndrome: unknown cause; no dark stained granules or pigmentation	
Post-hepatic *(always **conjugated hyperbilirubinemia**)*	**Biliary obstruction**			- Obstruction due to Cholelithiasis or tumors (↑ urine bilirubin, ↓ urine urobilinogen) - Acholic stool (pale/clay colored stool)		

BILIRUBIN DETERMINATION

SPECIMEN CONSIDERATIONS:

- ***AVOIDANCE OF LIGHT EXPOSURE***: because bilirubin is photosensitive; photo-oxidation of bilirubin must be avoided
- ***AVOID HEMOLYSIS***: it may interfere the colorimetric-spectrophotometric (C/S) methods, leading to falsely decreased results with the Evelyn-Malloy technique

METHODS:

Total Bilirubin	Bilirubin + Diazo + Accelerator (to measure B1) → Azodipyrroles (all fractions, including B1)
Direct Bilirubin	Bilirubin + Diazo → Azodipyrroles (B2 and delta only; No accelerator)
Indirect Bilirubin	Calculated → TB - DB

COLORIMETRIC-SPECTROPHOTOMETRIC - VAN DEN BERGH REACTION

- there is accelerator
- Based on the Van den Bergh reaction (diazotization with incorporated accelerator/solubilizer)

	EVELYN-MALLOY	JENDRASSIK-GROF
Reagents	Diazo reagent or Diazotized sulfanilic acid (0.1% sulfanilic acid in HCl, 0.5% sodium nitrite)	Diazo reagent Ascorbic acid (stopping reagent) Alkaline tartrate solution (alkalinizing reagent)
Accelerator/ Solubilizer	Methanol	Caffeine benzoate
pH of the reaction	Acidic	Alkaline (basic)
Color reaction	Red-purple	Blue
Absorbance maximum	560 nm	600 nm
Comments	Affected by hemolysis or by high hemoglobin level	Better than Evelyn-Malloy because it is not affected by pH changes, high protein concentration, and hemolysis (up to 750 mg/dL of hemoglobin level)

DIRECT SPECTROPHOTOMETRIC

- No reagent is added because bilirubin is pigmented
- Absorbance measurement at **450nm** to determine the total bilirubin concentration in the serum of newborns or testing the amniotic fluid of pregnant women to determine the risk for HDN (Rh incompatibility between the mother and the fetus)
  - Strasinger: 450nm (bilirubin in amniotic fluid)
- Not done on adult serum due to the presence of *carotenoid pigments* that would affect the absorbance at 455nm

ENZYMATIC

- *bilirubin oxidase* is used to convert bilirubin back to biliverdin
- Oxidation of bilirubin to a colorless product and measurement of the change (decrease) in absorbance

Reference values:

B1	0.2-0.8 mg/dL
B2	0-0.2 mg/dL
Total	0.2-1.0 mg/dL

- **Conversion factor** (mg/dL to mmol/L): 17.1
- **> 3 mg/dL:** jaundice appears/manifests
- **≥ 15mg/dL**: Panic value/critical value (especially true for newborns with Crigler-Najjar syndrome or Physiologic jaundice of the newborn)
- **≥ 20 mg/dL**: Kernicterus

DIFFERENTIAL DIAGNOSIS OF LIVER DISORDERS

Condition	AST	ALT	LD	ALP	TP	Albumin	Bilirubin	Ammonia
1. Hepatitis	↑	↑	↑	↑	N		↑	N
2. Cirrhosis	N	N	N	N or Slightly ↑	↓		↑	↑
3. Biliary obstruction	N	N	N	↑	N		↑	N
4. Space-occupying lesion	N or ↑	N or ↑	↑	↑	N		N or ↑	N
5. Passive congestion	Slightly ↑	Slightly ↑	Slightly ↑	N or Slightly ↑	N		N or Slightly ↑	N
6. Fulminant failure	Very ↑	↑	↑	↑	↓		↑	↑

- Markers of hepatocellular injury: AST, ALT, LP, ALP
- Markers of hepatobiliary disease: ALP, GGT, 5'-Nucleotidase
- Elevated liver enzymes and bilirubin; Normal TP, albumin, and ammonia: ***Hepatitis***
- Hepatic functions = Synthetic (TP, Albumin, Bilirubin), Conjugation (Bilirubin), Excretory (Bilirubin), Detoxification (Ammonia)

CLINICAL ENZYMOLOGY

CLASSES OF ENZYMES

EC CODE	ENZYME CLASS	REACTION CATALYSED	EXAMPLES
1	**Oxidoreductases/** *dehydrogenases*	Oxidation-reduction	• Lactate dehydrogenase (LD) • Glucose 6 phosphate dehydrogenase (G-6-PD) • Glutamate dehydrogenase (GLDH) • Malate dehydrogenase (MDH)
2	**Transferases**	Transfer of functional groups	• Aminotransferases/Transaminases: Aspartate aminotransferase [SGOT: *serum glutamic oxaloacetic transaminase*], Alanine aminotransferase [SGPT: *serum glutamic pyruvic transaminase*] • Kinases: Creatine kinase (CK), Pyruvate kinase (PK) • Transpeptidases: Gamma Glutamyl Transferases (GGT)
3	**Hydrolases**	Hydrolysis	• Phosphatases: ALP, ACP • Markers of pancreatitis: Amylase, Lipase • Markers of pancreatic insufficiency: Trypsin, Chymotrypsin, elastase, cholinesterases, 5'-Nucleotidase
4	**Lyases**	Removal of groups to form double bonds	• Aldolase
5	**Isomerases**	Interconversion of geometric, optical, or positional isomers	• Triose phosphate isomerase
6	**Ligases**	Bond formation coupled with ATP hydrolysis	• Glutathione synthetase

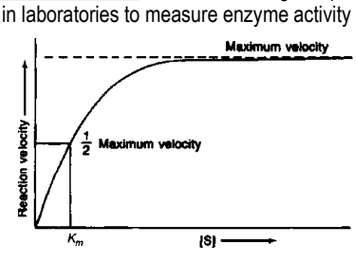
ENZYME KINETICS

1. Enzymatic Reaction/Catalytic Mechanism: **E + S → ES → P + E**
   - The active site of the enzyme accommodates a specific substrate forming the enzyme-substrate complex
   - The reaction may proceed forward leading to the formation of the product (conversion of substrate to product) and liberation of enzyme, or if the requirements are not met, then the ES complex dissociates
   - The enzyme is not consumed in the reaction
   - Enzyme concentration is not measured but enzyme activity
2. **Michaelis-Menten equation**:

V = Velocity of enzyme catalyzed reaction
V_{max} = Maximum velocity
K_m = Michaelis-Menten constant; substrate concentration at ½ of V_{max}
S = Substrate concentration

3. **Lineweaver-Burke equation:**
   - Double reciprocal plot of the Michaelis-Menten equation

4. Effect of substrate and enzyme concentration
   - **First-order kinetics**
     - point at which the enzyme concentration exceeds that of the substrate (E>S)
     - the more substrate is added, the faster the reaction goes
     - substrate concentration is the rate limiting component
   - **Zero-order kinetics**
     - substrate is greater than the available enzyme or active sites are all saturated (S>E)
     - no further increase but the reaction continues because once product is formed, the enzyme is liberated to catalyze another reaction
     - enzyme concentration is the rate limiting component
     - used in laboratories to measure enzyme activity

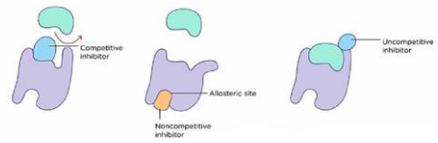


MEASUREMENT OF ENZYME ACTIVITY

1. Methods
   a. **Fixed-time/endpoint**
      - the reactants are combined, the reaction proceeds for a designated time, the reaction is stopped (usually by inactivating the enzyme with a weak acid), and a measurement is made of the amount of reaction that has occurred.
   a. **Continuous-monitoring/Kinetic**
      - multiple measurements of absorbance change as a function of time, either at specific time intervals or continuously by a continuous-recording spectrophotometer.
- Activity Units
   a. **IU (International unit)**: amount of enzyme that catalyzes the reaction of *1 umol of substrate per minute*
   b. **Katal**: amount of enzyme that catalyzes the reaction of *1 mol of substrate per second*
      - 1 IU = 17 nkat (0.0167 ukat)

OTHER FACTORS THAT INFLUENCE ENZYME REACTIONS

1. **pH**
   - Most reactions occur in the pH range of ***7-8***
   - except ACP (pH 4-5) and ALP (pH 9-10); buffers are used to allow maximal activity in the lab
2. **Temperature**
   - For every 10°C increase in temperature, the rate of the reaction doubles (velocity doubles) up to denaturation point (***40-50°C***)
   - enzyme assays are performed at *room temperature or 37°C*
3. **Cofactors**: Non-protein entities that must bind to particular enzymes before a reaction occurs.
   - **Activators** (inorganic cofactors):
      - may be essential for the reaction or may only enhance the reaction rate
      - include *metallic and nonmetallic ions* e.g. calcium (for amylase activity), magnesium (need in kinases and ALP assays), zinc, chloride (for amylase activity), bromide
   - **Coenzymes** (organic cofactors):
      - serve as second substrates for enzymatic reactions
      - called prosthetic groups when bound tightly to the enzyme
      - include *nucleotide phosphates* (NAD/NADH and NADP/NADPH) and *vitamins* such as pyridoxal phosphate (Vitamin B_6 for transaminases such as AST and ALT)
      - **Apoenzyme**: coenzyme that serves as a prosthetic group while the complex is called holoenzyme
4. **Inhibitors**
   a. **Competitive inhibitor**
      - Competes with the substrate for binding to the active site
      - Reversed by: increasing the substrate concentration or by diluting the sample (to dilute the inhibitor and weaken the inhibition)
   b. **Non-competitive inhibitor**
      - Binds to sites other than the active site (allosteric sites) and cause the conformation of the enzyme to change such that the binding site changes and the substrate no longer fits, or the enzyme denatures
      - Reversible: if it doesn't cause denaturation of the enzyme or cause the conformation to change
      - Irreversible: if the inhibitor causes a change in the conformation leading to denaturation of the enzyme
   c. **Uncompetitive inhibitor**
      - Binds to the enzyme after it has accommodated the substrate
      - Binds to the ES substrate complex
      - Irreversible



MACROENZYMES

- High-molecular-mass forms of the serum enzymes that can be bound to either:
   - an immunoglobulin (macroenzyme type 1)
   - a non-immunoglobulin substance (macroenzyme type 2)
- Enzymes with macroforms:
   - Macro-CK, macroamylase, ACP, ALP, AST, ALT, GGT, Lipase, and LD

ENZYMES OF CLINICAL IMPORTANCE

CREATINE KINASE

Tissue Sources:

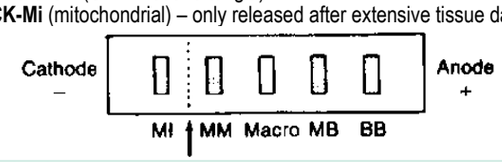
- Brain, heart (cardiac muscles), and skeletal muscles

Diagnostic Significance

- **Pronounced elevation (≥5x ULN)**: ***Duchenne's muscular dystrophy*** (source of highest elevation of CK), polymyositis, dermatomyositis, myocardial infarction
- **Mild to moderate elevation (≥2-4x ULN)**: acute agitated psychosis, alcoholic myopathy, severe exercise, delirium tremens, severe ischemic injury, pulmonary infarction, intramuscular injections, hypothyroidism, muscular trauma

Isoenzymes

- CK isoenzymes consist of two subunits: **M** for muscle and **B** for brain.
- Each CK isoenzyme is a dimer with three possible types: CK-MM (or CK-3), CK-MB (or CK-2), and CK-BB (or CK-1).
- In serum, healthy individuals have ***CK-MM as the major isoenzyme*** and a small amount of CK-MB (less than 6% of total CK), whereas CK-BB is not normally detectable.
- **CK-BB (CK1)** - elevated in the presence of macro-CK (combined with IgG); *fastest and most anodal* in electrophoresis
   - elevated in central nervous system disorders and tumors of various organs, including the prostate gland.
- **CK-MB (CK2)** - present in significant amounts in the cardiac tissue; <6% of total CK in serum
   - increases are associated with heart muscle damage, and elevations are indicative of AMI when used in conjunction with other markers, such as troponin. However, CK-MB also increases in other disorders, such as skeletal muscle damage.
- **CK-MM (CK3)** – major isoenzyme found in striated and skeletal muscle and normal serum (94-100%); *slowest and most cathodal* in electrophoresis
   - increases are associated with skeletal muscle and heart muscle disorders.
- Abnormal isoenzymes
   - **Macro-CK** (CK-BB bound to IgG)
   - **CK-Mi** (mitochondrial) – only released after extensive tissue damage



CK determination

- **Specimen consideration**:
   - the use of *chelating agents must be avoided* (use serum or heparinized plasma)
   - avoid *excessive/prolonged storage* because CK is labile when stored (CK1 is the most labile)
   - *hemolysis* must be avoided because it would falsely increase the reverse CK assay

METHODS (FOR TOTAL CK ASSAY)	
Tanzer-Gilvarg *Forward reaction*	**Oliver-Rosalki** *Reverse reaction*
$creatine + ATP \overset{CK}{\rightarrow} creatine\ PO_4 + ADP$ $ADP + PEP \overset{PK}{\leftrightarrow} pyruvate + ATP$ $pyruvate + NADH \overset{LDH}{\longleftrightarrow} lactate + NAD$ *NAD is measured as the ↓ in absorbance at 340nm	$creatine\ PO_4 + ADP \overset{CK}{\rightarrow} creatine + ATP$ $ATP + glucose \overset{HK}{\leftrightarrow} ADP + glucose6phosphate$ $G6P + NADP \overset{G6PD}{\longleftrightarrow} 6PG + NADPH$ *NAD is measured as the ↑ in absorbance at 340nm

- Reference values (Total CK):
   - Male: 15-160 U/L (37°C)
   - Female: 15-130 U/L (37°C)

LACTATE DEHYDROGENASE

Tissue Sources

- Heart, RBCs, renal cortex, lungs, liver, skeletal muscle (arranged according to isoenzyme predominance)
   - LD1 and 2 = Heart, RBCs, renal cortex
   - LD3 = Lungs
   - LD 4 and 5 = Liver and skeletal muscles
- Inside the cardiac myocytes: LD1 is more abundant than LD2 (LD1>LD2)
- In normal serum: LD2 is greater than LD1 (LD2>LD1)
   - **Flipped pattern** = LD1>LD2 (associated with myocardial infarction, hemolytic disorders, and renal infarction; hemolyzed sample – Artifactual hemolysis)

Isoenzymes

- LD isoenzymes consist of *four subunits* (polypeptide chains) derived from two types of polypeptides designated M (muscle/liver) and H (heart).
- LD-1 (HHHH), LD-2 (HHHM), LD-3 (HHMM), LD-4 (HMMM), LD-5 (MMMM)
- Each LD isoenzyme is a tetramer with five isoenzyme types: LD-1 through LD-5.
- **LD-1 and LD-2** are associated with acute myocardial infarction (AMI) and erythrocyte destruction.
- **LD-3** is associated with pulmonary disorders, pancreatitis, and lymphocytosis.
- **LD-4 and LD-5** are associated with liver and skeletal muscle disorders.

Diagnostic Significance

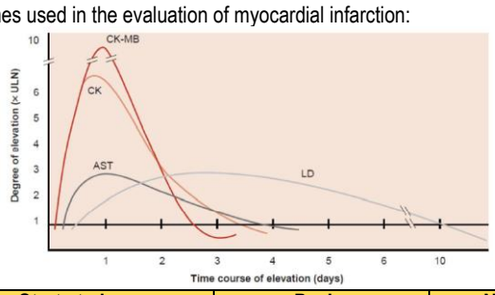
- **Pronounced elevation (>5x ULN):** ***Megaloblastic/pernicious anemia*** (source of highest elevation – because of the destruction of RBC precursors in the bone marrow through intramedullary hemolysis), renal infarction, systemic shock and hypoxia, hepatic metastases, hepatitis
- **Moderate elevation (3-5x ULN):** Myocardial infarction, hemolytic conditions, pulmonary infarction, muscular dystrophy, delirium tremens, leukemias, IM
- **Slight elevation (<3x ULN):** Most liver diseases, nephrotic syndrome, hypothyroidism, cholangitis

LD Determination

- Specimen considerations:
  - Avoidance of *hemolysis* (most greatly affected by hemolysis)
  - Avoid *cold storage* or do not freeze the sample because LD is cold labile
    - loses activity at cold temperature
    - LD-4 and LD-5 are labile at 4°C
    - LD5 most cold-labile
    - Recommended storage temperature: 25°C
  - Do not use *chelating agents* such as EDTA
- Methods:

Wacker *Forward reaction*	Wroblewski-Ladue *Reverse reaction*
$lactate + NAD \overset{LDH}{\longleftrightarrow} pyruvate + NADH + H^+$ *measured as the ↑ in absorbance at 340nm Optimal pH: 7.1-7.4	$pyruvate + NADH + H^+ \overset{LDH}{\longrightarrow} lactate + NAD^+$ *measured as the ↓ in absorbance at 340nm Optimal pH: 8.3-8.9

- Reference value: 100-225 U/L at 37°C
- Enzymes used in the evaluation of myocardial infarction:



	Starts to increase	Peaks	Normalize
CK	4-6 hours	12-24 hours	2-3 days
AST	6-8 hours	18-24 hours	4-5 days
LD	12-24 hours	48-72 hours	10 days

AMINOTRANSFERASES/TRANSAMINASES

	AST/SGOT	ALT/SGPT
1. Tissue sources	• Liver, heart, skeletal muscles	• Liver *(more liver specific than AST; more useful in diagnosing hepatocellular disorders)*
2. Diagnostic Significance	• **Pronounced ↑ (>5x ULN)**: ***Acute hepatocellular disorders***, circulatory collapse, myocardial infarction • **Moderate ↑ (3-5x ULN)**: muscular dystrophy, hepatic tumor, biliary obstruction, CHF, cardiac arrhythmia • **Slight ↑ (<3x ULN)**: cirrhosis, pericarditis, pulmonary infarction, cerebrovascular accident According to Ciulla: • >100 times ULN in viral hepatitis • >20 times ULN in infectious mononucleosis • >4 times ULN in cirrhosis • >8 times ULN in skeletal muscle disorders • >3 times ULN in pulmonary emboli	• ***Acute hepatocellular disorders*** *(more specific, higher and more sustained elevation because it has a longer half-life; De Ritis ratio [AST/ALT] <1)* **De Ritis ratio of less than one has a predominance of ALT* • Hepatocellular disorders (hepatitis cirrhosis) exhibit higher ALT levels than intra- or extrahepatic obstruction.
3. Isoenzymes	• **Cytoplasmic AST** – routinely determined in AST assays • **Mitochondrial AST** - ↑ in necrotic processes such as in alcoholic hepatitis or cirrhosis and fulminant hepatic failure where the De Ritis ratio is >3 or has predominance of AST; has an even longer half-life than ALT	

AST/ALT Determination

- Specimen Considerations:
  - Hemolysis will cause false increase
  - Refrigeration is allowed for 3-4 days
- Methods
  - **Reitman-Frankel (colorimetric/spectrophotometric)** – uses 2,4-dinitrophenylhydrazine or DNPH (serves as color developer) and 0.4N NaOH (serves as color intensifier)
  - **Coupled Enzymatic Reaction** – based on Karmen method

$$aspartate\ (alanine) + \alpha - ketoglutarate \overset{AST\ (ALT)}{\longleftrightarrow} oxaloacetate\ (pyruvate) + glutamate$$

$$oxaloacetate\ (pyruvate) + NADH + H^+ \overset{MDH\ (LDH)}{\longleftrightarrow} malate\ (lactate) + NAD^+$$

- positive = brown; but blue in Calbreath's textbook
- NAD is measured as the ↓ in absorbance at 340nm
  - References
    - AST: 5-35 U/L
    - ALT: 7-45 U/L

ALKALINE PHOSPHATASE

Tissue Sources

- Liver, bone, placenta (in pregnant women), intestine
- According to Ciulla:
  - Highest concentrations are found in liver, bone, intestines, spleen, kidney, and placenta. ALP is found on cell surfaces, in sinusoidal and bile canalicular membranes in the liver, and in bone osteoblasts.
- In normal adult serum, ALP is mainly of liver origin, with a small amount from bone.

Diagnostic Significance

- **Pronounced elevation (>5x ULN):** Hepatobiliary disorders such as Bile duct obstruction and biliary cirrhosis, ***Paget's disease/Osteitis deformans*** (source of highest elevation), osteogenic sarcoma, hyperparathyroidism
- **Moderate elevation (3-5x ULN):** Granulomatous or infiltrative diseases of the liver, IM, metastatic bone tumors, rickets, osteomalacia
- **Slight elevation (<3 ULN):** Viral hepatitis, cirrhosis, healing bone fractures, growing children, pregnancy
- **Decreased serum ALP** levels are seen in hypophosphatasia because of lack of ALP bone isoenzyme. This disorder is characterized by insufficient bone calcification.
- ALP levels are ***normally higher in children*** than adults because of bone growth.
- ALP levels are ***normally higher in women during pregnancy*** because the placenta is a source of ALP.

Isoenzyme analysis

- Electrophoresis: [Anode] **L**iver, **B**one, **P**lacental, **I**ntestinal [Cathode]
- Heat stability: **P**lacental [most heat stable]> **I**ntestinal> **L**iver> **B**one [most heat labile]
- Chemical inhibition:
  - **Urea** (strongly and selectively inhibits bone ALP)
  - **Phenylalanine** (placental, intestinal, carcinoplacental isoenzyme)
  - **Levamisole** (liver and bone)
  - **Leucine** (Nagao isoenzyme)
- Atypical isoenzymes (not routinely used as tumor markers):
  - **Kasahara isoenzyme**: associated with hepatoma and GI tumors
  - **Regan** – carcinoplacental isoenzyme; most heat stable; associated with gynecologic, ovarian, lung, and breast cancers
  - **Nagao** – carcinoplacental isoenzyme, a variant of the Regan isoenzyme; associated with metastatic carcinomas of pleural surfaces
- **Intestinal ALP**
  - isoenzyme whose presence in serum depends on the blood group and secretor status of the individual
  - more likely to be found in B or O secretor individuals
  - bound by group A erythrocytes
  - level increases after consumption of a fatty meal

ALP Determination

- Specimen Considerations:
  - *Hemolysis* will falsely increase ALP
  - Avoid *prolonged storage* (especially uncapped tubes; alkalinization = increased levels)
- Reference Method:
  - **Bowers-McComb method** (Colorimetric/Spectrophotometric)
    - Substrate = paranitrophenylphosphate (PNPP)
    - Positive = Yellow; pH = 9-10
    - Assay requires addition of Magnesium (cofactor/activator of ALP)
- Other methods:
  - Bessy, Lowry, Brock (PNPP)
  - King-Armstrong (phenylphosphate)
  - Moss (α naphthylphosphate)
  - Bodansky, Sinowara, Jones, Reinhart (ß-glycerophosphate)
  - Huggins-Talalay (phenolphthalein diphosphate)
  - Klein, Babson, Reed (buffered phenolphthalein phosphate)
- Reference Values: 30-90 U/L

PANCREATIC ENZYMES

	AMYLASE	LIPASE
1. Tissue sources	• Found in pancreas, salivary glands, small intestine, fallopian tubes, and other tissues *(less tissue specific)*	• Pancreas *(more specific; produces higher and more sustained elevations; has longer half-life)* • Lesser amounts in gastric mucosa, intestinal mucosa, adipose tissue
2. Diagnostic Significance	• Acute pancreatitis *(occur in 2-12 hours after the onset of pain, with peak values in 24 hours, and return to normal in 3-4 days)*. • Increased in: Mumps, parotitis, perforated peptic ulcer, intestinal obstruction, cholecystitis, ruptured ectopic pregnancy, mesenteric infarction, acute appendicitis, macroamylasemia *(amylase bound to a protein; not excreted in urine because it cannot be filtered by the glomeruli)*	• **Acute pancreatitis** *(occur in 4-8 hours after the onset of pain, with peak values in 24 hours, and return to normal in 8-14 days)*. • Increased in: Perforated peptic ulcer, duodenal ulcers, intestinal obstruction, cholecystitis
3. Isoenzymes	• Pancreatic amylase • Salivary amylase *(identified by chemical inhibition assay and is inhibited by wheat germ lectin)*	• None

Amylase Determination

- Specimen Considerations
  - Patient must refrain from taking *pain relievers* (induce the release of salivary amylase [false ↑])
  - Avoid lactescent or *lipemic* samples = falsely ↓ results; triglycerides inhibit amylase
- Methods
  - **Amyloclastic method** - decrease in the intensity of color of starch-iodine complex
  - **Saccharogenic method** – ↑ reducing sugars as starch is hydrolyzed by AMS; classic reference method reported in Somogyi units
  - **Chromogenic method** – ↑ color intensity of soluble dye-substrate solution
  - **Coupled-enzymatic/Continuous-monitoring method** - ↑ Absorbance of NADH at 340 nm; most commonly used
- Reference Values:
  - Serum, 25–130 U/L
  - Urine, 1–15 U/h (has a faster clearance due to its small size)

Lipase Determination

- Specimen Considerations
  - Avoid *hemolysis* (Falsely decrease)
  - Storage: stable for up to 3 weeks at 4°C
- Methods
  - **Cherry-Crandall method**
    - uses *olive oil or triolein* as substrate
    - $TG + H_2O \xleftrightarrow{LPS}$ glycerol + fatty acids
    - Reagent: Triglycerides
    - Fatty acids are measured via titration
  - **Turbidimetric method**
    - measures the rate of clearing as fats are hydrolyzed by LPS
  - **Colorimetric methods**
    - based on coupled reactions with peroxidase or glycerol kinase
- Reference Values: 0-1.0 U/mL (<38 U/L)

MISCELLANEOUS ENZYMES

Gamma Glutamyl Transferase	- Found in liver (canaliculi of hepatic cells and epithelial cells lining biliary ductules), kidneys, pancreas, intestine, and many other tissues. - GGT is not found in skeletal muscle tissue or bone. - Used to identify the source of ↑ ALP - Clinical significance: Hepatobiliary disorders, **alcoholic liver disease**, microsomal induction by drugs and alcohol - Reference ranges: Male, up to 55 U/L; female, up to 38 U/L at 37°C

5'-Nucleotidase	- **Intrahepatic cholestasis**
	- Used to identify the source of ↑ ALP
	  ○ ↑ ALP, ↑ GGT or 5'NT = Liver
	  ○ ↑ ALP, Normal GGT or 5'NT = bone

Acid Phosphatase (ACP)	- Tissue location: Highest concentration in *prostate gland*, with lesser amounts in bone (osteoclasts), liver, spleen, erythrocytes, platelets
	- Clinical significance: Increased in prostate cancer (**prostatic carcinoma/hyperplasia**), benign prostatic hypertrophy, bone disease, Paget disease, breast cancer with bone metastases, Gaucher disease, platelet damage, idiopathic thrombocytopenic purpura
	- ACP is useful in forensic cases involving rape because vaginal washings containing seminal fluid would exhibit ACP activity.
	- Avoid hemolysis when assaying and maintain anaerobic collection
	- Tested using Roy method which uses thymolphthalein monophosphate
	- To determine the level of prostatic ACP, you may perform chemical inhibition using tartrate inhibition (prostatic ACP is inhibited by tartrate) or formaldehyde & copper (inhibit RBC ACP)
	- **Reference ranges:**
	  ○ Total ACP:
	  ▪ Male 2.5-11.7 U/L at 37°C
	  ▪ Female 0.3-9.2 U/L at 37°C
	  ○ Prostatic ACP:
	  ▪ Male 0.2-5.0 U/L at 37°C
	  ▪ Female 0.0-0.8 U/L at 37°C

Glucose-6-phosphate dehydrogenase (G-6-PD)	- Found in erythrocytes, adrenal glands, thymus, lymph nodes, spleen - Important in the Hexose monophosphate shunt - Decreased: Primary importance of G6PD is in cases of deficiency, inherited as a sex-linked trait (X-chromosome). In G6DP deficiency, a **drug-induced hemolytic anemia** occurs when an individual is administered antimalarial drugs or primaquine. Hemolysis may also be caused by infections and after ingestion of fava beans. - Increased: Megaloblastic anemias and AMI - Specimen is RBC or whole blood
Acetylcholinesterase *(AChE; True cholinesterase)*	- Found in red blood cells, lungs, spleen, nerve endings, gray matter of brain

Acylcholine acylhydrolase *(PChE; Pseudocholinesterase)*	- Found in liver, pancreas, heart, white matter of brain, serum
	- Clinically significant when ↓ in serum because it is normally high in plasma
	  ○ **Pesticide** (organophosphate and carbamate) **poisoning**
	  ○ Liver disease
	  ○ Genetic variants (mutant gene) associated with abnormal response to muscle relaxants
	- Reference ranges (PChE serum): Male, 40-78 U/L; female, 33-76 U/L at 37°C

Aldolase	- ↑ in **Skeletal muscle disorders**
Glutamate dehydrogenase **Glutamate-S-transferase**	- **Hepatic disorders**
Angiotensin-converting enzyme	- Blood pressure regulation
Pyruvate kinase	- Hemolytic anemia
Trypsin	- ↑ in Acute pancreatitis
Chymotrypsin/ Elastase	- ↓ in chronic pancreatic insufficiency
Glycogen phosphorylase	- Acute myocardial infarction

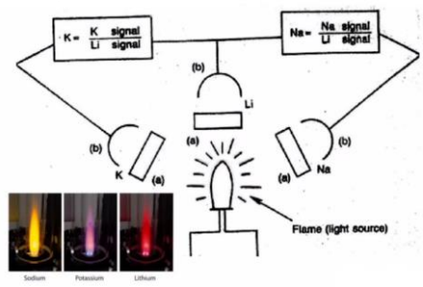
ELECTROLYTES

FUNCTIONS OF ELECTROLYTES

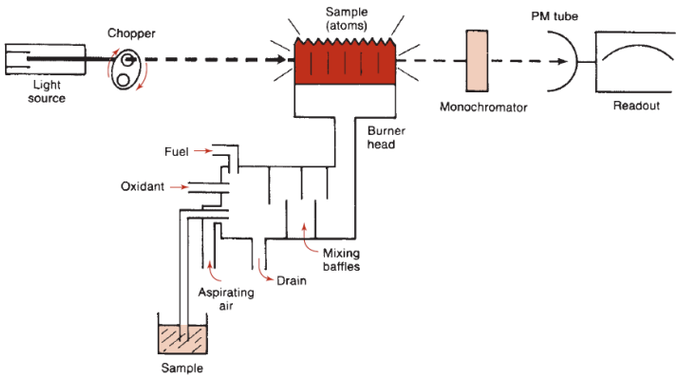
1. Volume and osmotic regulation (Na^+, K^+, Cl^-)
2. Myocardial rhythm and contractility and neuromuscular excitability (K^+, Ca^{2+}, Mg^{2+})
3. Cofactors in enzyme activation (Ca^{2+}, Mg^{2+}, Zn, Cl^-)
4. Regulation of ATPase-ion pumps (Mg^{2+})
5. Blood coagulation (Ca^{2+}, Mg^{2+})
6. Production and use of ATP from glucose (Mg, PO_4)
7. Maintenance of acid-base balance (K^-, Cl^-, HCO_3^-)

INSTRUMENTATION

1. **Flame Photometry/Atomic Emission Spectrophotometry**
   - Measurement of light emitted by atoms following excitation by heat energy (flame)
   - The flame serves as the light source and the cuvette or sample holder
   - Has been replaced by ion selective electrodes
   - Traditionally used to determine the concentration of sodium (yellow), potassium (violet), and Lithium (red)
   - Employs Lithium as internal standard for sodium and potassium assay
     - Cesium is used as internal standard for Lithium assay in TDM of manic depressive patients (lithium is used as treatment for bipolar disorders)



2. **Atomic Absorption Spectrophotometry**
   - Measurement of the amount of light absorbed by ground state atoms
   - When atoms are excited they *emit light*, afterwards, they return to ground state and *absorb light*
   - The photomultiplier must only measure light from the source not absorbed by ground state atoms or analytes
     - The chopper supplies light from the source in pulses (pulsed signal)
     - The system is programmed to detect pulsed signals only in order to distinguish between the light emitted by the hollow-cathode lamp and that emitted by excited atoms in the flame
   - Light source: Hollow cathode lamp
   - Reference method for calcium and magnesium assay and other trace metals



3. **Potentiometry**
   - Measurement of electrical potential (voltage) due to the activity of free ions
     - pH meter
       - Indicator electrode: $Ag/AgCl_2$
       - Reference electrode: Calomel
     - pCO2 electrode (Severinghaus electrode)
     - Ion-selective electrodes (more commonly used for electrolytes)
4. **Amperometry/Polarography**
   - measurement of the current flow produced by an oxidation-reduction reaction
     - pO2 electrode (Clark) - uses a gas-permeable membrane which allows dissolved O_2 to pass through
5. **Coulometry**
   - measurement of the amount of electricity in coulombs at a fixed potential
     - Cotlove chloridometer - used in serum and sweat chloride analysis (for the diagnosis of Cystic fibrosis)

OSMOLALITY AND OSMOLAL GAP

- **Colligative properties** refer to the properties of a solution that are influenced by the number of molecules in solution, but not their individual composition. There are four types of colligative properties: ***boiling point, freezing point, osmotic pressure,*** *and* ***vapor pressure.***
- **Osmolality** is the measure of the number of dissolved particles in solution expressed as osmoles per kilogram of water. Serum osmolality is expressed as milliosmoles/kg (mOsm/kg).
- Osmolality is regulated by the **hypothalamus** through the sensation of thirst and the signaling to secrete *antidiuretic hormone (ADH)*. When the osmolality of the blood is increased, two processes occur:
  - Consuming more water will decrease the osmolality.
  - Posterior pituitary secretion of ADH will cause renal reabsorption of water and decrease the osmolality.
- Methods
  - **Osmometry** (measured osmolality)
    - based on the measurement of a decrease in freezing point or vapor pressure as the number of osmotic particles increase
  - **Calculated osmolality**

$$= 2\,Na + \frac{glucose\ (mg/dL)}{20} + \frac{BUN\ (mg/dL)}{3}$$

$$= 1.86\,Na + \frac{glucose\ (mg/dL)}{18} + \frac{BUN\ (mg/dL)}{2.8} + 9$$

  - **Osmolal gap**
    - Effective screening method in detecting the presence of toxic compounds
    - Formula: Measured osmolality minus calculated osmolality (Osm_m-Osm_c)
- Reference values
  - Serum osmolality: 275-295 mOsm/kg
  - Osmolal gap: 5-10 mOsm/kg
- Clinical significance (OG>12 mOsm/kg)
  - Indicative of the presence other osmotically active particles other than urea or glucose
  - Renal failure (increased NPNs, phosphates, sulfates), Diabetic ketoacidosis (ketone bodies), Lactic acidosis (lactate), and metabolites of toxic compounds such as in salicylate poisoning, alcohol poisoning, and ethylene glycol poisoning
- **Sample problem!** Electrolytes, glucose, BUN, Creatinine, and osmolality examinations have been performed on a patient suspected of ingesting ethanol. What can be done to help reveal this condition without additional test?
  a. Anion gap
  b. Glycemic index
  c. *Osmolal gap*
  d. Serum osmolality

ANION GAP

- Formulas
  - AG = (Na^+ + K^+) – (HCO_3^- + Cl^-)
  - AG = Na^+ - (HCO_3^- + Cl^-)
  - Note! You can calculate for the anion gap to perform quality control for electrolyte determination by ISE instruments
- Reference values:
  - 10-20 mmol/L (if formula used included potassium)
  - 7-16 mmol/L (if formula used did not include potassium)
- Clinical significance:
  - ↑ **Anion gap**
    - ↑ in unmeasured anions (anions not tested in routine panel): in conditions associated with increased osmolal gap (because substances that contribute to osmolality are primarily anions) such as in renal failure, lactic acidosis, diabetic ketoacidosis, intoxication, etc.
    - ↓ in unmeasured cations: hypocalcemia
    - Laboratory errors: falsely ↑ sodium or falsely ↓ chloride or bicarbonate
  - ↓ **Anion gap**
    - ↓ unmeasured anions: hypoproteinemia, hypoalbuminemia (albumin is negatively charged at physiologic pH)
    - ↑ unmeasured cations: hypercalcemia, multiple myeloma (↑ IgG proteins; cathodal and cationic)
    - Laboratory errors (falsely ↓ sodium or falsely ↑ chloride or bicarbonate

IMPORTANT ELECTROLYTES

ELECTROLYTE	ANALYSIS	CLINICAL SIGNIFICANCE

1. SODIUM

- Major extracellular cation (90%)
- Major contributor to *plasma osmolality*
- Renal threshold: ***110-130mmol/L (120 mmol/L)***

Methods

- **ISE** (most commonly used method); the tip of the electrode is made of *glass* which makes it selective to the ion of interest
  - Direct ISE: no sample dilution
  - Indirect ISE: requires sample dilution
    - prone to pseudohyponatremia when plasma contains increased proteins or lipids
- Flame photometry (emit yellow light)
- ***Albanese-Lein*** (traditional colorimetric reaction; positive = yellow)

Reference values

Normal	136-145 mmol/L	
Panic	≤120 mmol/L	≥160 mmol/L

Hyponatremia

- ↑ **Na^+ loss** *(depletional hyponatremia):* Addison's disease (hypoadrenalism), hypoaldosteronism, potassium deficiency, salt-losing nephropathy, ketonuria (DKA), prolonged vomiting or diarrhea, severe burns, diuretics.
- ↑ **water retention** *(dilutional hyponatremia; relative decrease):* renal failure, CHF, nephrotic syndrome, hepatic cirrhosis
- **Water imbalance:** increased water intake, SIADH, pseudohyponatremia

Hypernatremia

- ↑ **Na^+ intake or retention:** $NaHCO_3$ (sodium bicarbonate) excess, hyperaldosteronism, dialysis fluid excess
- ↑ **water loss** *(concentrated; relative increase)***:** dehydration, profuse sweating, prolonged diarrhea, Diabetes insipidus, renal tubular disorder, severe burns
- ↓ **water intake**: older persons, infants, mental impairment

2. POTASSIUM

- Major intracellular cation
- 20x higher inside the cells
- Electrolyte which is most greatly affected by hemolysis (K^+ > Mg^{2+} > PO_4^{2-})
- Plasma potassium abnormalities require immediate treatment

Methods

- ISE (tip of the electrode contains *valinomycin*)
- Flame photometry (emit violet light)
- ***Lockhead-Purcell*** (traditional colorimetric reaction; positive = blue)

Reference values

Normal	3.5-5.1 mmol/L (serum – slightly higher due to platelet release of potassium during clotting)	
	3.5-4.5 mmol/L (plasma)	
Panic	≤2.8 mmol/L	≥6.2 mmol/L

Hypokalemia

- **Cellular shift (influx/intracellular shift)**: alkalosis (↑pH = ↓H^+ = ↓K), insulin overdose
- **GI loss**: vomiting, diarrhea, gastric suction, intestinal tumor, laxatives, malabsorption, cancer therapy
- **Renal loss**: nephritis, renal tubular acidosis, hyperaldosteronism, thiazide diuretics, Cushing's syndrome, Hypomagnesemia, Acute leukemia
- ↓ **Intake**

Hyperkalemia

- **Cellular shift (efflux/extracellular shift)**: acidosis (0.1 ↓pH = ↑H^+ = 0.2-1.7 ↑K), chemotherapy, muscle injury, leukemia, hemolysis
- ↑ **intake**: oral or IV potassium replacement
- ↓ **renal excretion**: renal failure (acute or chronic), hypoaldosteronism (aldosterone promotes excretion of K^+), Addison's disease, K^+-sparing diuretics
- **Artifactual**: hemolysis, venous stasis through prolonged tourniquet application or excessive fist clenching, thrombocytosis (plasma should be used not serum)

3. CHLORIDE

- Major extracellular anion
- Passively follows Na^+
- Counter ion of sodium
- Inverse relationship with bicarbonate (chloride shift: in order to maintain electric neutrality)

Methods

- ***ISE*** ($Ag/AgCl_2$ or Ion exchange membrane)
- ***Cotlove chloridometry*** (amperometric-coulometric; titration is coulometric while detection is amperometry)
- ***Schales-Schales method*:** mercurimetric titration method which uses diphenylcarbazone as indicator and the end product is blue violet
- ***Gibson and Cooke's pilocarpine iontophoresis*** (Gold standard for sweat chloride analysis)

Reference values

Normal	98-107 mmol/L (serum)	
Panic	≤80 mmol/L	≥ 120 mmol/L
Sweat chloride	<40 mmol/L	

Hypochloremia

- **Increased chloride loss:** aldosterone deficiency, salt-losing nephropathy, DKA, prolonged vomiting
- **Bicarbonate excess:** metabolic alkalosis, compensated respiratory acidosis

Hyperchloremia

- **Increased bicarbonate loss:** GI loss of HCO_3^-, renal tubular acidosis
- **Bicarbonate depletion:** Metabolic acidosis
- Cystic fibrosis is characterized by sweat chloride which is ***≥60 mmol/L***

4. MAGNESIUM

- 2nd most abundant intracellular cation
- 10x higher in RBCs
- Essential cofactor of >300 enzymes

Distribution

- 55% Ionized (physiologically active form)
- 30% bound to proteins; 15% bound to ions

Methods

- ***AAS*** (Gold standard for divalent ions)
- ***Dye-binding*** (more routinely performed primarily using Calmagite producing red-violet complex whose intensity is measured spectrophotometrically)
- ***Dye-lake method***: uses Titan/Clayton/Thiazole yellow; positive result is red-lake

Reference values

Normal	1.26-2.1 mEq/L (1.51-2.52 mg/dL)	
Panic	≤1 mg/dL	≥ 4.7mg/dL

Hypomagnesemia

- ↓ **intake**: poor diet, starvation, prolonged Mg-deficient IV therapy, chronic alcoholism
- ↓ **absorption**: malabsorption syndrome, intestinal resection, nasogastric suction, prolonged vomiting, diarrhea, laxatives
- ↑ **excretion**: renal, endocrine, drug-induced

Hypermagnesemia

- ↑ **intake**: antacids (ex. milk of magnesia), enemas, cathartics
- ↓ **excretion**: renal failure (primarily), hypoaldosteronism

5. CALCIUM

- Plasma levels regulated by **parathyroid hormone** (parathyroid gland), **vitamin D** (1,25-dihydroxy vitamin D) synthesized in the skin but activated in the kidneys, **Calcitonin** (thyroid)
- PTH and Calcium increase calcium by increasing intestinal absorption, renal reabsorption, and bone resorption
- Calcitonin inhibits the action of PTH and lowers calcium

Distribution

- 50% Ionized (physiologically active form)
- 40% bound to proteins; 10% bound to ions

Methods

- ***AAS*** (gold standard)
- ***ISE*** (for ionized calcium; requires anaerobic collection)
- ***Dye-binding*** (dyes used are Cresolphthalein complexone [CPC] or arsenazo dye)

Reference values

Normal	4.3-5 mEq/L (8.6-10 mg/dL)	
Panic	≤ 6 mg/dL	≥ 13 mg/dL
Conversion factor	0.25 (mg/dL → mmol/L) 0.5 (mEq/L → mmol/L)	

Hypocalcemia

- ↓ **calcium, ↓ PTH:** primary hypoparathyroidism
- ↓ **calcium but normal or ↑ PTH:** secondary hyperparathyroidism, chronic renal failure and renal disease, ↓ Magnesium, pseudohypoparathyroidism, ↓ vitamin D, pancreatitis, hypoalbuminemia

Hypercalcemia

- ↑ **Calcium and ↑ PTH:** primary hyperparathyroidism
- ↑ **Calcium but PTH is normal or ↓:** hypercalcemia of malignancy, multiple myeloma, increased vitamin D (hypervitaminosis D), prolonged immobilization characterized by bone wasting

6. PHOSPHATE

- 85% in bones, 15% in soft tissues, <1% in serum or plasma
- Major intracellular anion
- Component of several essential biomolecules (ATP, ADP, 2'3 BPG, DNA, RNA)
- PTH has opposite effects on calcium and phosphate (increase renal reabsorption of calcium and renal excretion of phosphate)

Distribution

- 55% Ionized (physiologically active form)
- 10% bound to proteins; 35% bound to ions

Method

- Usually measured as inorganic phosphorus
- ***Fiske-Subbarow*** (phosphorus combines with ammonium molybdate to produce ammonium phosphomolybdate complex; reduced by ascorbic acid to phosphomolybdenum blue for C/S [600-700nm])
- UV: increased absorbance at 340nm

Reference values

Normal (P_i)	2.2-4.4 mg/dL	
Panic	≤1 mg/dL	≥8.9 mg/dL

Hypophosphatemia

- Diabetic ketoacidosis, long-term total parenteral nutrition (TPN), Inflammatory bowel disease, anorexia nervosa, alcoholism
- Hyperparathyroidism (calcium is ↑ but phosphate is ↓), vitamin D deficiency (Rickets or Osteomalacia)

Hyperphosphatemia

- Renal failure (increased retention), milk or laxatives (increased intake)
- Conditions characterized by cell destruction which release intracellular ions: neoplastic disorders, intravascular hemolysis, lymphoblastic leukemia

BLOOD GASES

ACID-BASE HOMEOSTASIS

BUFFER SYSTEMS:

1. **Hemoglobin**: second most important blood buffer (most important intracellular buffer), contains 38 histidine residues that bind H^+
2. **Plasma proteins**: have free carboxyl and amino groups that are able to bind H^+
3. The **phosphate buffer system** uses HPO_4^{2-} (hydrogen phosphate) and $H_2PO_4^-$ (Dihydrogen phosphate) to minimize pH changes in plasma and erythrocytes.
4. **Bicarbonate/Carbonic Acid**
   - Major blood buffer system or Most important blood buffer system
   - In blood CO_2 binds with water to form carbonic acid which may dissociate back into CO_2 and water or form bicarbonate and H^+ which is a reaction catalyzed by carbonic anhydrase
   - Normal ratio = 20 bicarbonate:1 carbonic acid

$$CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow HCO_3^- + H^+$$

Henderson-Hasselbalch equation

$$pH = pKa\ (6.1) + log\frac{HCO_3^-}{H_2CO_3\ (or\ pCO_2\ x\ 0.03)}$$

- When the ratio is 20:1, the pH is = 7.4
- The numerator (HCO_3^-) denotes metabolic function of the kidneys
- The denominator (pCO_2, which represents H_2CO_3) denotes respiratory function of the lungs
- pH may be ↑ by ↑ bicarbonate or ↓ pCO_2 level
- pH may be ↓ by ↓ bicarbonate or ↑ pCO_2 level
- pH is directly proportional to bicarbonate and inversely proportional to pCO_2 or carbonic acid (ROME: Respiratory opposite, metabolic equal)

DEFINITION OF TERMS

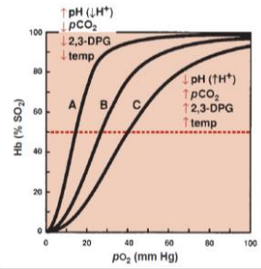
1. **Base/Alkali:** chemical that gives off hydroxyl ions in a solution
2. **Buffer:** consists of a weak acid and a salt of its conjugate base, and it resists the change in pH upon adding acid or base
3. **Respiration:** Process to supply cells with oxygen for metabolic processes and remove the carbon dioxide produced during metabolism
4. **Partial pressure:** In a mixture of gases, partial pressure is the amount of pressure contributed by each gas to the total pressure exerted by the mixture.
5. **Acidemia** occurs when arterial blood pH <7.35.
6. **Alkalemia** occurs when arterial blood pH >7.45.
7. **Hypercapnia** is increased blood PCO_2
8. **Hypocapnia** is decreased blood PCO_2
9. **Partial pressure of carbon dioxide (PCO_2):** Measured in blood as mm Hg
10. **Concentration of dissolved carbon dioxide ($cdCO_2$):** Includes undissociated carbonic acid (H_2CO_3) and carbon dioxide dissolved in blood **(represented by *PCO_2*)**
11. **Concentration of total carbon dioxide ($ctCO_2$):** Includes bicarbonate (primary component), carbamino-bound CO_2, carbonic acid, and dissolved carbon dioxide

ARTERIAL BLOOD GAS PARAMETERS

PARAMETER	METHOD/DERIVATION	REFERENCE RANGE	PANIC VALUES
pH	pH electrode (potentiometry)	7.35-7.45	≤7.2; ≥7.6
pCO_2	pCO_2 electrode (potentiometry)	35-45 mm Hg	≤20; ≥ 60
pO_2	pO_2 electrode (amperometry)	80-110 mm Hg	≤40
HCO_3^-	Total CO_2 – 1	22-26 mmol/L	≤10; ≥40
Total CO_2	HCO_3^- + H_2CO_3 + CO_2 (HCO_3^- + 1)	23-27 mmol/L	
Base excess	Calculated	-2 to 2	
O_2 saturation	Co-oximetry/Differential spectrophotometry	95-100%	

FACTORS THAT AFFECT ABG VALUES

1. **Temperature**
   - For every 1°C increase: pCO_2 ↑ by 3%, pO_2 ↓ by 7%, pH ↓ by 0.015
2. **Specimen exposure to air**
   - Presence of bubbles, tube not tightly sealed
   - Atmospheric air contains high amount of oxygen
   - ↑ pO_2, ↓ pCO_2, ↑ pH
3. **Prolonged storage of specimen**
   - Bacterial contamination
   - Continued metabolism causing depletion of oxygen, CO_2 is a byproduct of metabolism
   - ↓ pO_2, ↑ pCO_2, ↓ pH
4. **Excess anticoagulant (heparin)**
   - ↓ pH (because heparin is an acidic mucopolysaccharide)



- Shift to the right = CADET face right! Caron dioxide, Acidosis (pH), 2'3 DPG, exercise, temperature
- Low pH means high hydrogen ion concentration

ACID-BASE DISORDERS

Condition	Major cause/s	pH	HCO_3^- *(metabolic)*	pCO_2 *(respiratory)*	Compensation
Metabolic acidosis	DKA, Lactic acidosis (hypoxic states)	↓	↓	N/↓	Hyperventilation
Metabolic alkalosis	Vomiting, sodium bicarbonate or corticosteroid overdose, hypokalemia, GI suction	↑	↑	N/↑	Hypoventilation
Respiratory acidosis	Chronic Obstructive Pulmonary Disorders (hypoventilation), acute airway obstruction, circulatory failure	↓	N/↑	↑	↑ Reabsorption of bicarbonate
Respiratory alkalosis	Anxiety, acute pain (hyperventilation), pulmonary embolism, pulmonary edema, certain drugs used in CNS disorders	↑	N/↓	↓	↑ Excretion of bicarbonate or decreasing its reabsorption

CAUSES

- **Metabolic acidosis** may be caused by organic acid production or when ingestion exceeds the excretion rate. Disorders include:
  - Diabetic ketoacidosis due to the production of acetoacetic acid and β-hydroxybutyric acid;
  - Lactic acidosis due to the production of lactic acid;
  - Poisonings such as salicylate, ethylene glycol, and methyl alcohol;
  - Reduced acid excretion due to renal failure or tubular acidosis; and
  - Loss of bicarbonate due to diarrhea or excessive renal excretion.
- **Metabolic alkalosis** may be caused by ingestion of excess base, decreased elimination of base, or loss of acidic fluids. Disorders include:
  - ingestion of excess alkali (antacids);
  - intravenous administration of bicarbonate;
  - renal bicarbonate retention;
  - prolonged diuretic use;
  - loss of hydrochloric acid from the stomach after vomiting, intestinal obstruction, or gastric suction;
  - glucocorticoid excess as in Cushing syndrome; and
  - mineralocorticoid excess as in hyperaldosteronism.
- **Respiratory acidosis** may be caused by:
  - chronic obstructive pulmonary disease, such as chronic bronchitis and emphysema
  - ingestion of narcotics and barbiturates, and
  - severe infections of the central nervous system such as meningitis.
- **Respiratory alkalosis** may be caused by:
  - hypoxia, anxiety, nervousness, excessive crying, pulmonary embolism, pneumonia, congestive heart failure, salicylate overdose, and so on.

SAMPLE ABG PROBLEMS!

	1	2	3	4	6
pH [7.35-7.45]	7.3 *(acidotic)*	7.29 *(acidotic)*	7.48 *(alkalotic)*	7.44 *(normal)*	7.66 *(alkalotic)*
HCO_3^- *(metabolic function)* [22-26]	19 *(acidotic)*	30 *(alkalotic)*	29 *(alkalotic)*	18 *(acidotic)*	22 *(normal)*
pCO2 *(respiratory function)* [35-45]	38 *(normal)*	52 *(acidotic)*	42 *(normal)*	30 *(alkalotic)*	20 *(alkalotic)*
Interpretation	Uncompensated metabolic acidosis	Partially compensated respiratory acidosis	Uncompensated metabolic alkalosis	Fully compensated respiratory alkalosis	Uncompensated respiratory alkalosis

OXYGEN METABOLISM

1. **Oxygen** is **transported bound to hemoglobin** present in red blood cells and in a **physically dissolved** state.
   - Three factors control oxygen transport: the PO_2, free diffusion of oxygen across the alveolar membrane, and affinity of hemoglobin for oxygen.
   - Release of oxygen to the tissues is **facilitated by** an increase in H^+ concentration and PCO_2 levels at the tissue level.
   - Under normal circumstances, the saturation of hemoglobin with oxygen is 95%. When the PO_2 is >110 mm Hg, greater than 98% of hemoglobin binds to oxygen.
   - When a person's oxygen saturation falls below 95%, either the individual is not getting enough oxygen or does not have enough functional hemoglobin available to transport the oxygen.
   - The amount of functional hemoglobin available in the blood can be altered due to decreased red blood cells or presence of nonfunctional hemoglobin (e.g., carboxyhemoglobin or cyanmethemoglobin).
2. **Clinical significance of *PO_2* levels in blood**
   - Increased values (>95%) are observed with supplemental oxygen.
   - **Hypoxemia:** Causes include decreased pulmonary diffusion, decreased alveolar spaces due to resection or compression, and poor ventilation/perfusion (due to obstructed airways—asthma, bronchitis, emphysema, foreign body, secretions)

MICRONUTRIENTS

IRON

- Component of heme-containing substances such as hemoglobin and myoglobin
- Transport protein: *transferrin*; storage form: *ferritin/hemosiderin*
- Clinical Significance
  - **Iron deficiency:** Anemia due to blood loss, decreased intake, decreased release from ferritin (anemia of chronic inflammation)
  - **Iron overload:** Hemochromatosis (primary/genetic or acquired or metabolic)

Iron Determination

- **Serum Iron**
  - Specimen considerations:
    - Diurnal variation: affected by circadian rhythm hence early morning collection is recommended because there is significant (30%) decrease of iron in the afternoon
    - Heparinized plasma or serum
    - Deionized water should be used (falsely ↑)
  - Procedure:
    - **Acidification**: to allow the release of ferric iron from transferrin
    - **Reduction**: ferric is reduced to ferrous (most common reducing agent used is ascorbic acid)
    - **Colorimetry**: ferrous form of iron reacts with ferrozine or bathophenanthroline and spectrophotometric determination is then carried out to measure the intensity of the blue color
  - ***Reference range = 50-160 ug/dL***
- **Total Iron-Binding Capacity**
  - Amount of iron that could be bound by saturating transferrin and other minor iron-binding proteins present in the sample
  - Measure unbound iron binding capacity (UIBC) + serum iron
  - Tf (mg/dL) = TIBC (ug/dL) x 0.7
  - ***Reference range = 250 to 350 ug/dL***
- **Percent (%) Saturation or Transferrin saturation**
  - An index of iron storage and determined using the following formula
  - Serum iron divided by TIBC multiplied by 100

$$\%\ saturation = \frac{Serum\ iron}{TIBC}\ x\ 100$$

  - ***Reference range = 20-50%***

VITAMIN DEFICIENCIES

		VITAMIN	CLINICAL MANIFESTATIONS OF DEFICIENCY
Fat-soluble	A	Retinol	- Night blindness, Growth retardation, dermatitis
	D	Cholecalciferol	- Rickets, osteomalacia
	E	Tocopherol	- Mild hemolytic anemia - Red blood cell fragility, ataxia
	K	Phylloquinone/ Menaquinone	- Hemorrhage or bruising - Post traumatic bleeding - Hemorrhagic disease of infancy (prolonged CT)
Water-soluble	C	Ascorbic acid	- Scurvy - Vague aches and pains
	B_1	Thiamine	- Beriberi - Wernicke-Korsakoff syndrome
	B_2	Riboflavin	- Angular stomatitis - Dermatitis, photophobia, neurologic changes
	B_3	Niacin/Nicotinic acid	- Pellagra - Dermatitis, mucus membrane inflammation
	B_5	Pantothenic acid	- Depressed immune system - Paresthesia
	B_6	Pyridoxine	- Facial seborrhea, peripheral neuropathy - Irritability, seizures, anemia, vomiting
	B_7	Biotin	- Dermatitis
	B_9	Folic acid/ Pteroylglutamic acid	- Megaloblastic anemia, neural tube defect
	B_{12}	Cyanocobalamin	- Megaloblastic anemia with neurologic involvement

CLINICAL ENDOCRINOLOGY

HORMONES

- Chemical messengers synthesized and secreted by endocrine glands, organs, or isolated cells that have specific regulatory effects on the activity of target cells
  - **Endocrine hormones**: produced by ductless glands; secretions reach the target cells via the circulation (distant target sites)
- Generally produced at one site in the body and exert their action(s) at distant sites, except:
  - **Paracrine hormones**: exert their action locally; affect adjacent target cells
  - **Autocrine hormones:** exert their action on their cell of origin regulating their own synthesis and secretion; self-regulatory molecule: target is their own self to stimulate their own growth, differentiation and activation (ex. Lymphokines, cytokines/interleukins, growth factors)

FUNCTIONAL CLASSES OF HORMONES

1. **Releasing hormones** – stimulate the release of anterior pituitary hormones (ex. GnRH, TRH, CRH, GHRH)
2. **Inhibitory hormones** - inhibit the release of certain anterior pituitary hormones (ex. somatostatin, dopamine)
3. **Tropic hormones** – secreted by the anterior pituitary and stimulate the activity of other endocrine glands (produced by an endocrine organ and has another endocrine organ as target)
   - Different from the term Trophic: promote hypertrophy
   - FLAT:
     - Gonadotropins: FSH & LH
     - Corticotropin: ACTH
     - Thyrotropin: TSH
4. **Effector hormones** – secreted by all endocrine organs other than the hypothalamus (does not target another endocrine gland)

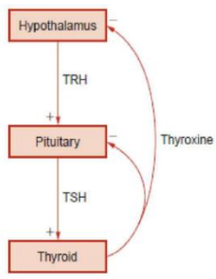
STRUCTURAL CLASSES OF HORMONES		
PROTEINS/PEPTIDES	**STEROIDS**	**AMINES**
- Water-soluble - No carrier protein - Has short half-life (10-30 minutes) - Exert their effect by binding to cell surface receptors on the surface of target cells - **Glycoproteins** such as: o Dimeric: *FSH, LH, TSH, hCG* o Alpha sub-unit is the same for all four but the beta subunit is unique - **Peptide hormones** such as: *GH, prolactin, ADH, oxytocin, calcitonin, PTH, insulin, glucagon, releasing hormones, Inhibiting hormones* o Most are produced as prohormones which has to undergo post-translational post-modification via cleavage of a part	- Synthesized from: cholesterol - Lipid-soluble - Water-insoluble/ Hydrophobic - Require a carrier protein (ex. Cortisol binding globulin, Sex hormone binding globulin, and albumin) - Bind to intracellular receptors - Has a half-life of 30-90 minutes - *Cortisol, aldosterone, estrogens, progesterone, testosterone*	- Synthesized from: tyrosine - **Thyroid hormones** (*T_3 and T_4*) require carrier protein (ex. Thyroxine-binding globulin, Thyroxine-binding pre-albumin, and albumin) - Half-life: 7-10 days - **Catecholamines** (*epinephrine, norepinephrine*) do not require a carrier protein; HF: <1 minute

***Fatty acid derivatives** include the prostaglandins; derived from arachidonic acid

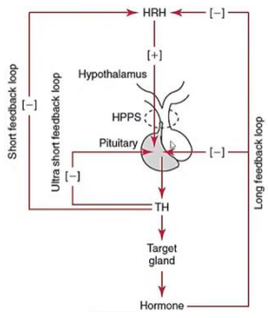
THE HYPOTHALAMIC-PITUITARY-TARGET ORGAN (HPT) AXIS

Feedback Loop

- **Positive feedback**
  - begins when the hypothalamus receives input to produce a releasing factor that acts on the pituitary gland which in turn releases tropic hormones that act on a specific target gland to promote hormone synthesis and release
    - Releasing hormones from the hypothalamus
    - Tropic hormones from the pituitary
    - Target organ hormones



- **Negative feedback**
  - occurs when elevated levels of a hormone turn off the secretion of a releasing hormone from the hypothalamus and a stimulating hormone from the pituitary gland
    - **Long feedback loop** - negative feedback of the target cell hormone at the pituitary gland and hypothalamus
    - **Short feedback loop** – involves an anterior pituitary tropic hormone feeding back at the hypothalamus
    - **Ultra-short feedback loop** - involves an anterior pituitary hormone feeding back at the anterior pituitary (self-regulation)

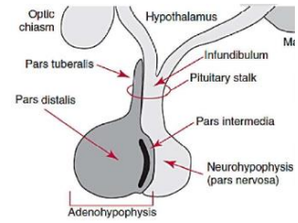


Disorders of Hormone Secretion

- Either hypersecretion or hyposecretion
- **Primary disorders** refer to a defect in the target organ
- **Secondary disorders** refer to a defect in the anterior pituitary gland (Adenoma or panhypopituitarism)
- **Tertiary disorders** refer to a defect in the hypothalamus (rare)

HYPOTHALAMUS

- Consists of several nuclei including clusters of neurosecretory cells that are responsible for producing the releasing and inhibiting hormones
- Hypothalamus and pituitary gland is connected by the pituitary stalk
- **Adenohypophysis** (anterior pituitary)
  - populated by granular or hormone producing cells
- **Neurohypophysis** (posterior pituitary)
  - has no hormone producing cells
  - axons of neurosecretory cells from the hypothalamus terminate here
  - the only significant role is as a repository of the hormones produced by the neurosecretory cells of the hypothalamus (does not produce its own hormones, but only stores and secretes)



HORMONE	ACTION
Gonadotropin-releasing hormone	Stimulates release of FSH and LH
Corticotropin-releasing hormone	Stimulates release of ACTH
Thyrotropin-releasing hormone	Stimulates release of TSH
Growth-hormone releasing hormone	Stimulates release of GH
Somatostatin	Inhibits release of GH, TSH
Dopamine *(PIF – Prolactin Inhibiting Factor)*	Inhibits release of Prolactin

PITUITARY GLAND/HYPOPHYSIS

- Previously referred to as the master gland
- Connected to the hypothalamus by the pituitary stalk/infundibulum
- Endocrine systems that operate independently of the anterior pituitary gland: [1] Renin-Angiotensin-Aldosterone System (RAAS), [2] Calcium-Parathyroid axis (Ca-PTH), and [3] Glucose-Insulin axis

ANTERIOR PITUITARY GLAND (ADENOHYPOPHYSIS)

- *5 distinct populations of glandular cells: somatotrophs (secrete GH), lactotrophs (secrete PRL), thyrotrophs (secrete TSH), gonadotrophs (secrete FSH and LH), and corticotrophs (secrete ACTH)*

	HORMONE	TARGET TISSUE	ACTION	CLINICAL SIGNIFICANCE
Tropic hormones	FSH	Gonads (testes and ovaries)	Follicle development; secretion of estrogen; sperm production	• Secondary hypergonadism or hypogonadism
	LH		Stimulates ovulation and secretion of progesterone/ testosterone	
	ACTH	Adrenal cortex	Release of cortisol	• Secondary hypercortisolism (Cushing's disease) • Cushing syndrome = primary hypercortisolism • No significant hyposecretory state
	TSH	Thyroid	Release of thyroid hormones	• Secondary hyperthyroidism or hypothyroidism
Effector hormones	PRL	Breasts	Production of milk	• Prolactinoma (pituitary adenoma) leading to hyperprolactinemia • Galactorrhea = inappropriate production of milk • Females: amenorrhea, anovulation, infertility, galactorrhea • Males: impotence, galactorrhea
	GH	Whole body	Growth of muscles and long bones; stimulates release of Insulin-like Growth Factor 1 (IGF-1)	• Pituitary giantism = Hypersecretion of GH in children • Acromegaly = hypersecretion of GH in adults *(Suppression test = Oral glucose loading; normally hyperglycemia suppresses GH)* • Dwarfism = hyposecretion of GH *(Insulin stimulation test = hypoglycemia stimulates GH)* • Panhypopituitarism

TABLE 20-3 OTHER MODIFIERS OF GROWTH HORMONE SECRETION	
STIMULATE GROWTH HORMONE SECRETION	**INHIBIT GROWTH HORMONE SECRETION**
Sleep	Glucose loading
Exercise	β-Agonists (e.g., epinephrine)
Physiologic stress	α-Blockers (e.g., phentolamine)
Amino acids (e.g., arginine)	Emotional/psychogenic stress
Hypoglycemia	Nutritional deficiencies
Sex steroids (e.g., estradiol)	Insulin deficiency
α-Agonists (e.g., norepinephrine)	Thyroxine deficiency
β-Blockers (e.g., propranolol)	

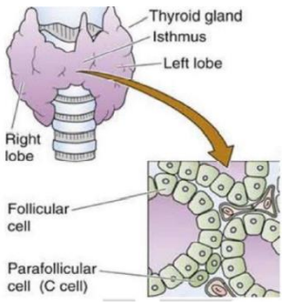
POSTERIOR PITUITARY GLAND (NEUROHYPOPHYSIS)

- *Pars nervosa*
- *Serves as repository of hormones produced by the supraoptic and paraventricular nuclei of the hypothalamus and releases them on demand*

HORMONE	TARGET TISSUE	ACTION	CLINICAL SIGNIFICANCE
ADH/AVP *(Vasopressin)*	Renal tubules and arterioles	Increases water reabsorption and BP	↑ = **Syndrome of Inappropriate Antidiuretic Hormone Secretion** (↓ *plasma osmolality & plasma sodium, ↑ SG & Urine osmolality)* ↓ = **Diabetes insipidus** *(↑ plasma osmolality & plasma sodium, ↓ SG & Urine osmolality)*
Oxytocin	Uterus and breasts	Uterine contraction, ejection of milk	None

THYROID GLAND

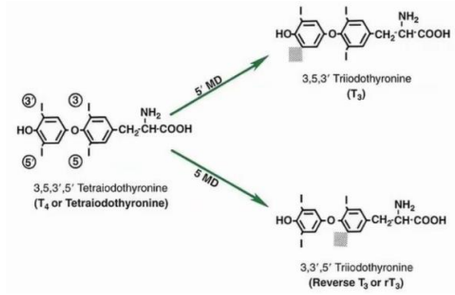
- The thyroid gland is a small organ consisting of right and left lobes bridged by a band of thyroid tissue called isthmus
- Main cell types:
  - **Follicular cells** produce *T_4 and T_3* which are stored in the colloid of the follicles
  - **Parafollicular or C cells** produce *calcitonin* which is important in calcium homeostasis



- Transport Proteins:
  - 100% of circulating T_4 and 20% of T_3 is of thyroidal origin (thyroid follicles)
  - 80% of T_3 is produced from 5'monodeionization of T_4 in non-thyroidal tissues.
  - 70% of T_4 is bound to TBG; 20% to TBPA/Transthyretin; and 10% to Albumin
  - Biologically active forms: FT_4 (0.03%) and FT_3 (0.3%)
- Regulation/Control:
  - TRH from the hypothalamus enters the HPPS to release TSH from anterior pituitary thyrotrophs.
  - TSH stimulates the release of T_4 and T_3 from the thyroid gland. More than 99% of T_4 and T_3 is bound to various thyroid hormone-binding proteins.
  - T_4 and T_3 negatively feedback on the hypothalamus and, more powerfully, the pituitary. They enter target tissues, where T_4 is converted to T_3 and then binds to the thyroid hormone receptor.

THYROID HORMONES

HORMONE	FUNCTION	CLINICAL SIGNIFICANCE
Thyroxine (T_4) – *3,5,3',5'-tetraiodothyronine*	Regulation of oxygen consumption, heat production, growth, sexual maturity, metabolism	• **Primary hyperthyroidism** (Graves' disease and T_3 thyrotoxicosis) • Graves' disease: TSH/Thyrotropin receptor antibodies which mimic TSH • **Primary hypothyroidism** (Hashimoto's thyroiditis, Congenital hypothyroidism, and Myxedema)
3, 5, 3' – Triiodothyronine (T_3)		
Reverse T3 *(3, 3', 5' – Triiodothyronine)*	Has no biological function because it is an abnormal metabolite of T_4	• Non-thyroidal illness (Euthyroid sick syndrome)
Calcitonin	Regulation of calcium level (↓ plasma calcium level; counters the action of PTH and Vitamin D)	• Tumor marker of Medullary thyroid carcinoma



LABORATORY EVALUATION

TEST	NORMAL VALUES	ADDITIONAL NOTES
TSH	0.5-5.0 mIU/L	- Most important test of thyroid function. - 1st test for screening - If normal, no further testing is needed.
T_4	F: 0.9-2.3 ng/dL T: 5.5-12.5 ug/dL	- 2nd step in screening if TSH is abnormal. - **FT_4** (more useful, reliable, and less subject to interferences) is the biologically active form of T_4. - Requires separation of the hormone from its carrier protein using *ultrafiltration or equilibrium dialysis* (to measure the free fraction only) - Reference method: equilibrium dialysis
T_3 UPTAKE (T_3U)	25-38%	- Reflects the quantity of TBG present and the quantity of hormone attached to it - Residual radioactivity representing the % of labeled T_3 left unbound by TBG in serum - Usual test for TBG activity (inversely proportional) - ↓ in ***pregnancy and estrogen replacement*** - ↑ in ***nephrotic syndrome***
FTI *(Free Thyroxine Index)*	5.4-9.7	- FT_4I or T7 - Relative estimate of free T_4 concentration - Product of TT4 and THBR. - Formula: Patient's T_3U divided by T_3U of the reference population multiplied by total T_4
T_3	F: 210-440 pg/dL T: 80-200 ng/dL	- FT_3 is the biologically active form of T_3. - Usually not helpful in the diagnosis of hypothyroidism because it is the last test to become abnormal. - Usually only tested when TSH is low & FT_4 is normal: ***T_3 Thyrotoxicosis***
RAIU	5-30% at 24 h	- Measures thyroidal uptake of radioactive iodine. - Radioactivity is determined by external counting at intervals after an oral dose of iodine 131 (^{131}I) - Useful in assessing the ***metabolic activity of the thyroid*** and in the evaluation and treatment of ***thyroid cancer*** (cold nodules).
rT_3	38-44 ng/dL	- Major metabolite of thyroxine produced by 5-monodeiodination of T_4 - High in patients with ***non-thyroidal illness***
Anti-TPO	Negative	- Detected as thyroid microsomal antibodies using agglutination or immunofluorescence - Very sensitive marker for ***Hashimoto's thyroiditis***
TRAb/ TSHRAb (TSI or LATS)	Negative	- Thyroid-stimulating antibodies that mimic the action of TSH - Positive in 85% of patients with ***Graves' disease***

INTERPRETATION OF THYROID SCREENING TESTS

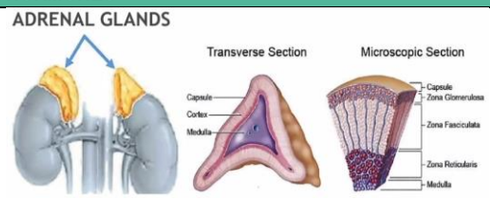
	Low Free T_4	Normal Free T_4	High Free T_4
Low TSH	Secondary hypothyroidism	Subclinical hyperthyroidism	Primary hyperthyroidism
Normal TSH	Non thyroidal illness	Euthyroid	Artifactual hyperthyroxinemia
High TSH	Primary hypothyroidism	Subclinical hypothyroidism	Secondary hyperthyroidism

LABORATORY FINDINGS IN THYROID DISORDERS

DISORDER	Total T_4/T_3	Free T_4	T_3U	TSH
Primary hyperthyroidism	↑			↓
Primary hypothyroidism	↓			↑
Secondary hyperthyroidism	↑			
Secondary hypothyroidism	↓			
Euthyroid with abnormally high TBG *(Pregnancy or estrogen replacement)*	↑	N	↓	N
Euthyroid with abnormally low TBG *(Nephrotic syndrome)*	↓	N	↑	N

- Total T4/T3: Directly proportional to TBG
- T3U: Inversely proportional to TBG

ADRENAL HORMONES



- Medulla: produces catecholamines
- Cortex: produces steroid hormones

HORMONE	SOURCE	TARGET SITE	MAJOR FUNCTIONS
Aldosterone *(mineralocorticoid)*	Zona glomerulosa	Kidneys *Independent of the pituitary gland	Blood pressure and electrolyte regulation (under the control of RAAS)
Cortisol *(glucocorticoid)*	Zona fasciculata	General (whole body)	Carbohydrate, protein, and lipid (glucocorticoid) metabolism
Sex steroids (DHEAS, Androgens)	Zona reticularis	Gonads, bones, and skeletal muscles	Precursor to active androgens Growth and maturation
Epinephrine "fight or flight hormones"	Medullary region	Sympathetic nervous system, liver, muscle, adipose	Stimulates sympathetic nerves Hyperglycemic hormone Promotes glycogenolysis (in the liver and muscles) and lipolysis (in adipose tissues)
Norepinephrine		Sympathetic nervous system	Stimulates sympathetic nerves

CLINICAL SIGNIFICANCE OF ADRENAL HYPERSECRETION AND HYPOSECRETION	
CLINICAL SIGNIFICANCE	**LABORATORY TESTS**
Primary aldosteronism *(Conn's syndrome)*	- Urinary potassium excretion - Captopril suppression (confirmatory test) - Upright PA/PRA ratio - 18-Hydroxycorticosterone (OHCS)

Cushing's syndrome *(Primary hypercortisolism)*
 	
Cushing's disease *(Secondary hypercortisolism)*
	- Urine free cortisol *collected over 24 hours* (on 2 occasions to be considered positive)
	- Plasma and serum collected between 8am and 4pm due to diurnal variation/circadian rhythm
	- Midnight salivary cortisol
	- Dexamethasone suppression test (1 mg overnight or 2 mg over 48 hours)
	  ○ Primary = not suppressed
	  ○ Secondary = suppressed
	- Plasma ACTH (for distinguishing Cushing syndrome and disease) - requires use of *pre-chilled tube or container*
	  ○ Primary = ↑ cortisol, ↓ ACTH
	  ○ Secondary = ↑ cortisol and ↑ ACTH

Addison's disease *(hypoadrenalism)*	- ACTH stimulation (Cortrosyn) test - Insulin tolerance test - Metyrapone test

Congenital adrenal hyperplasia	- 21-Hydroxylase
	  ○ most common enzyme deficiency leading to CAH
	  ○ Confirmatory test
	  ○ Part of newborn screening
	  ○ Necessary for aldosterone and cortisol synthesis
	- Plasma 17-hydroxyprogesterone and pregnanetriol
	  ○ Patients have low aldosterone and cortisol but have high androgen levels
	- Urinary pregnanetriol and 17-ketosteroids
	- Scale used for grading hirsutism: ***Ferriman-Gallway scoring system***

Phaeochromocytoma	- A tumor characterized by hyperproduction of catecholamines - Plasma free metanephrine (MN) /normetanephrine (nMN) - Clonidine suppression test
Neuroblastoma	- Urine vanillylmandelic acid (VMA) and homovanillic acid (HVA)

GONADAL/REPRODUCTIVE HORMONES

HORMONE	SOURCE	CLINICAL SIGNIFICANCE
Testosterone	Testes	• **Primary hypergonadism** = testicular tumors • **Primary hypogonadism** = Klinefelter syndrome
Estradiol *(abundant in premenopausal women)*	Ovaries	• **Primary hyperestrinism** = ovarian tumors • **Primary hypoestrinism** = menopause
Progesterone	Ovaries and placenta	• Increased in 2nd to 3rd trimester of pregnancy • Important in the assessment of placental function and fertility studies
Estriol *(abundant in pregnant women)*	Placenta	• Assessment of fetoplacental units
hPL		• Important in the assessment of placental function
hCG		• Pregnancy (routine test kits) • Also a tumor marker (hCG-producing tumors)

*Estrone: abundant in post-menopausal women

TUMOR MARKERS

CLINICAL UTILITIES

1. Tumor staging
2. Monitoring therapy
3. Predicting patient outcomes
4. Detecting cancer recurrence

Note!
- Tumor markers cannot be used for screening or not as a stand-alone test for diagnosing cancer
- No tumor marker has satisfied all the characteristics of an ideal tumor marker

CHARACTERISTICS OF AN IDEAL TUMOR MARKER

1. Cancer-sensitive: present in elevated amounts only in the serum of patients with a malignant tumor; MAb-defined tumor markers
2. Cancer-specific (such as her2/neu): absent in the serum of healthy individuals and those with non-malignant diseases
3. Tissue-specific (such as PSA)
4. Readily detectable in the blood
5. Shows elevation at the early stage of cancer
6. Serum levels correlate with tumor burden

METHODS

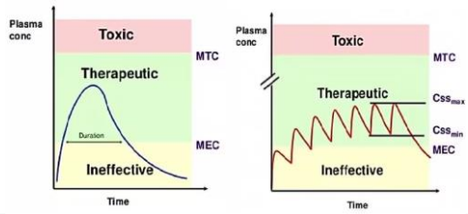
1. Immunoassay: subject to "hook effect" (immunoassays based on sandwich reaction wherein both the capture and label antibodies are saturated resulting in a lack of sandwich reaction)
2. HPLC: most widely used to detect catecholamines
3. Enzyme assays: carcinoplacental isoenzymes of ALP
4. Immunohistochemistry: requires biopsy specimens to detect tumor markers direct within solid tissues

CLINICALLY USEFUL TUMOR MARKERS	
AFP	- Hepatocellular carcinoma (HCC or Hepatoma)
CEA	- Colorectal carcinoma
CA 15-3	- Breast carcinoma
CA 19-9	- Pancreatic carcinoma
CA 72-4	- Gastric carcinoma
CA 125	- Ovarian carcinoma
CYFRA 21-1	- Lung carcinoma
HER-2/neu	- Breast carcinoma
Catecholamines	- Phaeochromocytoma (MN/nMN) - Neuroblastoma (VMA/HVA)
PSA	- Prostatic carcinoma
Nuclear matrix protein	- Urinary bladder carcinoma
β-hCG	- Gestational trophoblastic disorders - Choriocarcinoma - Teratoma - Hydatidiform mole - Testicular carcinoma or seminoma

TOXICOLOGY

THERAPEUTIC DRUG MONITORING

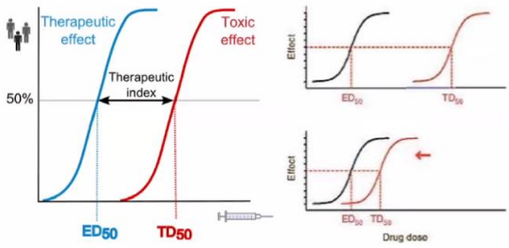
- TDM ensures that a given drug produces maximal therapeutic benefit and minimal side effects; to achieve a constant serum level of the drug that will be therapeutic
- Indications:
  - Monitoring compliance
  - Detecting suspected adverse effects (pharmacovigilance)
  - Tailor-fitting doses based on the individual's clinical response



Basic Pharmacologic Concepts

- **Peak**
  - highest point; amount of drug absorbed and distributed is greater than the amount metabolized and excreted
  - must not be above the MTC (Minimum Toxic Concentration)
  - 1 hour after oral administration
- **Trough**
  - lowest concentration achieved just before the next dose
  - most not be below the MEC (Minimum Effective Concentration)
- **Half-life**
  - time required for the concentration of the drug to decrease by half
  - affected by protein binding and renal and hepatic functions
  - considered when determining the appropriate time of subsequent dose administration and blood collection
  - has bearing in the attainment of a steady state
- **Steady state**
  - amount of drug absorbed and distributed equals amount metabolized and excreted
  - applies to multiple dosing regimens
  - the point at which the optimal peak and trough concentrations are less than the MTC and above the MEC
  - usually reached after 5-7 half-lives
- **TD_{50}**: drug dose that produces adverse effect in 50% of the population
- **ED_{50}**: drug dose that produces beneficial effect in 50% of the population

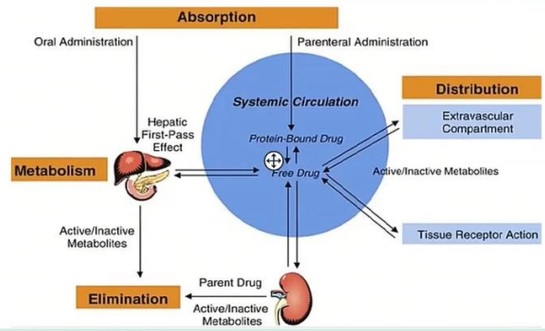
- **Therapeutic index**: ratio between TD_{50} and ED_{50}
  - The wider the safer
  - TDM is only recommended for drugs with narrow TI



- **MTC**: lowest concentration of drug in the blood that will produce adverse response
- **MEC** - lowest concentration of drug in the blood that will produce desired effect
- **Therapeutic range** - range of values between the MTC and MEC that produce therapeutic effect

Instrumentation

- **Thin-layer Chromatography**
  - used as a semi-quantitative screening test
  - confirmatory: GC or LC coupled with MS
  - **R_f =** distance migrated by a sample component divided by the distance migrated by the solvent
- **Liquid Chromatography (LC)/HPLC**
  - For insufficiently volatile and thermolabile compounds.
  - **t_R** = retention time; time it takes for a compound to elute in GC or HPLC
  - **Q** = peak area or peak height ratio; used to quantify
- **Gas Chromatography (GC)**
  - useful for compounds that are naturally volatile or can easily be converted into a volatile form (various organic molecules including many drugs)
  - GC/MS: Gold standard for drug analysis
- **Mass spectrometry (MS)**
  - common detector system in GC or HPLC (for insufficiently volatile and thermolabile compounds)
- **Immunoassays**
  - EIA, RIA, CLIA
  - Immunoturbidimetry (PETINIA: Particle-enhanced Turbidimetric Inhibition Immunoassay)
  - Homogenous assays such as EMIT and FPIA



Pharmacokinetics (ADME)

- what the body does to the drugs
- drug activity or fate of drugs in the body as influenced by:
  - **Absorption**
    - Transport of the drug from the site of administration to the blood
    - through the GI tract for orally administered drugs
    - dependent on many factors and is related to the drug's bioavailability
    - 100% bioavailability if the route is parenteral administration/intravenous
      - IV is indicated for drugs susceptible to chemical modification by the liver – first pass metabolism by the liver
    - Drugs in the circulation may bind to proteins such as albumin and α1 acid-glycoprotein; it is the free form that is biochemically active
    - Protein abnormalities may lead to altered drug response
  - **Distribution** - diffusion out of the vasculature into interstitial and intracellular spaces; dependent on the lipid solubility of the drug
  - **Metabolism** - hepatic uptake and enzymatic biotransformation during passage through the liver (first-pass metabolism)
  - **Excretion** - elimination through hepatic or renal clearance or a combination of the two

Pharmacodynamics

- What the drug does to the body
- Biochemical and physiological effects of drugs and the mechanisms of their actions
- Involves receptor binding, post-receptor effects, and chemical interactions

Specimen Considerations

- Steady state must be established before sampling (5-7 half-lives).
- STAT sampling when toxicity is suspected.
- **Trough specimen:** collected before the next dose is administered
- **Peak specimen**
  - Oral: collected 1 hour after administration (Half-life: ≥2 hours)
  - IV: collected 15-30 minutes after administration
  - IM: collected 30-60 minutes after administration
- Gel separator tubes such as SST and PST: may falsely ↓ antiarrhythmics and tricyclic antidepressants due to drug absorption by the gel
- **Appropriate specimen/Collection tubes**
  - **Serum (plain tube)** - preferred for most drug assays
  - **Heparinized plasma** - suitable for most drugs; except when lithium is the analyte or in free drug assays (may affect protein binding)
  - **EDTA** whole blood - appropriate for immunosuppressants
  - **Hirudin** - recommended for the impedance method of aspirin determination

DRUGS THAT REQUIRE TDM	
	Representative drugs
Analgesics	- **Acetylsalicylate (Aspirin)** ○ Has analgesic, antipyretic, and anti-inflammatory effects ○ Acts by inhibiting cyclooxygenase which decreases thromboxane and prostaglandin synthesis ○ Toxic effects include: ▪ Mixed acid-base disorder (metabolic acidosis followed by respiratory alkalosis) ▪ It is also a direct stimulator of the respiratory center, and hyperventilation can result in respiratory alkalosis ▪ Reye syndrome (among children with post-viral infection) - **Acetaminophen (Tylenol)** ○ Toxic effects include: ▪ Severe hepatotoxicity ▪ Some intermediates are free radicals causing toxic effect to the cell that leads to necrosis of the liver
Cardioactive drugs/ Cardiotropics	- **Cardiac glycosides (Digoxin)** – used to treat Atrial fibrillation and Congestive heart failure *(acts by inhibiting the sodium-potassium ATPase pump)* - **Antiarrhythmics** – used to treat tachyarrythmias or ventricular tachycardias (action depends on the blockage of one of the channels or receptors involved in the generation of action potential) ○ **Class I** (Lidocaine, Quinidine, Procainamide) ▪ block sodium channels ▪ NAPA is a metabolite of Procainamide that is also monitored; when a drug has a metabolite that mimics its action or enhances its toxicity, that metabolite must also be measured ○ **Class II** (Propranolol) – blocks adrenergic receptors ○ **Class III** (Amiodarone) – blocks potassium channels ○ **Class IV** (Verapamil) – block calcium channels

Antiepileptics/ Anticonvulsants / Antiseizures	- **Gabaergic drugs** (increase inhibition of hyperexcitatory states)
	  ○ used to treat grand mal seizures (tonic-clonic seizures), generalized, and simple seizures:
	  ▪ *Phenobarbital* - slow-acting barbiturate; inactive form = primidone which is quickly converted to a phenobarbital and must be monitored also
	  ▪ *Benzodiazepines*
	- **Glutamatergic drugs** (decrease nervous stimulation of hyperexcitatory states)
	  ○ *Ethosuximide* - for petit mal seizures
	  ○ *Phenytoin* - for tonic-clonic, generalized, and simple seizures
	  ○ *Carbamazepine* - for tonic-clonic, generalized, and simple seizures
	- **Valproic acid**
	  ○ Has multiple actions; for treatment of petit mal seizures, grand mal seizures, generalized, and partial seizures
	- **New generation anticonvulsants** (has wider therapeutic index; safer; do not usually need TDM): *Tiagabine, Vigbatrin, Gabapentin, Topiramate, Felbamate*

Psychoactive drugs/ Antidepressants	- **Neuroleptics/Antipsychotics** (used to treat schizophrenia)
	  ○ Chlorpromazine, Haloperidol, Clozapine
	- **Antidepressants and Mood Stabilizers**
	  ○ *Lithium* - used to treat bipolar/Manic-depressive disorders; overdose can lead to hypercalcemia
	  ○ *TCAs* (amitriptyline [nortriptyline], imipramine [desipramine], doxepin) - used to treat clinical depression

Bronchodilators	- Theophylline - used to treat asthma and other Chronic-Obstructive Pulmonary Disorders (COPDs)

Antibiotics	- **Aminoglycosides**
	  ○ Tobramycin, Amikacin, Gentamicin, Kanamycin
	  ○ used to treat gram negative infections via disruption of protein synthesis
	  ○ toxic effects include autotoxicity and nephrotoxicity
	  ○ both peak and trough measurement are done
	- **Glycopeptides**
	  ○ *Vancomycin*
	  ○ Used to treat gram positive infections via inhibition of cell wall synthesis
	  ○ May cause autotoxicity, nephrotoxicity, and red man syndrome (erythemic flushing of the extremities); renal function is monitored via GFR (serum creatinine)

Immunosuppressants	- **Calcineurin inhibitors**
	  ○ Reduce production of IL-2 and inhibits proliferation of T-lymphocytes
	  ○ *Cyclosporine* - used to prevent graft rejection and GVHD
	  ○ *Tacrolimus* - same indication as cyclosporine but is 100x more potent
	- **Proliferation signal inhibitors**
	  ○ Sirolimus, Everolimus, Mycophenolate

Antineoplastics	- **Antimetabolites**
	  ○ such as *Methotrexate*
	  ▪ Folic acid antagonist, inhibits DNA synthesis
	  ▪ used for the treatment of leukemias (CML) and lymphomas (Hodgkin's)
	  ▪ requires leucovorin rescue to prevent toxic effects
	  ▪ Leucovorin rescue: administration of leucovorin to offset the effect of methotrexate and spare the normal cells of its toxic effect; cytolytic or cytotoxic and attack normal cells at high concentrations
	- **Alkylating agents**
	  ○ such as *Busulfan*
	  ▪ used to treat CML and used for myeloablation

ILLICIT DRUGS	
	Representative drugs
Stimulants *(Uppers)*	- **Cocaine** – therapeutically used as local anesthetic; primary metabolite is ***benzoylecgonine*** and has a short half-life of 1-2 hours (0.5-2 hours) - **Amphetamines** (MDMA – methylenedioxymethamphetamine or ecstasy, methamphetamine or shabu) - used to treat narcolepsy and attention-deficit disorder
Hallucinogens	- **Marijuana** (*Cannabis sativa*) - primary cannabinoid component is ***THC (Tetrahydrocannabinol)*** which is detected in urine (THC-COOH) - **Phencyclidine** is also called as angel hair or angel dust; lipophilic or rapidly distributes into fatty tissues and the brain which makes elimination slow; effects: auditory and visual hallucinations - **Lysergic acid diethylamide** is the most potent hallucinogen; causes undulating/blurred visions and panic reactions/bad trip; just 20 ug can cause potent effects
Opiates/Narcotics *used for sedation, analgesia, anesthesia*	- **Naturally occurring**: Opium, Morphine, and Codeine (antitussive) - **Chemically modifie**d: Heroin, Oxycodone, Hydromorphone - **Fully synthetic**: Meperidine (Demerol), methadone (Dolophine), propoxyphene (Darvon), pentazocine (Talwin), and fentanyl (Sublimaze) ○ Fentanyl is 80x more potent than morphine

Sedative/Hypnotics *CNS depressants*	- **Barbiturates**
	  ○ Intermediate-acting: Amobarbital
	  ○ Short-acting: Secobarbital, Pentobarbital
	  ○ Long-acting: Phenobarbital
	- **Benzodiazepines**: Diazepam (Valium), Chlordiazepoxide, Lorazepam
	- **Methaqualone** (has similar properties with barbiturates)

OTHER SUBSTANCES AND POISONS

- **Acute toxicity**: associated with a single, short-term exposure to a substance, the dose of which is sufficient to cause immediate toxic effects
- **Chronic toxicity**: bioaccumulation; results from repeated frequent exposure for extended periods at doses insufficient to cause an immediate response; may affect different systems

1. **Ethanol**
   - Also a CNS depressant
   - Chronic consumption: chronic hepatitis progressing to alcoholic cirrhosis
   - Pathologic effects due to formation of acetaldehyde products
   - Indicators of ethanol intoxication: GGT, AST, AST/ALT (De Ritis ratio)
   - Not legally intoxicated: <100 mg/dL (0.1% w/v)

STAGES OF IMPAIRMENT	
BAC *(Blood alcohol concentration)*	**LEVEL OF IMPAIRMENT**
0.01 -0.05% w/v	No obvious impairment
0.03-0.12% w/v	Mild euphoria; some impairment of motor skills
>0.1% w/v	Legal intoxication (in the USA)
0.09-0.25% w/v	Loss of critical judgment; memory impairment
0.18-0.30% w/v	Mental confusion; strongly impaired motor skills
0.27-0.40 % w/v	Impaired consciousness
0.35-0.50% w/v	Coma and possibly death

   - Methods: Osmometry, enzymatic assay, and gas chromatography
   - Alcohol should not be used as antiseptic during blood collection because it falsely increases the estimate when using osmometers
   - Reference method: Gas liquid chromatography
2. **Methanol**: causes acidosis, blindness due to optic neuropathy, and death due to formation of formaldehyde which is metabolized to formic acid
3. **Isopropanol**: produces severe, acute ethanol-like symptoms that persist for a long period of time
4. **Ethylene glycol**: ingestion produces severe metabolic acidosis (oxalic and glycolic acid; calcium oxalate, monohydrate crystals in urine) and renal tubular damage
5. **Carbon monoxide**: has 245x greater affinity for Hemoglobin compared to oxygen; this shifts the oxyhemoglobin dissociation curve to the left (hypoxia)
   - screening test = spot test (cherry-red appearance of the blood)
   - spectrophotometry
   - Gas chromatography
   - Co-oximetry
6. **Cyanide** - commonly used as rodenticide/insecticide; characteristic odor of bitter almonds; binds to heme iron and mitochondrial cytochrome oxidase; methods: urinary thiocyanate assay, ISE, and photometry
7. **Heavy metals**
   a. **Arsenic**
      - component of insecticides, pesticides, and herbicides
      - has high affinity for keratin (keratinophilic heavy metal)
      - Mees' lines: white lines in the nail beds
      - binds to thiol groups or sulfhydryl groups in protein coenzymes disrupting metabolic systems e.g. oxidative phosphorylation and ATP production
      - "garlic breath" odor and metallic taste
   b. **Cadmium**: metal food containers or industrial exposure; may cause renal tubular dysfunction characterized by proteinuria, glycosuria, and aminoaciduria
   c. **Lead**: inhibits many enzymes and affects vitamin D metabolism and heme synthesis pathway; binds ALA dehydratase and ferrochelatase or heme synthase; coarse basophilic stippling, increased urine ALA, increased free erythrocyte protoporphyrin
   d. **Mercury**: acquired through inhalation and ingestion; may also cause renal tubular damage
      - Forms of mercury compounds:
        - **Elemental or metallic** (Hg^o): liquid at room temperature; can be ingested without significant effects
        - **Cationic/inorganic** (mercurous, Hgt and mercuric, Hg2+) - moderately toxic
        - **Organic** (alkyl mercury such as methyl mercury): environmental pollutants; extremely toxic/most toxic form

- Methods for Heavy metals
  - Reinsch test: screening test for heavy metals; arsenic = black, mercury = silvery gray
  - AAS: most commonly used method; gold standard for divalent cations and trace metals
  - Anodic stripping voltammetry: polarography; lead determination